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Investor Presentation | Japan

Electronic Components: Renewed Focus on Value-Added N. American Smartphone Components Plus AI Computing

In addition to proliferation of devices leveraging AI computing, we expect F3/27 earnings for suppliers of high value-added components in N. American smartphones to top consensus.

Key Takeaways

- OW on TDK: Top Pick. Earnings expansion for rechargeable batteries and HDD-related products.
- OW on Murata Mfg: Growing demand and utilization rates for high-value-added MLCCs.
- OW on Alps Alpine: High-performance camera actuators to ramp up from Apr-Jun.
- OW on Niterra: Continued earnings growth for replacement plugs and SPE electrostatic chucks.
- OW on Hirose Electric: Earnings expansion for general industrial machinery/AI server connectors.

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ELECTRONIC COMPONENTS

Japan

Industry View

In-Line

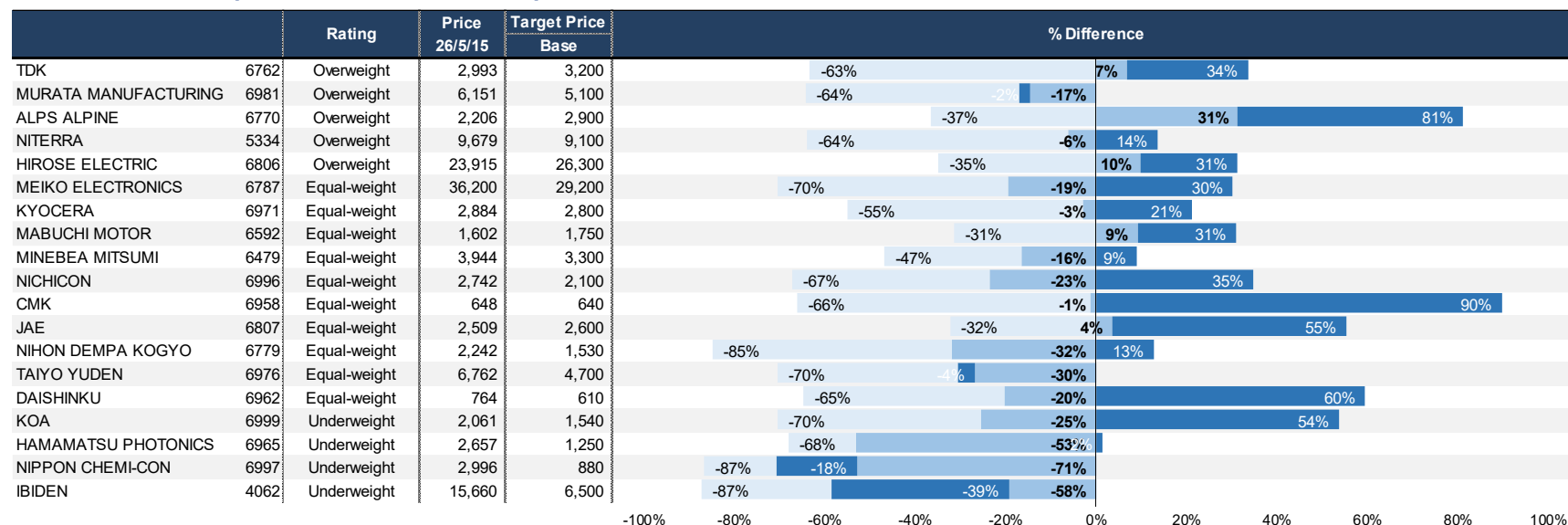
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Renewed Focus on Value-Added N. American Smartphone Components Plus AI Computing

■ Risk-Reward Snapshot: Ranked in order of preference



Note: Not included Iriso Electronics (6908) data in this material as full-year F3/26 results have not yet been announced

Source: FactSet, Morgan Stanley Research

Contents

Electronic Components: Valuation Comparison and Earnings Forecast Comparison

Sales Trends (Quarterly, Semi-annual, Annual)

Operating Profit Trends (Quarterly, Semi-annual, Annual)

Capex

Depreciation

Inventory and Inventory Days

Production & Inventory

Stock Price Performance

FX Sensitivities of Electronic Components Makers/Changes in the Competitiveness of Core Products Under a Weak Yen Environment

Sales and Production by Region

Preparing for Smartphone Price Compression and Slowdown in Technological Innovation

Memory Price Outlook

End-product Market and Major Components Market (Calendar Year)

End-product Market and Major Components Market (Quarterly)

MLCC Market Data

Capacitor Market: 016008 MLCCs are 75% smaller than 0201 MLCCs

MLCC Sales, Market Share, OP

Supply Chains of MLCC and Aluminum Capacitors

Capex by Hyperscalers has Increased Significantly

We Expect Ibiden Will Maintain Nearly 100% Market Share in FC Packages for NVIDIA

Ibiden: Medium-term Results and Forecasts

Connector Market

Connector Coverage Companies: Medium-term Results and Forecasts

Global Market Share Trend and Split for Spark Plugs / Exhaust Gas Oxygen Sensors

Niterra: Medium-term Results and Forecasts

Small Size LiB Market Overview

TDK: LiB Earnings Expansion, Major Acquisition and Transfer of Business

TDK: Actions for Business Portfolio Improvement

TDK: Mid-term Results and Forecasts

iPhone Sell-in Forecasts / iPhone Camera Actuators by Model

Alps Alpine: We Expect New Smartphone Camera Actuators to Contribute to Earnings

PWB Market

Meiko Electronics: Mid-term Results and Forecasts

HDD Market by Company (Quarterly, Fiscal Year)

HDD Head Market

HDD Motor Market

Supply Chain of HDDs

Electronic Components: Valuation Comparison and Earnings Forecast Comparison

		Rating	Price 26/5/15	Mkt Cap (JPY bn)	Net F.Asset (JPY bn)	EV (JPY bn)	P/E			Ex. Cash P/E			EV/EBITDA			P/B			ROE		
							F26e	F27e	F28e	F26e	F27e	F28e	F26e	F27e	F28e	F26e	F27e	F28e	F26e	F27e	F28e
IBIDEN	4062	Underweight	15,660	4,392.1	135.5	4,256.7	87.0	61.6	45.9	81.7	57.8	43.1	27.2	21.1	18.4	7.9	7.1	6.2	9.5%	12.3%	14.6%
NITERRA	5334	Overweight	9,679	1,922.0	26.8	1,895.2	16.3	14.5	12.4	16.5	15.1	13.2	9.6	8.8	7.9	2.4	2.2	1.9	15.2%	15.6%	16.4%
MINEBEA MITSUMI	6479	Equal-weight	3,944	1,597.0	-227.5	1,824.6	18.4	16.9	16.1	21.2	19.4	18.5	9.8	9.1	8.7	1.8	1.7	1.6	10.4%	10.5%	10.1%
MABUCHI MOTOR	6592	Equal-weight	1,602	412.8	146.8	266.0	17.7	15.8	14.7	11.9	11.2	10.5	6.5	6.1	5.8	1.2	1.2	1.2	6.8%	7.5%	8.3%
TDK	6762	Overweight	2,993	5,679.8	525.9	5,153.9	24.5	21.8	19.8	22.2	19.8	18.0	9.9	9.2	8.5	2.7	2.5	2.3	11.5%	11.9%	12.0%
ALPS ALPINE	6770	Overweight	2,206	432.4	129.0	303.4	12.0	8.8	7.6	8.3	6.2	5.5	3.3	2.8	2.5	1.0	0.9	0.8	8.7%	11.0%	11.6%
NIHON DEMP KOGYO	6779	Equal-weight	2,242	49.3	-16.4	65.7	22.4	19.1	19.1	28.6	24.3	24.3	8.0	7.1	6.8	1.6	1.5	1.4	7.4%	8.2%	7.8%
MEIKO ELECTRONICS	6787	Equal-weight	36,200	931.5	-92.6	1,024.1	37.2	27.1	21.3	41.0	29.9	23.4	19.1	14.3	11.5	6.3	5.2	4.3	18.2%	21.0%	22.1%
HIROSE ELECTRIC	6806	Overweight	23,915	785.9	225.5	560.4	21.8	20.0	18.2	16.1	15.2	14.2	8.2	7.6	7.0	2.0	1.9	1.9	9.5%	10.0%	10.5%
JAE	6807	Equal-weight	2,509	173.0	10.1	163.0	23.2	20.6	19.2	22.3	19.9	18.5	5.4	5.1	4.9	1.2	1.2	1.1	5.2%	5.7%	6.0%
CMK	6958	Equal-weight	648	43.7	-22.8	66.5	28.9	20.1	15.4	41.6	28.9	22.2	6.0	5.4	4.9	0.6	0.6	0.6	2.1%	3.0%	3.8%
DAISHINKU	6962	Equal-weight	764	24.3	-17.2	41.5	40.5	24.3	16.2	69.2	41.5	27.7	6.6	5.9	5.3	0.7	0.7	0.7	1.7%	2.8%	4.2%
HAMAMATSU PHOTONICS	6965	Underweight	2,657	821.1	31.0	790.1	68.2	52.4	43.4	68.0	52.1	43.2	20.7	16.6	15.5	2.6	2.6	2.6	3.7%	5.0%	6.0%
KYOCERA	6971	Equal-weight	2,884	4,062.5	1,797.8	2,264.8	40.6	36.8	31.1	24.4	22.7	19.8	9.6	8.8	7.9	1.4	1.5	1.6	3.2%	3.7%	4.6%
TAIYO YUDEN	6976	Equal-weight	6,762	849.2	-71.1	920.3	37.3	30.6	27.2	37.0	30.3	26.9	11.3	10.1	9.6	2.7	2.6	2.5	7.5%	8.7%	9.3%
MURATA MANUFACTURING	6981	Overweight	6,151	11,344.7	663.0	10,681.7	38.4	31.4	26.9	36.5	29.8	25.6	19.3	16.6	14.7	4.2	4.0	3.8	11.1%	13.1%	14.4%
NICHICON	6996	Equal-weight	2,742	185.9	-0.3	186.1	29.7	25.9	23.9	30.0	26.2	24.2	10.5	9.7	9.3	1.5	1.5	1.4	5.2%	5.7%	6.0%
NIPPON CHEMI-CON	6997	Underweight	2,996	76.9	-48.7	125.5	29.3	24.1	21.0	44.8	36.9	32.2	10.2	9.5	8.9	1.3	1.2	1.2	4.5%	5.3%	5.8%
KOA	6999	Underweight	2,061	70.0	-6.6	76.6	23.2	20.7	17.8	23.2	20.7	17.8	6.1	5.7	5.3	1.0	0.9	0.9	4.2%	4.5%	5.1%
Average (or total)				33,854.4		30,666.1	32.4	25.9	22.0	33.9	26.7	22.6	10.9	9.5	8.6	2.3	2.2	2.0	7.7%	8.7%	9.4%

(JPY bn, JPY/Shares)		Sales					OP					Net Profits					EPS					BPS
		F26e	Ce	FSe	F27e	FSe	F26e	Ce	FSe	F27e	FSe	F26e	Ce	FSe	F27e	FSe	F26e	Ce	FSe	F27e	FSe	F26e
IBIDEN	4062	499.7	500.0	501.4	602.5	594.0	77.0	90.0	93.7	106.4	127.2	52.1	58.0	62.5	73.6	86.2	180.0	207.7	221.7	254.3	305.7	1,988.7
NITERRA	5334	754.6	790.0	764.4	788.6	803.7	156.6	150.0	152.7	171.7	168.8	114.6	105.0	109.8	125.5	122.5	592.4	535.0	559.4	665.8	629.4	4,013.6
MINEBEA MITSUMI	6479	1,651.1	1,690.0	1,664.2	1,703.3	1,730.2	116.9	120.0	118.5	127.5	130.9	86.0	83.0	84.8	93.8	94.1	214.0	206.7	210.5	233.7	234.6	2,143.7
MABUCHI MOTOR	6592	213.4	213.0	208.4	227.3	218.2	27.2	26.0	24.5	29.3	26.8	22.3	21.5	20.3	23.8	21.7	90.7	89.6	83.7	101.5	90.4	1,376.4
TDK	6762	2,648.5	2,580.0	2,604.1	2,869.4	2,764.9	309.8	295.0	298.6	348.1	336.9	231.7	225.0	221.8	260.4	252.3	122.1	118.5	116.4	137.2	132.6	1,107.6
ALPS ALPINE	6770	1,082.0	1,045.0	1,044.0	1,179.9	1,090.6	50.8	48.5	43.8	68.1	51.6	36.3	30.0	28.3	48.6	35.0	184.5	150.4	142.8	250.3	178.4	2,169.4
NIHON DEMP KOGYO	6779	56.1	60.6	56.1	58.0	58.0	3.8	4.0	3.8	4.4	4.4	2.3	2.3	2.3	2.7	2.7	100.0	99.9	100.0	117.3	117.3	1,393.5
MEIKO ELECTRONICS	6787	305.1	320.0	285.5	368.7	333.9	33.6	38.0	30.7	45.5	38.7	25.0	27.0	24.1	34.3	31.2	974.1	1,039.7	938.1	1,336.5	1,215.6	5,770.9
HIROSE ELECTRIC	6806	224.2	230.0	221.8	238.6	235.7	46.6	46.0	47.3	49.6	52.6	34.8	34.0	34.7	36.9	38.7	1,096.5	1,039.1	1,054.7	1,198.4	1,193.7	11,835.7
JAE	6807	231.4	240.0	234.1	241.4	243.6	10.3	9.5	14.1	11.5	16.6	7.3	6.0	9.6	8.2	11.0	108.3	89.0	141.0	121.6	163.9	2,090.3
CMK	6958	104.9	104.0	103.5	109.9	108.5	2.9	3.2	4.0	3.9	4.5	1.6	2.0	1.9	2.3	2.4	22.4	28.1	26.7	32.3	33.7	1,086.7
DAISHINKU	6962	41.3	41.0	42.5	43.3	43.3	1.3	1.4	1.7	1.8	1.8	0.6	0.1	0.7	1.0	1.0	18.9	3.1	20.7	31.4	31.4	1,117.8
HAMAMATSU PHOTONICS	6965	216.9	212.8	212.1	233.6	224.7	14.9	18.0	16.2	19.9	18.8	11.6	14.0	14.2	15.2	15.5	39.0	46.9	47.3	50.7	52.4	1,009.7
KYOCERA	6971	1,895.2	1,940.0	1,983.0	1,981.5	2,051.4	121.5	130.0	119.7	138.0	140.0	92.9	141.0	124.0	99.7	135.3	71.1	102.7	95.0	78.4	108.8	2,059.5
TAIYO YUDEN	6976	383.0	384.0	382.5	414.7	417.3	31.8	30.0	37.3	39.1	55.7	24.9	18.0	25.4	30.3	37.2	181.3	143.9	196.1	221.1	289.9	2,462.3
MURATA MANUFACTURING	6981	1,944.9	1,960.0	1,971.1	2,195.1	2,199.2	372.8	380.0	394.3	459.3	504.6	292.5	293.0	311.1	358.2	389.2	160.0	161.0	170.6	195.9	213.6	1,461.4
NICHICON	6996	182.6	185.0	189.5	187.8	202.9	8.4	8.7	9.8	9.7	12.3	6.2	6.7	7.6	7.1	9.3	92.3	99.8	113.0	105.7	138.3	1,806.8
NIPPON CHEMI-CON	6997	147.6	160.0	149.4	153.5	158.9	5.6	8.0	7.0	6.4	9.2	2.8	4.0	3.9	3.4	5.1	102.3	142.2	147.1	124.2	192.4	2,322.6
KOA	6999	78.0	77.4	78.0	82.2	82.1	5.2	2.8	3.0	5.8	6.0	3.3	4.8	4.1	3.7	4.1	88.9	128.7	110.4	99.6	110.4	2,166.9
Average (or total)		12,660.6	12,732.8	12,695.5	13,679.3	13,561.0	1,396.9	1,409.1	1,420.3	1,645.9	1,707.4	1,048.8	1,075.4	1,091.1	1,228.7	1,294.6						

Note: For Mabuchi Motor, fiscal year ends in December; For Hamamatsu Photonics, fiscal year ends in Sep; Diluted EPS for our forecasts and basic EPS for company and consensus estimates; e = Morgan Stanley Research estimates; FSE = FactSet consensus; Net Financial Asset is based on current FY end actuals (Net Financial Asset = Cash & cash equivalents + Short-term marketable securities + Investments in securities – Interest-bearing debt)

Source: FactSet, Company data, Morgan Stanley Research

Sales Trends (Quarterly, Semi-annual, Annual)

(JPY bn)	F24 1Q	2Q	3Q	4Q	F25 1Q	2Q	3Q	4Q	F24 1H	2H	F25 1H	2H	F21	F22	F23	F24	F25	F26e	F27e	F28e	F26Ce 1He	2He	F26Ce	F26e FSE	F27e FSE	F28e FSE	+1Q FSE			
Sales																														
USD/JPY:Average Rate	155.9	149.3	152.4	152.3	144.5	147.5	154.0	156.9	152.6	152.4	146.0	155.5	112.4	135.4	144.5	152.5	150.6	150.0	150.0	150.0										
IBIDEN	4062	88.2	93.4	88.8	99.1	97.5	98.0	106.7	117.6	181.6	187.9	195.5	224.3	401.1	417.5	370.5	369.4	416.2	499.7	602.5	711.4	230.0	270.0	500.0	501.4	594.0	715.4	117.4		
NITERRA	5334	165.5	159.1	161.6	167.3	169.9	181.2	179.5	205.2	324.6	328.4	351.2	384.6	491.7	562.6	614.5	653.0	731.2	754.6	788.6	848.9	391.0	399.0	790.0	760.8	825.5	870.7	190.5		
MINEBEA MITSUMI	6479	355.5	422.8	369.6	374.8	366.9	411.4	420.1	432.2	778.2	744.5	778.3	852.3	1,124.1	1,292.2	1,402.1	1,522.7	1,664.4	1,651.1	1,703.3	1,763.6	846.5	843.5	1,690.0	1,664.2	1,730.2	1,804.2	396.7		
MABUCHI MOTOR	6592	45.1	50.2	50.8	50.1	47.0	48.0	52.1	47.3	95.3	100.9	94.9	99.4	134.6	156.7	178.7	196.2	200.4	213.4	227.3	242.2	104.6	108.4	213.0	208.4	218.2	228.5	53.2		
TDK	6762	518.8	570.7	581.0	534.3	535.8	647.6	621.0	646.3	1,089.5	1,115.3	1,183.4	1,267.2	1,902.1	2,180.8	2,103.9	2,204.8	2,504.8	2,648.5	2,869.4	3,116.3	-	-	2,580.0	2,604.1	2,764.9	2,914.9	623.7		
ALPS ALPINE	6770	233.6	256.7	250.3	249.8	238.9	268.8	247.5	258.3	490.3	500.1	505.7	505.8	802.9	933.1	964.1	990.4	1,019.5	1,082.0	1,179.9	1,246.6	508.0	537.0	1,045.0	1,044.0	1,090.6	1,066.8	246.2		
NIHON DENPA KOGYO	6779	12.6	13.7	13.4	13.4	12.7	14.0	13.8	14.8	26.3	26.8	28.7	28.6	45.4	52.5	50.3	53.1	54.6	56.1	58.0	59.8	29.5	31.1	60.6	56.1	58.0	59.8	12.0		
MEIKO ELECTRONICS	6787	47.0	51.1	53.6	55.0	53.1	58.4	60.6	68.5	98.2	108.6	111.5	129.1	151.3	167.3	179.5	206.8	240.6	305.1	368.7	429.5	-	-	320.0	285.5	333.9	429.5	58.5		
HIROSE ELECTRIC	6806	44.9	49.6	49.9	45.0	49.0	53.1	53.0	54.7	94.5	94.9	102.0	107.7	163.7	183.2	165.5	189.4	211.3	224.2	238.6	254.2	115.0	115.0	230.0	221.8	235.7	248.8	52.8		
JAE	6807	55.6	57.9	53.0	55.2	51.6	58.8	55.6	61.1	113.5	108.2	110.4	116.7	225.1	235.9	225.8	221.6	227.9	231.4	241.4	252.0	120.0	120.0	240.0	234.1	243.6	261.0	56.5		
CMK	6958	22.9	23.4	24.4	24.8	22.8	24.4	25.1	27.2	46.3	49.2	47.2	52.3	81.5	83.8	90.6	95.5	100.2	104.9	109.9	115.3	-	-	104.0	103.5	108.5	115.3	25.1		
DAISHINKU	6962	9.8	9.6	10.1	9.1	9.4	10.2	10.6	10.0	19.4	19.2	19.6	20.6	41.3	38.4	39.3	38.6	39.6	41.3	43.3	45.3	-	-	41.0	42.5	43.3	45.3	9.8		
HAMAMATSU PHOTONICS	6965	53.5	50.5	46.6	53.4	50.6	56.2	48.7	56.6	104.0	100.0	108.6	108.6	169.0	208.8	221.4	204.0	212.1	216.9	233.6	252.0	112.5	119.5	232.0	224.7	240.2	254.2	54.8		
KYOCERA	6971	498.9	499.7	493.5	522.4	478.0	513.3	490.3	548.2	998.6	1,015.9	991.4	1,038.5	1,838.9	2,025.3	2,004.2	2,014.5	2,070.2	1,895.2	1,981.5	2,107.4	-	-	1,940.0	1,983.0	2,051.4	2,184.7	463.5		
TAIYO YUDEN	6976	81.1	86.7	86.9	86.7	84.8	92.8	91.2	89.2	167.9	173.6	177.6	180.4	349.6	319.5	322.6	341.4	355.3	383.0	414.7	449.3	-	-	384.0	382.5	417.3	463.2	90.0		
MURATA MANUFACTURING	6981	421.7	461.8	448.0	411.9	416.2	486.6	444.4	460.6	883.5	859.9	902.8	905.0	1,812.5	1,686.8	1,640.2	1,743.4	1,830.9	1,944.9	2,195.1	2,485.0	960.0	1,000.0	1,960.0	1,971.1	2,199.2	2,455.7	459.9		
NICHICON	6996	41.2	43.4	48.2	43.0	39.6	41.1	47.6	45.5	84.5	91.2	80.7	93.0	142.2	184.7	181.6	175.8	169.7	182.6	187.8	192.7	-	-	185.0	189.5	202.9	223.6	44.9		
NIIPPON CHEMI-CON	6997	29.3	30.6	30.9	31.9	30.9	33.4	33.1	36.7	59.9	62.8	64.2	69.8	140.3	161.9	150.7	122.7	136.8	147.6	153.5	157.5	77.0	83.0	160.0	149.4	158.9	157.5	35.8		
KOA	6999	16.2	15.7	15.9	16.4	16.8	17.8	17.8	19.2	31.9	32.2	34.6	37.0	65.0	75.1	64.8	64.1	72.3	78.0	82.2	85.0	38.6	38.8	77.4	78.0	82.1	86.0	19.5		
Total	2,741.2	2,946.6	2,876.0	2,843.5	2,771.4	3,113.2	3,018.6	3,199.0	5,687.9	5,719.5	5,886.4	6,220.9	10,082.4	10,966.2	10,970.4	11,407.4	12,257.9	12,660.6	13,679.3	14,813.9	-	-	12,752.0	12,704.5	13,598.3	14,585.1	3,010.7			
Sales																														
(JPY bn)	F00	F01	F02	F03	F04	F05	F06	F07	F08	F09	F10	F11	F12	F13	F14	F15	F16	F17	F18	F19	F20	F21	F22	F23	F24	F25	F26e	F27e	F28e	
Sales																														
USD/JPY:Average Rate	110.5	124.9	121.9	113.1	107.5	113.3	116.9	114.4	100.7	92.9	85.7	79.0	82.9	100.1	109.8	120.1	108.3	110.8	110.9	108.7	106.1	112.4	135.4	144.5	152.5	150.6	150.0	150.0	150.0	
IBIDEN	4062	205.4	212.0	210.2	220.5	247.6	319.0	398.6	413.5	309.3	274.2	305.0	300.9	285.9	310.3	318.1	314.1	266.5	300.4	291.1	296.0	323.5	401.1	417.5	370.5	369.4	416.2	499.7	602.5	711.4
NITERRA	5334	224.3	221.4	228.9	228.8	241.2	284.9	344.9	345.6	292.1	243.9	269.2	284.7	302.8	329.8	347.6	363.3	372.9	409.9	425.0	426.1	427.5	491.7	562.6	614.5	653.0	731.2	754.6	788.6	848.9
MINEBEA MITSUMI	6479	287.0	279.3	272.2	268.6	294.4	318.4	331.0	334.4	256.2	228.4	269.1	251.4	282.4	371.5	500.7	609.8	638.9	879.1	884.7	978.4	988.4	1,124.1	1,292.2	1,402.1	1,522.7	1,664.4	1,651.1	1,703.3	1,763.6
MABUCHI MOTOR	6592	116.2	105.1	116.4	105.7	99.3	93.9	100.5	107.6	92.6	70.4	82.8	78.9	85.3	108.4	122.5	143.1	140.7	146.9	143.1	131.8	116.4	134.6	156.7	178.7	196.2	200.4	213.4	227.3	242.2
TDK	6762	689.9	575.0	608.9	658.9	657.9	795.2	862.0	866.3	727.4	808.9	875.7	814.5	851.6	984.5	1,082.6	1,152.3	1,178.3	1,271.7	1,381.8	1,363.0	1,479.0	1,902.1	2,180.8	2,103.9	2,204.8	2,504.8	2,648.5	2,869.4	3,116.3
ALPS ALPINE	6770	573.1	540.3	601.8	619.7	643.6	709.6	708.1	692.7	539.0	493.6	550.7	526.5	546.4	684.4	748.6	774.0	753.3	858.3	851.3	810.6	718.0	802.9	933.1	964.1	990.4	1,019.5	1,082.0	1,179.9	1,246.6
NIHON DENPA KOGYO	6779	74.1	49.3	49.6	55.9	60.7	63.6	73.3	78.1	59.2	52.6	54.9	50.8	50.6	48.7	44.9	43.8	44.0	42.5	39.5	39.2	45.4	52.5	50.3	53.1	54.6	56.1	58.0	59.8	
MEIKO ELECTRONICS	6787	23.0	17.2	21.6	29.2	40.2	50.3	65.3	78.5	75.8	69.8	74.7	63.0	60.7	79.2	90.9	95.3	95.9	108.5	118.9	115.5	119.3	151.3	167.3	179.5	206.8	240.6	305.1	368.7	429.5
HIROSE ELECTRIC	6806	83.5	61.1	67.9	79.0	89.6	104.4	115.5	118.0	89.9	84.5	92.4	94.8	95.9	125.0	125.7	120.3	115.1	125.3	124.6	121.8	133.5	163.7	183.2	165.5	189.4	211.3	224.2	238.6	254.2
JAE	6807	120.4	102.6	105.6	113.6	115.8	126.3	137.6	156.1	123.0	104.3	110.6	112.4	129.6	159.8	191.2	179.5	209.5	253.9	222.1	208.1	209.7	225.1	235.9	225.8	221.6	227.9	231.4	241.4	252.0
CMK	6958	122.3	104.5	111.7	115.4	116.0	119.2	128.2	138.5	107.6	90.1	87.0	80.6	72.7	71.0	73.7	73.1	75.4	86.9	90.2	82.6	70.0	81.5	83.8	90.6	95.5	100.2	104.9	109.9	115.3
DAISHINKU	6962	48.0	32.6	31.5	30.2	34.8	35.9	40.9	41.9	36.5	32.0	38.0	35.2	32.9	33.8	31.1	32.2	31.0	30.3	28.5	29.9	33.2	41.3	38.4	39.3	38.6	39.6	41.3	43.3	45.3
HAMAMATSU PHOTONICS	6965	51.6	62.6	60.7	60.9	70.5	73.9	87.0	96.3	98.8	72.0	91.0	101.9	98.1	102.2	112.1	120.7	121.9	130.5	144.3	145.9	140.3	169.0	208.8	221.4	204.0	212.1	216.9	233.6	252.0
KYOCERA	6971	1,285.1	1,034.6	1,069.8	1,140.8	1,180.7	1,181.5	1,283.9	1,290.4	1,128.6	1,073.8	1,266.9	1,190.9	1,280.1	1,447.4	1,526.5	1,479.6	1,422.8	1,577.0	1,623.7	1,599.1	1,526.9	1,838.9	2,025.3	2,004.2	2,014.5	2,070.2	1,895.2	1,981.5	2,107.4
TAIYO YUDEN	6976	190.0	152.1	153.7	163.3	172.3	186.5	221.2	238.3	185.5	195.7	210.4	183.8	192.9	208.2	227.1	240.4	230.7	244.1	274.3	282.3	300.9	349.6	319.5	322.6	341.4	355.3	383.0	414.7	449.3
MURATA MANUFACTURING	6981	584.0	394.8	395.0	414.2	424.5	490.8																							

Operating Profit Trends (Quarterly, Semi-annual, Annual)

	F24				F25				F24				F25				F21	F22	F23	F24	F25	F26e	F27e	F28e	F26Ce	F26e	F27e	F28e	+1Q	
(JPY bn)	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q	1H	2H	1H	2H	1H	2H									1He	2He	FSE	FSE	FSE	FSE		
OP																														
USD/JPY:Average Rate	155.9	149.3	152.4	152.3	144.5	147.5	154.0	156.9	152.6	152.4	146.0	155.5	112.4	135.4	144.5	152.5	150.6	150.0	150.0	150.0										
IBIDEN	4062	11.3	17.2	6.3	12.8	17.6	14.9	12.0	17.5	28.5	19.1	32.6	29.5	70.8	72.4	47.6	47.6	62.0	77.0	106.4	137.0	38.0	52.0	90.0	93.7	127.2	185.7	19.7		
NITERRA	5334	39.1	33.2	31.0	26.3	33.8	38.1	38.9	27.3	72.3	57.4	72.0	66.2	75.5	89.2	107.6	129.7	138.2	156.6	171.7	196.6	72.0	78.0	150.0	152.3	174.2	196.0	39.8		
MINEBEA MITSUMI	6479	18.9	27.9	26.2	21.5	17.4	27.0	30.8	28.8	46.8	47.7	44.4	59.6	90.3	76.5	69.0	91.6	102.7	116.9	127.5	134.0	53.0	67.0	120.0	118.5	130.9	133.0	26.1		
MABUCHI MOTOR	6592	3.9	5.3	6.8	5.7	6.4	5.7	7.4	5.9	9.2	12.5	12.1	13.4	13.8	10.8	15.5	21.6	25.5	27.2	29.3	31.6	12.8	13.2	26.0	24.5	26.8	28.8	6.4		
TDK	6762	57.9	75.4	75.8	15.1	56.4	91.2	83.1	41.7	133.3	90.9	147.6	124.8	166.8	168.8	172.9	224.2	272.4	309.8	348.1	383.6	-	-	295.0	298.6	336.9	372.5	61.1		
ALPS ALPINE	6770	1.2	10.0	14.1	8.8	3.7	17.5	12.0	8.8	11.2	22.9	21.2	20.8	35.2	33.6	19.7	34.1	42.0	50.8	68.1	77.8	9.0	39.5	48.5	43.8	51.6	50.3	8.9		
NIHON DENPA KOGYO	6779	1.2	1.2	1.3	1.0	0.8	0.7	0.8	1.2	2.4	2.3	1.4	1.9	5.2	8.3	4.3	4.6	3.4	3.8	4.4	4.4	1.4	2.6	4.0	3.8	4.4	4.4	0.6		
MEIKO ELECTRONICS	6787	3.3	6.0	5.4	4.4	5.6	5.8	6.1	7.1	9.3	9.8	11.4	13.2	13.3	9.6	11.7	19.1	24.6	33.6	45.5	57.4	-	-	38.0	30.7	38.7	57.4	6.6		
HIROSE ELECTRIC	6806	10.3	12.0	12.0	8.5	9.8	10.6	12.0	10.5	22.2	20.5	20.5	22.5	40.8	46.8	34.0	42.7	43.0	46.6	49.6	53.5	23.0	23.0	46.0	47.3	52.6	57.0	11.5		
JAE	6807	3.8	5.4	2.2	4.2	1.5	3.3	1.2	3.0	9.2	6.4	4.8	4.2	18.0	17.6	14.4	15.6	8.9	10.3	11.5	12.4	3.8	5.7	9.5	14.1	16.6	20.5	2.9		
CMK	6958	0.7	1.0	1.1	1.0	0.2	0.3	0.9	1.5	1.8	2.0	0.4	2.3	3.0	2.6	3.5	3.8	2.8	2.9	3.9	4.9	-	-	3.2	4.0	4.5	4.9	0.9		
DAISHINKU	6962	0.4	0.0	0.3	0.2	0.1	0.1	0.3	0.6	0.4	0.5	0.2	1.0	5.2	4.2	2.1	0.9	1.1	1.3	1.8	2.5	-	-	1.4	1.7	1.8	2.5	0.1		
HAMAMATSU PHOTONICS	6965	11.5	8.6	5.2	6.8	4.3	6.5	1.5	3.9	20.1	12.1	10.8	5.4	34.3	57.0	56.7	32.1	16.2	14.9	19.9	24.4	10.0	10.0	20.0	18.8	22.7	28.1	4.5		
KYOCERA	6971	21.0	16.9	-25.6	15.0	18.6	23.4	28.7	47.5	37.8	-10.5	41.9	76.2	148.9	128.5	92.9	27.3	118.1	121.5	138.0	166.8	-	-	130.0	119.7	140.0	145.5	26.6		
TAIYO YUDEN	6976	2.6	4.9	0.8	2.1	3.1	5.9	7.5	3.5	7.6	2.9	9.0	11.0	68.2	32.0	9.1	10.5	20.0	31.8	39.1	44.2	-	-	30.0	37.3	55.7	76.6	6.5		
MURATA MANUFACTURING	6981	66.4	91.8	76.0	45.5	61.6	103.5	37.9	78.8	158.2	121.5	165.1	116.7	424.1	297.9	215.4	279.7	281.8	372.8	459.3	537.3	177.0	203.0	380.0	394.3	504.6	609.5	82.5		
NICHICON	6996	0.2	1.6	2.0	1.4	0.7	0.8	2.3	2.6	1.8	3.4	1.5	4.9	6.4	12.7	8.9	5.2	6.5	8.4	9.7	10.5	-	-	8.7	9.8	12.3	15.1	2.0		
NIPPON CHEMI-CON	6997	0.5	1.3	0.4	1.5	0.2	0.8	1.0	1.4	1.9	1.9	1.0	2.4	8.8	12.9	9.4	3.7	3.4	5.6	6.4	7.1	3.0	5.0	8.0	7.0	9.2	7.1	1.2		
KOA	6999	0.2	0.2	0.1	0.6	0.5	0.8	1.8	0.5	0.5	0.7	1.3	2.3	5.7	10.2	3.3	1.2	3.6	5.2	5.8	6.6	1.1	1.8	2.8	3.0	6.0	7.0	0.5		
Total		254.3	320.0	241.4	182.4	242.3	357.0	286.2	292.0	574.3	423.8	599.3	578.2	1,234.3	1,091.6	898.1	995.2	1,176.2	1,396.9	1,645.9	1,892.6	-	-	1,411.1	1,422.6	1,716.6	2,001.8	308.3		
(JPY bn)	F00	F01	F02	F03	F04	F05	F06	F07	F08	F09	F10	F11	F12	F13	F14	F15	F16	F17	F18	F19	F20	F21	F22	F23	F24	F25	F26e	F27e	F28e	
OP																														
USD/JPY:Average Rate	110.5	124.9	121.9	113.1	107.5	113.3	116.9	114.4	100.7	92.9	85.7	79.0	82.9	100.1	109.8	120.1	108.3	110.8	110.9	108.7	106.1	112.4	135.4	144.5	152.5	150.6	150.0	150.0	150.0	
IBIDEN	4062	13.6	12.0	11.1	13.6	20.1	43.6	69.0	67.3	10.6	21.3	33.8	15.5	5.4	23.4	26.0	22.6	7.1	16.7	10.1	19.7	38.6	70.8	72.4	47.6	47.6	62.0	77.0	106.4	137.0
NITERRA	5334	25.5	11.8	15.8	20.7	26.1	41.5	52.4	35.0	-5.2	10.7	28.8	24.5	23.8	51.7	62.2	66.3	53.6	67.3	58.7	48.4	47.4	75.5	89.2	107.6	129.7	138.2	156.6	171.7	196.6
MINEBEA MITSUMI	6479	33.0	22.0	19.4	18.1	14.1	19.3	26.3	30.8	13.4	12.1	22.2	8.6	10.2	32.2	60.1	51.4	49.0	79.2	72.1	58.9	59.8	90.3	76.5	69.0	91.6	102.7	116.9	127.5	134.0
MABUCHI MOTOR	6592	28.6	22.1	29.1	24.6	17.3	8.1	10.7	11.7	5.7	3.0	6.6	3.1	5.1	9.3	16.9	23.0	24.2	24.1	21.2	17.5	12.9	13.8	10.8	15.5	21.6	25.5	27.2	29.3	31.6
TDK	6762	56.3	-43.7	22.1	54.3	59.8	60.5	79.6	87.2	-54.3	25.8	63.8	18.7	21.6	36.6	72.5	93.4	208.7	85.6	107.8	97.9	111.5	166.8	168.8	172.9	224.2	272.4	309.8	348.1	383.6
ALPS ALPINE	6770	24.6	13.2	41.1	36.2	31.1	45.5	22.1	19.9	-28.5	4.7	28.9	15.2	6.9	28.5	53.5	52.3	44.4	71.9	49.6	26.8	13.1	35.2	33.6	19.7	34.1	42.0	50.8	68.1	77.8
NIHON DENPA KOGYO	6779	12.6	0.5	0.1	2.5	4.5	5.3	9.4	9.6	-6.9	4.0	2.1	0.1	0.2	0.2	0.4	0.7	-9.6	0.4	-8.3	2.8	5.2	8.3	4.3	4.6	3.4	3.8	4.4	4.4	0.6
MEIKO ELECTRONICS	6787	1.3	-0.9	1.7	3.5	5.0	6.5	5.4	7.3	1.8	3.8	3.3	0.8	-0.6	0.9	-2.9	3.3	5.8	7.5	8.9	5.2	6.7	13.3	9.6	11.7	19.1	24.6	33.6	45.5	57.4
HIROSE ELECTRIC	6806	26.0	14.4	20.1	26.0	29.7	36.9	38.0	35.4	19.2	20.4	22.0	19.6	21.0	33.0	32.6	28.8	28.5	28.6	23.2	20.4	27.9	40.8	46.8	34.0	42.7	43.0	46.6	49.6	53.5
JAE	6807	10.6	5.2	6.6	10.1	7.9	11.0	11.7	14.4	-0.8	2.9	6.3	6.5	8.6	16.5	25.9	17.9	12.1	20.6	17.3	14.2	8.7	18.0	17.6	14.4	15.6	8.9	10.3	11.5	12.4
CMK	6958	9.3	-1.1	0.7	9.2	10.4	9.6	6.0	4.8	-2.6	-0.4	-1.0	0.8	0.4	-1.1	0.6	0.4	2.7	4.3	3.8	1.6	-1.7	3.0	2.6	3.5	3.8	2.8	2.9	3.9	4.9
DAISHINKU	6962	3.6	-3.5	-2.7	1.7	1.6	2.5	3.8	4.5	-0.4	1.2	1.4	0.3	0.4	0.8	-2.3	0.7	1.4	0.3	0.0	0.3	2.1	5.2	4.2	2.1	0.9	1.1	1.3	1.8	2.5
HAMAMATSU PHOTONICS	6965	5.9	8.7	5.9	4.1	9.7	11.7	15.7	19.9	20.1	3.7	15.8	21.8	17.6	16.8	21.7	23.6	20.5	22.8	27.3	25.4	21.8	34.3	57.0	56.7	32.1	16.2	14.9	19.9	24.4
KYOCERA	6971	207.2	51.6	83.4	109.0	101.0	103.2	135.1	152.4	43.4	63.9	155.9	97.7	76.9	120.6	93.4	92.7	104.5	95.6	94.8	100.2	70.6	148.9	128.5	92.9	27.3	118.1	121.5	138.0	166.8
TAIYO YUDEN	6976	35.4	6.2	8.2	11.0	2.1	6.7	22.0	21.3	-12.8	4.2	8.8	-8.0	5.0	11.4	13.2	23.4	12.4	20.2	35.2	37.2	40.8	68.2	32.0	9.1	10.5	20.0	31.8	39.1	44.2
MURATA MANUFACTURING	6981	174.2	51.0	59.2	74.2	69.5	89.8	113.4	115.8	-16.3	26.7	77.5	45.0	58.6	125.9	214.5	275.4	201.2	162.1	266.8	253.2	313.2	424.1	297.9	215.4	279.7	281.8	372.8	459.3	537.3
NICHICON	6996	10.3	5.1	7.7	4.6	6.3	6.0	6.7	4.6	-7.9	-4.5	5.1	2.1	-3.4	4.2	3.9	4.8	3.0	6.2	5.5	2.5	1.6	6.4	12.7	8.9	5.2	6.5	8.4	9.7	10.5
NIPPON CHEMI-CON	6997	13.6	-1.0	3.5	3.6	4.7	6.1	9.4	8.7	-6.9	-3.0	8.2	-2.6	-7.0	4.9	5.1	2.2	3.3	5.8	5.1	-2.9	3.0	8.8	12.9	9.4	3.7	3.4	5.6	6.4	7.1
KOA	6999	10.2	-0.9	1.5	2.6	2.4	4.1	6.2	3.8	-1.6</																				

Capex

		F00	F01	F02	F03	F04	F05	F06	F07	F08	F09	F10	F11	F12	F13	F14	F15	F16	F17	F18	F19	F20	F21	F22	F23	F24	F25	F26e	F27e	F28e	F26Ce
(JPY bn)																															
Capex																															
IBIDEN	4062	26.2	19.9	16.9	12.5	23.9	65.8	70.4	42.4	61.8	18.7	60.3	47.3	43.3	37.7	56.4	41.0	21.0	22.4	22.9	58.2	79.4	61.7	131.3	146.6	157.3	64.3	160.0	160.0	210.0	210.0
NITERRA	5334	23.5	25.5	10.8	10.4	14.0	26.9	29.3	63.2	24.2	11.0	8.8	14.0	24.0	41.0	36.4	45.3	36.3	37.6	56.6	51.0	33.4	29.6	31.3	41.2	37.5	49.0	40.0	45.0	45.0	47.3
MINEBEA MITSUMI	6479	39.9	26.2	16.4	18.8	31.1	21.9	17.0	24.9	19.6	10.8	27.0	26.9	42.8	19.8	35.0	41.6	29.8	53.2	53.1	48.8	44.2	71.9	144.8	81.4	88.7	96.5	99.1	102.2	105.8	95.0
MABUCHI MOTOR	6592	4.6	5.1	6.3	5.7	11.6	5.5	3.6	3.7	3.8	3.7	3.2	6.5	6.8	5.8	9.6	16.2	11.2	12.8	16.3	17.9	7.7	9.6	10.6	12.8	14.1	10.3	12.6	15.0	15.0	10.3
TDK	6762	99.5	58.8	41.5	44.9	61.0	73.9	70.4	84.3	98.4	64.4	78.6	99.7	85.6	68.6	102.5	160.7	167.6	178.6	173.6	173.4	212.4	291.3	275.7	218.6	225.3	298.6	290.0	290.0	300.0	370.0
ALPS ALPINE	6770	31.4	31.8	34.9	47.1	47.1	50.1	45.3	43.2	47.2	18.5	23.5	34.0	31.8	26.6	31.4	41.2	47.7	76.2	52.9	42.4	40.4	53.0	50.8	55.7	51.8	61.6	70.0	60.0	60.0	75.2
NIHON DEMP A KOGYO	6779	10.8	3.6	3.5	6.8	11.9	6.9	9.8	13.2	10.9	2.0	3.4	4.9	4.7	2.7	2.0	2.1	6.8	7.1	2.4	3.1	3.2	3.2	4.0	4.6	4.6	3.9	6.3	6.3	6.0	-
MEIKO ELECTRONICS	6787	3.5	3.2	2.0	4.2	10.0	10.1	16.6	11.2	13.7	7.0	17.4	9.6	6.6	6.6	4.1	3.1	3.1	9.6	15.8	11.8	10.7	14.4	17.4	19.9	29.8	-	55.0	55.0	55.8	-
HIROSE ELECTRIC	6806	7.5	4.3	3.5	6.6	5.5	9.0	13.0	10.2	7.3	7.2	9.0	11.3	11.4	11.3	14.6	13.3	15.2	19.6	21.3	18.2	14.6	15.2	19.6	32.8	21.2	19.9	28.0	28.6	30.5	25.0
JAE	6807	10.5	7.6	9.1	9.1	11.5	15.2	16.5	17.8	13.8	10.1	10.7	13.5	26.6	22.0	22.2	22.4	21.7	17.6	22.2	24.5	18.2	19.0	22.6	20.4	18.0	23.7	20.8	21.7	22.7	21.0
CMK	6958	9.7	10.1	8.5	2.9	4.6	6.2	12.3	14.7	8.0	2.4	1.9	1.8	3.5	2.6	5.0	3.1	4.6	7.0	7.5	8.3	3.7	6.1	6.7	16.0	18.9	9.2	6.5	6.0	6.3	8.4
DAISHINKU	6962	3.9	3.4	2.3	1.6	1.5	3.4	4.7	2.6	3.3	2.0	4.9	2.9	1.9	2.6	3.5	3.3	3.6	2.9	2.5	4.0	4.4	7.1	5.9	3.6	7.5	8.2	4.0	4.3	4.5	4.8
HAMAMATSU PHOTONICS	6965	9.7	13.7	8.3	6.2	5.8	9.8	11.7	9.7	8.6	9.4	9.2	7.3	8.6	9.2	15.5	14.3	9.3	13.6	14.2	17.4	20.3	13.0	20.4	31.2	30.9	34.8	42.3	15.0	15.0	-
KYOCERA	6971	105.9	54.6	40.6	54.9	63.2	90.3	69.9	85.1	63.1	37.9	70.7	66.4	56.7	56.6	56.7	68.9	67.8	86.5	117.0	106.0	117.1	151.8	173.9	161.7	141.9	149.1	160.0	160.0	160.0	225.0
TAIYO YUDEN	6976	54.0	16.8	7.4	14.4	22.5	15.8	30.2	44.6	27.0	9.4	17.5	29.1	19.7	16.9	18.8	37.4	31.6	26.5	42.6	44.1	43.9	51.6	63.3	79.9	64.2	40.4	50.0	60.0	60.0	40.0
MURATA MANUFACTURING	6981	107.8	34.6	18.2	33.1	48.0	51.0	99.7	125.6	65.4	22.9	56.8	68.4	77.7	68.2	101.2	172.5	158.6	306.6	291.6	196.7	152.8	208.1	219.5	180.5	247.8	220.0	210.0	220.0	250.0	
NICHICON	6996	25.9	16.0	8.1	2.6	12.4	11.0	11.9	9.4	7.5	2.2	6.6	15.4	5.4	2.3	2.4	2.6	7.5	6.8	8.6	7.1	6.5	10.1	11.2	16.2	11.1	7.6	9.1	9.4	12.5	10.0
NIPPON CHEMI-CON	6997	14.6	11.0	9.4	6.1	7.6	8.4	14.6	14.6	11.9	4.0	9.6	13.5	6.0	3.1	5.2	4.4	4.6	7.5	9.6	5.6	3.5	5.9	7.7	11.2	7.6	5.9	7.1	7.4	7.6	5.7
KOA	6999	1.4	2.8	1.9	1.4	1.3	2.8	2.3	2.9	2.7	0.9	2.7	2.4	4.7	3.1	2.3	3.7	2.3	5.1	6.2	4.0	2.2	6.9	11.9	19.1	24.2	5.9	7.4	7.8	8.1	4.1
Total		590.1	349.1	249.7	289.3	394.4	484.1	549.1	623.2	498.3	244.3	421.8	474.9	467.8	406.7	524.8	697.1	650.3	897.2	936.7	927.6	862.5	974.1	1,217.2	1,192.3	1,135.1	1,136.7	1,288.1	1,263.7	1,344.8	-
% of sales																															
IBIDEN	4062	13%	9%	8%	6%	10%	21%	18%	10%	20%	7%	20%	16%	15%	12%	18%	13%	8%	7%	8%	20%	25%	15%	31%	40%	43%	15%	32%	27%	30%	50%
NITERRA	5334	10%	12%	5%	5%	6%	9%	8%	18%	8%	5%	3%	5%	8%	12%	10%	12%	10%	9%	13%	12%	8%	6%	6%	7%	6%	7%	5%	6%	5%	6%
MINEBEA MITSUMI	6479	14%	9%	6%	7%	11%	7%	5%	7%	8%	5%	10%	11%	15%	5%	7%	7%	7%	5%	6%	6%	5%	4%	6%	11%	6%	6%	6%	6%	6%	6%
MABUCHI MOTOR	6592	4%	5%	5%	5%	12%	6%	4%	3%	4%	5%	4%	8%	8%	5%	8%	11%	8%	9%	11%	14%	7%	7%	7%	7%	7%	5%	6%	7%	6%	5%
TDK	6762	14%	10%	7%	7%	9%	9%	8%	10%	14%	8%	9%	12%	10%	7%	9%	14%	14%	14%	13%	13%	14%	15%	13%	10%	10%	12%	11%	10%	10%	15%
ALPS ALPINE	6770	5%	6%	6%	8%	7%	7%	6%	6%	9%	4%	4%	6%	6%	4%	4%	5%	6%	9%	6%	5%	6%	7%	5%	6%	5%	6%	5%	5%	5%	7%
NIHON DEMP A KOGYO	6779	15%	7%	7%	12%	20%	11%	13%	17%	18%	4%	6%	10%	9%	5%	4%	5%	15%	16%	6%	8%	8%	7%	8%	9%	9%	7%	11%	11%	10%	-
MEIKO ELECTRONICS	6787	-	-	-	14%	25%	20%	25%	14%	18%	10%	23%	15%	11%	8%	5%	3%	3%	9%	13%	10%	9%	10%	10%	11%	14%	-	18%	15%	13%	-
HIROSE ELECTRIC	6806	9%	7%	5%	8%	6%	9%	11%	9%	8%	9%	10%	12%	12%	9%	12%	11%	13%	16%	17%	15%	11%	9%	11%	20%	11%	9%	12%	12%	12%	12%
JAE	6807	9%	7%	9%	8%	10%	12%	12%	11%	11%	10%	10%	12%	21%	14%	12%	12%	10%	7%	10%	12%	9%	8%	10%	9%	8%	10%	9%	9%	9%	9%
CMK	6958	8%	10%	8%	2%	4%	5%	10%	11%	7%	3%	2%	2%	5%	4%	7%	4%	6%	8%	8%	10%	5%	8%	8%	18%	20%	9%	6%	5%	5%	8%
DAISHINKU	6962	8%	10%	7%	5%	4%	10%	11%	6%	9%	6%	13%	8%	6%	8%	11%	10%	12%	10%	9%	14%	13%	17%	15%	9%	19%	21%	10%	10%	10%	12%
HAMAMATSU PHOTONICS	6965	19%	22%	14%	10%	8%	13%	13%	10%	9%	13%	10%	7%	9%	9%	14%	12%	8%	10%	10%	12%	15%	8%	10%	14%	15%	16%	19%	6%	6%	-
KYOCERA	6971	8%	5%	4%	5%	5%	8%	5%	7%	6%	4%	6%	6%	4%	4%	4%	5%	5%	7%	7%	8%	8%	9%	8%	7%	7%	8%	8%	8%	11%	
TAIYO YUDEN	6976	28%	11%	5%	9%	13%	8%	14%	19%	15%	5%	8%	16%	10%	8%	8%	16%	14%	11%	16%	16%	15%	15%	20%	25%	19%	11%	13%	14%	13%	11%
MURATA MANUFACTURING	6981	18%	9%	5%	8%	11%	10%	18%	20%	12%	4%	9%	12%	11%	8%	10%	14%	14%	22%	19%	18%	12%	8%	12%	13%	10%	14%	11%	10%	9%	14%
NICHICON	6996	21%	15%	7%	3%	12%	10%	10%	8%	8%	3%	6%	14%	6%	2%	2%	2%	7%	6%	7%	6%	6%	7%	6%	9%	6%	4%	5%	5%	6%	6%
NIPPON CHEMI-CON	6997	11%	12%	10%	6%	7%	7%	11%	10%	10%	4%	8%	13%	6%	3%	4%	4%	4%	6%	7%	5%	3%	4%	5%	7%	6%	4%	5%	5%	5%	4%
KOA	6999	2%	5%	3%	2%	3%	6%	4%	6%	7%	3%	6%	6%	12%	7%	5%	8%	5%	10%	11%	8%	4%	11%	16%	29%	38%	8%	9%	9%	10%	6%
Total		12%	8%	6%	6%	8%	9%	10%	10%	10%	5%	8%	9%	9%	7%	8%	10%	9%	11%	11%	11%	10%	10%	11%	11%	10%	9%	10%	9%	9%	-

Note: For Mabuchi Motor, fiscal year ends in December; for Hamamatsu Photonics, fiscal year ends in September; e = Morgan Stanley Research estimates; Ce=Company estimates
Source: Company data, Morgan Stanley Research

Depreciation

		F00	F01	F02	F03	F04	F05	F06	F07	F08	F09	F10	F11	F12	F13	F14	F15	F16	F17	F18	F19	F20	F21	F22	F23	F24	F25	F26e	F27e	F28e	F26Ce	
(JPY bn)																																
Depreciation																																
IBIDEN	4062	14.6	17.5	16.6	13.8	14.4	22.3	31.5	39.2	43.5	38.8	40.7	42.4	42.7	35.7	39.4	44.1	33.1	24.6	25.1	24.2	35.4	52.7	54.9	46.0	54.2	66.2	79.5	95.0	94.0	81.0	
NITERRA	5334	18.1	20.0	18.5	15.9	14.5	18.0	18.9	25.5	31.8	18.8	16.6	15.4	12.8	11.6	13.3	16.8	19.6	20.9	24.0	32.0	34.7	37.9	39.2	39.8	39.7	45.0	40.0	43.0	43.0	42.1	
MINEBEA MITSUMI	6479	22.1	24.4	22.8	21.7	22.5	24.0	24.6	26.4	24.0	21.1	20.8	19.6	20.8	23.7	28.8	34.8	28.2	31.6	36.4	46.2	48.6	45.2	53.1	58.4	66.2	70.4	69.9	72.1	74.6	75.0	
MABUCHI MOTOR	6592	6.4	6.6	6.3	5.6	5.2	5.7	5.4	4.9	4.4	4.0	3.6	3.3	3.6	4.6	5.1	6.6	6.4	7.4	8.0	8.6	8.9	10.0	11.9	12.8	14.2	13.5	13.9	14.0	14.0	13.9	
TDK	6762	63.6	61.9	57.8	51.2	52.8	58.5	65.3	71.3	89.6	83.8	77.6	80.2	77.9	83.1	80.2	83.2	87.5	92.2	106.6	125.0	140.3	177.0	206.3	190.5	196.2	204.2	210.0	210.0	220.0	240.0	
ALPS ALPINE	6770	31.7	32.5	30.0	32.7	40.9	36.9	39.2	37.8	29.9	21.2	18.7	19.8	21.6	24.5	28.0	30.7	33.1	36.0	44.2	46.1	41.3	45.7	46.8	41.5	35.1	34.0	40.0	42.0	42.0	44.8	
NIHON DEMPA KOGYO	6779	5.7	5.9	5.9	6.2	6.3	7.1	7.5	7.8	9.0	3.1	3.2	3.0	3.4	3.9	3.8	3.6	3.6	4.1	3.5	3.7	3.1	3.0	3.1	3.3	3.4	3.9	4.4	4.8	5.2	4.4	
MEIKO ELECTRONICS	6787	1.0	1.0	1.1	1.4	2.1	3.2	4.7	6.0	7.1	6.8	6.5	6.4	6.3	6.2	7.2	6.5	5.5	5.8	6.3	6.6	7.4	8.0	9.8	11.2	12.5	13.7	20.0	26.0	31.5	-	
HIROSE ELECTRIC	6806	5.1	5.3	4.3	5.3	6.2	6.9	8.1	10.2	10.7	8.3	7.5	8.9	8.7	11.4	11.4	11.6	8.1	10.8	13.1	15.8	16.8	16.4	16.7	16.8	17.9	19.5	22.0	24.2	26.0	17.0	
JAE	6807	6.0	6.4	6.0	6.2	6.6	7.7	8.4	11.2	13.0	10.6	10.0	11.6	14.7	18.9	19.9	17.7	20.3	20.8	19.6	21.1	21.2	20.4	20.6	20.6	20.9	19.7	19.7	20.2	20.7	22.0	
CMK	6958	10.3	10.1	9.5	8.2	6.8	6.3	6.2	8.1	9.2	6.8	6.3	5.3	4.1	4.7	5.0	4.9	4.1	4.1	4.3	4.7	4.5	4.2	4.8	5.3	5.9	6.5	8.2	8.5	8.7	8.5	
DAISHINKU	6962	4.0	3.9	3.4	3.1	2.8	2.7	3.1	3.4	3.6	3.2	3.7	3.9	3.6	3.5	3.6	2.5	2.7	2.9	2.9	2.7	3.0	3.6	4.1	4.0	4.1	4.3	5.0	5.2	5.4	5.2	
HAMAMATSU PHOTONICS	6965	5.2	6.6	7.1	7.1	7.1	6.7	7.1	8.0	9.2	9.7	8.7	8.1	8.7	8.7	8.0	9.5	10.5	10.0	11.3	12.1	12.7	13.6	13.5	14.5	16.7	18.9	23.2	27.7	26.7	-	
KYOCERA	6971	74.2	88.5	75.3	70.3	65.9	73.2	82.2	87.0	97.6	72.8	71.5	73.1	73.6	76.2	73.0	76.6	77.4	82.8	63.9	92.7	109.1	129.0	149.6	157.0	160.9	158.2	115.0	120.0	120.0	120.0	
TAIYO YUDEN	6976	17.1	20.7	20.3	18.3	18.0	17.1	18.4	23.3	27.9	23.9	19.3	19.3	19.8	20.8	21.8	23.8	24.9	25.6	26.5	26.5	27.0	29.3	31.3	34.9	39.4	46.3	49.1	50.0	52.0	52.0	48.0
MURATA MANUFACTURING	6981	54.0	56.5	50.8	44.6	42.4	45.1	49.8	65.1	81.0	69.9	61.8	61.0	72.3	76.9	84.9	99.1	113.5	141.6	124.4	140.3	143.1	155.6	161.3	175.9	173.3	178.2	180.0	185.0	190.0	178.0	
NICHICON	6996	13.0	14.6	13.2	9.7	8.8	8.9	9.4	10.5	11.1	8.4	7.7	9.2	7.6	5.1	4.3	4.4	3.4	3.1	4.5	5.3	5.2	6.1	7.5	6.9	8.5	8.1	9.3	9.4	9.6	8.5	
NIPPON CHEMI-CON	6997	8.8	8.6	8.8	8.2	7.2	7.5	8.8	11.1	11.6	9.5	9.1	9.2	9.2	9.5	8.6	7.9	7.6	6.7	6.6	7.0	7.7	6.0	6.3	6.8	7.1	7.3	6.7	6.8	7.0	6.6	
KOA	6999	2.0	2.5	2.8	2.2	1.9	2.0	2.3	2.4	2.6	2.4	2.0	1.9	1.9	2.1	2.2	2.4	2.3	2.5	2.9	3.4	3.6	3.5	4.1	4.9	5.9	7.0	7.4	7.6	7.8	7.2	
Total		363.0	393.6	360.6	331.8	332.5	360.0	400.8	459.4	516.7	423.3	395.5	401.6	413.5	430.1	447.8	486.3	491.0	533.4	534.5	625.2	674.3	769.4	848.5	855.6	889.1	927.9	924.1	973.5	998.2	-	
% of sales																																
IBIDEN	4062	7%	8%	8%	6%	6%	7%	8%	9%	14%	14%	13%	14%	15%	12%	12%	14%	12%	8%	9%	8%	11%	13%	13%	12%	15%	16%	16%	16%	13%	19%	
NITERRA	5334	8%	9%	8%	7%	6%	6%	5%	7%	11%	8%	6%	5%	4%	4%	4%	4%	5%	5%	6%	7%	8%	8%	7%	6%	6%	6%	5%	5%	5%	6%	
MINEBEA MITSUMI	6479	8%	9%	8%	8%	8%	8%	7%	8%	9%	9%	8%	8%	7%	6%	6%	6%	4%	4%	4%	5%	5%	4%	4%	4%	4%	4%	4%	4%	4%	5%	
MABUCHI MOTOR	6592	6%	6%	5%	5%	5%	5%	6%	5%	5%	5%	6%	4%	4%	4%	4%	5%	5%	5%	5%	6%	7%	8%	7%	7%	7%	7%	6%	6%	7%	7%	
TDK	6762	9%	11%	9%	8%	8%	7%	8%	8%	12%	10%	9%	10%	9%	8%	7%	7%	7%	8%	9%	9%	9%	9%	9%	9%	9%	9%	8%	8%	7%	10%	
ALPS ALPINE	6770	6%	6%	5%	5%	6%	5%	6%	5%	6%	4%	3%	4%	4%	4%	4%	4%	4%	5%	6%	6%	6%	5%	6%	6%	6%	4%	3%	4%	3%	4%	
NIHON DEMPA KOGYO	6779	8%	12%	12%	11%	10%	11%	10%	10%	15%	6%	6%	6%	7%	8%	8%	8%	8%	8%	9%	8%	9%	8%	7%	6%	7%	6%	7%	8%	9%	8%	
MEIKO ELECTRONICS	6787	4%	6%	5%	5%	5%	6%	7%	8%	9%	10%	9%	10%	10%	8%	8%	7%	6%	5%	5%	6%	6%	5%	6%	6%	6%	6%	7%	7%	7%	-	
HIROSE ELECTRIC	6806	6%	9%	6%	7%	7%	7%	7%	9%	12%	10%	8%	9%	9%	9%	9%	10%	7%	9%	11%	13%	13%	10%	9%	10%	9%	10%	10%	10%	10%	8%	
JAE	6807	5%	6%	6%	5%	6%	6%	6%	7%	11%	10%	9%	10%	11%	12%	10%	10%	10%	8%	9%	10%	9%	10%	9%	9%	9%	9%	9%	8%	8%	10%	
CMK	6958	8%	10%	8%	7%	6%	5%	5%	6%	9%	8%	7%	7%	6%	7%	7%	7%	5%	5%	5%	6%	6%	5%	6%	6%	6%	7%	8%	8%	8%	8%	
DAISHINKU	6962	8%	12%	11%	10%	8%	8%	7%	8%	10%	10%	10%	11%	11%	10%	12%	8%	9%	10%	10%	9%	9%	11%	10%	11%	11%	12%	12%	12%	13%	13%	
HAMAMATSU PHOTONICS	6965	10%	11%	12%	12%	10%	9%	8%	8%	9%	14%	10%	8%	9%	8%	7%	8%	9%	8%	8%	8%	9%	8%	6%	7%	8%	9%	11%	12%	11%	-	
KYOCERA	6971	6%	9%	7%	6%	6%	6%	6%	7%	9%	7%	6%	6%	6%	5%	5%	5%	5%	4%	6%	7%	7%	7%	7%	8%	8%	8%	6%	6%	6%	6%	
TAIYO YUDEN	6976	9%	14%	13%	11%	10%	9%	8%	10%	15%	12%	9%	10%	10%	10%	10%	11%	10%	10%	10%	10%	10%	9%	11%	12%	14%	14%	13%	13%	12%	14%	
MURATA MANUFACTURING	6981	9%	14%	13%	11%	10%	9%	9%	10%	15%	13%	10%	10%	11%	9%	8%	8%	10%	10%	8%	9%	9%	10%	11%	10%	10%	9%	8%	8%	8%	10%	
NICHICON	6996	11%	14%	12%	10%	8%	8%	8%	9%	12%	10%	7%	9%	8%	5%	4%	4%	3%	3%	4%	4%	5%	4%	4%	4%	5%	5%	5%	5%	5%	5%	
NIPPON CHEMI-CON	6997	7%	9%	9%	8%	7%	6%	7%	8%	10%	9%	7%	9%	10%	8%	6%	6%	6%	5%	5%	7%	5%	4%	4%	4%	6%	5%	5%	4%	4%	5%	
KOA	6999	3%	5%	5%	4%	4%	4%	4%	5%	6%	7%	5%	5%	5%	5%	5%	5%	5%	5%	5%	7%	7%	5%	5%	8%	9%	10%	9%	9%	9%	10%	
Total		7.3%	9.4%	8.3%	7%	7%	7%	7%	8%	11%	9%	7%	8%	8%	7%	7%	7%	7%	6%	7%	8%	8%	8%	8%	8%	8%	8%	8%	7%	7%	7%	-

Note: For Mabuchi Motor, fiscal year ends in December; for Hamamatsu Photonics, fiscal year ends in September; e = Morgan Stanley Research estimates; Ce=Company estimates

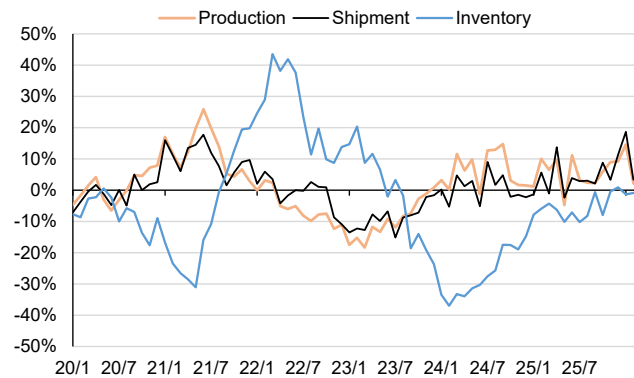
Source: Company data, Morgan Stanley Research

Inventory and Inventory Days

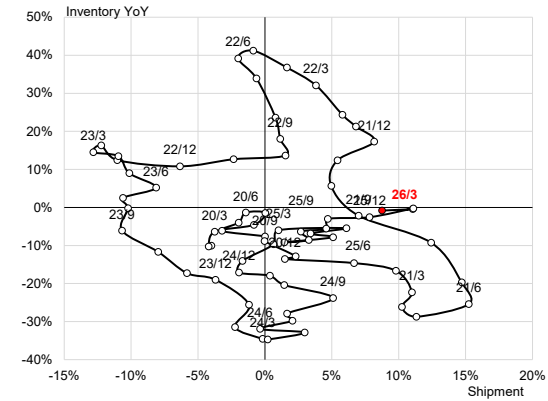
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(JPY bn)																														
Inventory																														
IBIDEN	4062	20.3	16.7	15.4	20.0	20.9	32.0	37.4	42.2	31.8	25.1	29.8	35.8	38.7	38.6	38.0	37.1	33.6	40.0	48.8	45.3	57.0	74.6	79.2	65.3	67.2	71.0	78.2	90.8	102.5
NITERRA	5334	51.7	51.1	46.0	42.2	45.4	51.1	65.4	76.5	52.7	59.7	62.1	62.1	67.8	71.1	78.4	82.1	86.9	95.0	113.1	116.7	126.8	156.8	192.3	189.6	183.9	217.7	180.5	174.8	180.7
MINEBEA MITSUMI	6479	52.8	49.9	43.2	41.5	47.0	48.9	45.9	42.4	38.7	35.9	44.0	49.0	58.2	63.7	92.2	103.0	120.4	150.1	141.4	169.8	171.4	219.3	263.1	294.9	350.9	391.3	377.9	388.8	403.0
MABUCHI MOTOR	6592	26.5	23.5	17.6	14.7	16.4	17.1	17.8	18.0	17.1	13.5	16.1	18.6	20.5	24.2	29.0	31.0	26.7	33.0	35.7	33.9	32.5	50.9	61.4	60.4	59.8	57.8	60.4	60.6	60.4
TDK	6762	116.4	91.1	73.9	77.3	74.9	89.0	89.8	88.8	105.7	105.1	121.7	137.2	137.9	136.4	151.0	157.1	154.5	207.5	226.9	236.5	288.9	437.0	443.0	406.1	410.0	585.4	429.0	440.8	453.0
ALPS ALPINE	6770	93.1	79.8	82.3	78.6	85.0	83.7	77.4	72.5	52.1	48.1	56.6	69.1	74.3	73.1	85.4	87.5	89.6	96.1	95.8	100.5	95.8	133.4	165.4	163.7	138.6	138.3	162.5	174.9	183.7
NIHON DEMP A KOGYO	6779	11.6	6.8	6.9	6.6	7.3	8.6	8.5	9.8	4.8	9.4	9.0	11.7	11.3	11.0	12.5	10.6	12.2	13.7	14.4	12.9	10.1	11.8	12.8	11.1	10.5	12.4	11.5	11.8	12.0
MEIKO ELECTRONICS	6787	3.1	2.8	4.1	4.6	4.3	5.9	8.8	9.8	8.6	8.9	11.3	9.1	10.5	11.6	13.7	11.6	12.5	14.3	16.0	19.0	22.2	29.8	32.6	35.6	38.5	49.5	60.2	69.7	77.7
HIROSE ELECTRIC	6806	7.6	5.0	4.7	5.7	5.5	6.6	6.7	6.6	5.6	6.3	7.3	7.2	7.9	8.4	9.1	8.2	8.9	11.7	13.9	14.5	16.9	21.7	27.7	24.4	25.1	30.2	28.3	29.4	30.6
JAE	6807	17.6	14.0	13.8	12.5	11.7	12.4	12.4	12.6	10.3	9.1	9.9	9.8	11.3	12.4	12.7	12.9	16.1	18.9	18.0	20.3	21.1	29.0	28.6	28.3	29.2	32.8	31.1	31.7	32.4
CMK	6958	9.9	10.3	9.9	9.6	8.5	9.5	10.3	12.4	7.9	9.6	9.0	8.0	6.7	7.9	9.2	8.5	8.5	9.8	10.3	10.6	10.3	16.1	17.3	17.3	18.6	18.4	19.5	20.2	20.8
DAISHINKU	6962	14.0	12.4	6.2	6.7	7.3	8.6	9.2	8.7	7.2	7.4	8.1	8.0	8.3	9.1	9.8	8.4	9.4	10.7	12.8	13.1	14.8	16.5	17.2	16.4	18.0	24.6	20.9	20.8	20.5
HAMAMATSU PHOTONICS	6965	13.0	13.0	12.8	12.8	13.3	13.7	15.4	17.8	20.2	20.0	20.7	22.3	24.4	25.1	26.9	29.5	30.0	31.8	34.5	37.5	43.5	44.4	59.1	75.6	80.6	77.0	75.1	75.0	74.9
KYOCERA	6971	252.8	205.8	183.2	197.2	213.4	190.6	209.2	205.2	199.6	177.4	232.9	270.3	296.5	335.8	354.5	327.9	331.2	364.9	343.9	344.3	345.4	452.5	539.4	540.2	521.8	522.0	457.7	468.0	506.3
TAIYO YUDEN	6976	36.9	28.3	25.1	23.5	25.5	27.3	32.2	40.5	30.3	28.6	32.5	30.5	35.9	37.6	44.1	48.7	48.8	52.1	60.9	62.6	67.9	96.8	104.8	102.3	110.0	125.8	109.9	114.5	120.2
MURATA MANUFACTURING	6981	92.0	59.2	53.3	54.8	59.0	66.4	86.1	114.6	94.1	89.2	110.5	132.0	160.9	170.6	186.3	217.5	211.4	290.3	349.3	334.4	361.3	464.7	575.0	513.0	482.8	520.5	481.8	535.0	605.6
NICHICON	6996	21.1	18.4	18.9	16.0	17.3	18.2	20.6	20.1	16.0	14.5	17.6	18.6	16.9	17.6	17.7	17.5	17.8	18.8	22.1	21.7	21.6	31.1	37.2	36.7	33.2	32.3	38.0	38.1	38.0
NIPPON CHEMI-CON	6997	23.1	18.2	18.9	18.8	18.7	21.9	23.5	24.4	22.4	19.5	21.0	27.1	21.8	20.9	22.7	20.8	20.9	23.0	27.9	26.5	25.5	31.6	33.0	29.2	33.1	31.2	36.0	37.0	37.5
KOA	6999	9.4	7.5	5.7	5.4	5.1	5.4	5.8	5.6	3.8	3.7	4.6	4.4	4.9	4.6	5.2	4.9	5.0	5.9	7.6	7.3	7.8	10.8	13.9	14.0	13.4	13.5	15.2	15.3	15.1
Total		873.0	713.8	641.8	648.6	686.5	717.0	782.5	828.5	728.9	690.9	824.5	930.9	1,014.7	1,079.7	1,198.6	1,224.8	1,244.6	1,487.5	1,593.4	1,627.3	1,740.8	2,328.8	2,703.1	2,624.1	2,625.3	2,951.9	2,673.5	2,797.2	2,975.1
Days in inventory (days)																														
IBIDEN	4062	38.6	38.2	33.6	36.1	38.1	41.0	44.7	49.6	54.2	49.8	45.1	50.8	58.0	59.5	59.0	57.5	61.3	57.3	69.6	75.3	78.4	85.4	96.8	98.4	94.4	88.6	79.1	75.3	74.4
NITERRA	5334	99.9	106.5	99.6	93.9	89.9	86.6	85.6	98.7	93.6	106.4	113.7	106.2	103.3	114.0	122.1	118.7	123.3	123.7	135.1	143.5	149.5	154.3	174.7	179.5	172.6	162.8	165.5	142.3	132.8
MINEBEA MITSUMI	6479	89.6	90.9	83.5	76.1	69.6	70.0	67.2	63.5	75.1	77.7	72.1	85.5	87.0	77.8	74.7	73.2	79.5	70.4	74.1	70.0	75.9	78.5	82.2	87.0	94.1	98.8	104.1	100.8	100.5
MABUCHI MOTOR	6592	122.7	137.7	107.6	93.0	89.6	91.9	88.5	84.8	93.1	107.8	90.0	105.8	111.1	99.9	111.7	110.0	111.4	109.4	126.9	139.9	148.5	155.1	171.9	165.3	153.1	152.6	144.7	137.7	128.1
TDK	6762	74.0	81.5	65.5	58.2	57.4	51.1	52.4	51.3	58.6	62.3	62.9	74.5	74.3	65.5	65.4	67.7	66.4	71.0	80.5	88.1	91.8	99.0	100.6	103.2	98.2	105.5	103.2	81.2	76.3
ALPS ALPINE	6770	64.6	69.9	61.9	58.4	56.4	53.0	49.3	46.6	47.0	44.1	43.3	53.0	57.1	48.2	48.7	50.7	53.7	50.6	50.8	53.4	59.3	63.0	71.2	75.3	67.7	60.3	62.6	65.3	66.1
NIHON DEMP A KOGYO	6779	72.9	82.9	62.0	56.3	55.6	61.4	59.6	58.9	49.1	68.1	78.8	97.1	102.4	99.3	110.9	117.0	120.2	125.3	144.7	139.8	129.1	121.6	124.5	121.4	106.3	107.5	111.3	104.2	102.6
MEIKO ELECTRONICS	6787	55.1	69.5	73.3	71.3	52.2	47.5	49.6	52.1	50.0	54.0	58.2	67.9	67.1	58.4	55.5	56.2	55.2	54.3	55.9	64.6	73.8	76.5	80.6	83.6	80.9	84.5	83.7	83.3	82.1
HIROSE ELECTRIC	6806	59.5	65.6	48.2	48.4	46.3	44.1	41.7	37.9	41.9	45.4	47.0	47.0	49.7	43.1	46.3	47.9	50.9	55.7	66.9	73.8	74.6	79.4	94.9	102.2	86.7	82.5	82.9	76.7	75.2
JAE	6807	76.2	73.6	63.4	57.4	50.5	46.5	43.4	38.1	39.8	41.4	39.6	40.3	37.0	35.0	31.8	33.5	30.1	30.3	37.0	40.8	42.0	49.9	54.2	55.8	58.4	59.1	60.4	57.1	55.9
CMK	6958	36.7	39.0	37.0	37.3	34.7	33.1	32.3	34.0	37.1	39.1	42.4	43.2	41.7	41.5	48.0	49.9	47.7	45.2	47.6	53.1	60.4	69.5	85.6	82.9	81.9	79.2	77.7	78.3	77.7
DAISHINKU	6962	125.0	159.5	119.4	100.5	93.1	107.1	107.7	107.6	96.8	106.2	93.0	102.0	111.5	119.1	129.0	131.6	139.1	155.7	192.0	201.2	204.5	196.6	229.6	207.4	214.6	256.2	263.0	231.3	221.4
HAMAMATSU PHOTONICS	6965	144.4	157.6	152.4	150.9	139.8	141.2	128.3	136.5	149.8	186.8	163.0	160.2	178.2	177.1	177.5	178.6	178.6	171.7	171.7	182.7	206.1	187.4	195.9	242.2	284.7	259.8	242.8	219.8	200.0
KYOCERA	6971	83.8	105.3	89.1	80.7	87.6	87.9	81.0	85.6	88.3	87.3	84.2	105.5	108.6	108.0	110.8	113.9	114.6	105.8	111.5	108.5	112.4	109.9	124.0	135.8	133.2	130.2	141.1	127.4	121.1
TAIYO YUDEN	6976	96.9	104.2	85.2	73.2	64.1	64.7	65.8	73.9	79.0	67.6	67.8	73.2	78.2	83.5	85.5	95.7	99.2	101.1	107.4	114.5	112.4	133.8	158.5	147.0	143.6	157.4	148.9	131.1	125.7
MURATA MANUFACTURING	6981	96.1	108.9	83.8	76.6	81.2	77.3	82.9	94.4	95.7	87.4	87.8	106.9	111.7	109.2	103.5	103.5	111.3	98.7	119.7	131.0	126.0	144.4	187.7	197.9	177.2	173.4	179.5	159.3	153.1
NICHICON	6996	68.5	81.5	75.2	75.5	70.4	73.4	71.8	73.1	75.4	71.1	65.6	70.2	78.1	71.1	71.5	71.3	77.9	71.8	73.8	78.8	79.6	81.7	82.6	89.6	86.5	85.6	86.1	91.4	89.5
NIPPON CHEMI-CON	6997	<																												

Production & Inventory

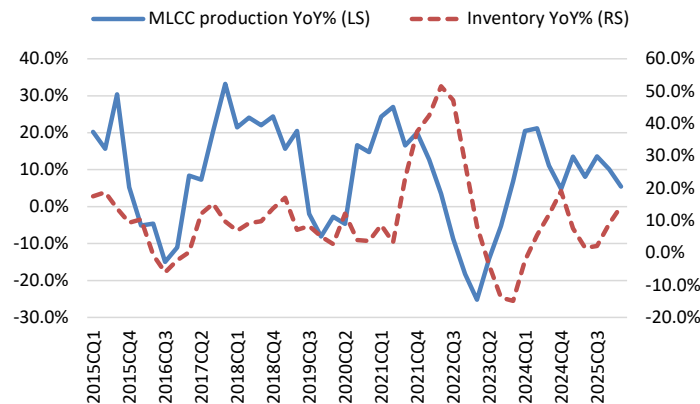
Industrial production: Production, shipments, inventories for Electronic parts and devices* (1)



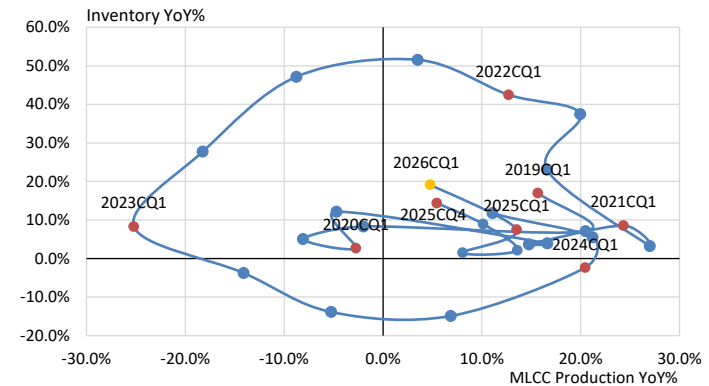
Industrial production: Production, shipments, inventories for Electronic parts and devices* (2)



Taiyo Yuden: MLCC production & inventory (1)



Taiyo Yuden: MLCC Production & inventory (2)



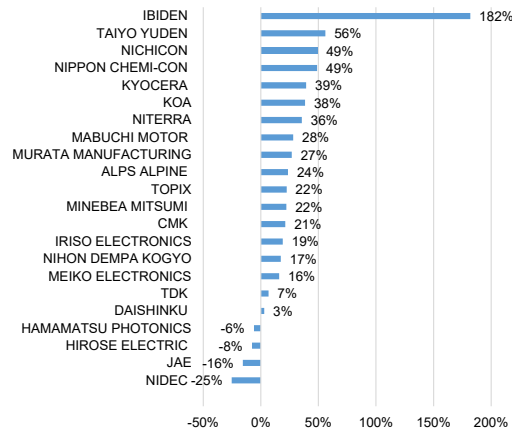
Note: *3-month average

Source: METI, Company data, Morgan Stanley Research

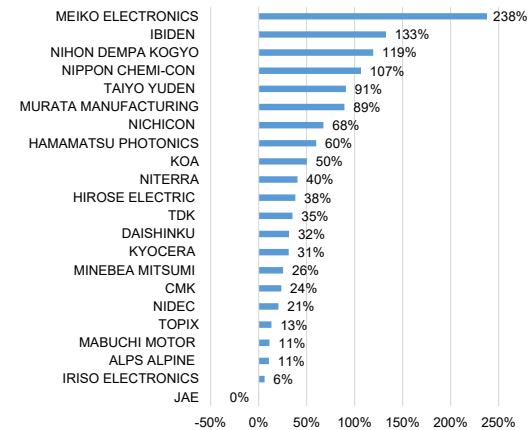
Stock Price Performance

Rank	2020	2021	2022	2023	2024	2025	2026 Based on 5/15 Closing Price
1	IBIDEN 85%	NIHON DEMP KOGYO 184%	NITERRA 22%	CMK 80%	MEIKO ELECTRONICS 122%	IBIDEN 182%	MEIKO ELECTRONICS 238%
2	NIDEC 73%	MEIKO ELECTRONICS 147%	KOA 17%	MEIKO ELECTRONICS 73%	TDK 54%	TAIYO YUDEN 56%	IBIDEN 133%
3	DAISHINKU 51%	DAISHINKU 135%	ALPS ALPINE 10%	IBIDEN 63%	NITERRA 52%	NICHICON 49%	NIHON DEMP KOGYO 119%
4	TAIYO YUDEN 44%	CMK 72%	JAE 6%	TDK 55%	ALPS ALPINE 31%	NIPPON CHEMI-CON 49%	NIPPON CHEMI-CON 107%
5	MURATA MANUFACTURING 38%	MINEBEA MITSUMI 59%	MABUCHI MOTOR -2%	JAE 53%	TOPIX 18%	KYOCERA 39%	TAIYO YUDEN 91%
6	HAMAMATSU PHOTONICS 31%	IBIDEN 42%	IRISO ELECTRONICS -2%	MINEBEA MITSUMI 47%	HIROSE ELECTRIC 17%	KOA 38%	MURATA MANUFACTURING 89%
7	TDK 25%	TAIYO YUDEN 37%	NICHICON -3%	NITERRA 37%	NIDEC 0%	NITERRA 36%	NICHICON 68%
8	NIHON DEMP KOGYO 19%	JAE 26%	TDK -3%	MURATA MANUFACTURING 36%	MABUCHI MOTOR -4%	MABUCHI MOTOR 28%	HAMAMATSU PHOTONICS 60%
9	NICHICON 14%	HAMAMATSU PHOTONICS 24%	TOPIX -5%	KYOCERA 26%	JAE -8%	MURATA MANUFACTURING 27%	KOA 50%
10	HIROSE ELECTRIC 12%	HIROSE ELECTRIC 24%	KYOCERA -9%	MABUCHI MOTOR 25%	MINEBEA MITSUMI -11%	ALPS ALPINE 24%	NITERRA 40%
11	MABUCHI MOTOR 8%	NITERRA 14%	NIPPON CHEMI-CON -13%	TOPIX 25%	MURATA MANUFACTURING -14%	TOPIX 22%	HIROSE ELECTRIC 38%
12	KOA 8%	KYOCERA 14%	NIHON DEMP KOGYO -13%	DAISHINKU 24%	NICHICON -15%	MINEBEA MITSUMI 22%	TDK 35%
13	TOPIX 5%	TOPIX 10%	HAMAMATSU PHOTONICS -14%	NICHICON 6%	KYOCERA -23%	CMK 21%	DAISHINKU 32%
14	IRISO ELECTRONICS -8%	KOA 9%	HIROSE ELECTRIC -14%	ALPS ALPINE 3%	NIPPON CHEMI-CON -26%	IRISO ELECTRONICS 19%	KYOCERA 31%
15	MINEBEA MITSUMI -10%	NIDEC 4%	MURATA MANUFACTURING -28%	TAIYO YUDEN -3%	IRISO ELECTRONICS -27%	NIHON DEMP KOGYO 17%	MINEBEA MITSUMI 26%
16	NIPPON CHEMI-CON -13%	MURATA MANUFACTURING -2%	IBIDEN -30%	HIROSE ELECTRIC -4%	NIHON DEMP KOGYO -31%	MEIKO ELECTRONICS 16%	CMK 24%
17	KYOCERA -15%	NIPPON CHEMI-CON -2%	CMK -34%	HAMAMATSU PHOTONICS -8%	DAISHINKU -35%	TDK 7%	NIDEC 21%
18	NITERRA -17%	IRISO ELECTRONICS -3%	MINEBEA MITSUMI -40%	IRISO ELECTRONICS -13%	KOA -36%	DAISHINKU 3%	TOPIX 13%
19	MEIKO ELECTRONICS -22%	NICHICON -3%	TAIYO YUDEN -42%	NIPPON CHEMI-CON -16%	IBIDEN -38%	HAMAMATSU PHOTONICS -6%	MABUCHI MOTOR 11%
20	JAE -29%	TDK -13%	NIDEC -49%	NIDEC -17%	HAMAMATSU PHOTONICS -39%	HIROSE ELECTRIC -8%	ALPS ALPINE 11%
21	CMK -40%	MABUCHI MOTOR -15%	MEIKO ELECTRONICS -49%	KOA -18%	TAIYO YUDEN -39%	JAE -16%	IRISO ELECTRONICS 6%
22	ALPS ALPINE -46%	ALPS ALPINE -20%	DAISHINKU -55%	NIHON DEMP KOGYO -19%	CMK -48%	NIDEC -25%	JAE 0%

■ Stock price performance (CY2025)



■ Stock price performance (CY2026 YTD)



Source: FactSet

FX Sensitivities of Electronic Components Makers/Changes in the Competitiveness of Core Products Under a Weak Yen Environment

	F26e	FX assumption (by company)		FX sensitivity (JPY bn/year)	Assumption¥145/USD		Assumption¥155/USD		
		OP (JPY bn)	USD/JPY		USD/JPY	OP (JPY bn)	Change (%)	OP (JPY bn)	Change (%)
IBIDEN	4062	77.0	150.0	150.0	0.8	73.0	-5%	81.0	5%
NITERRA	5334	156.6	150.0	150.0	1.0	151.6	-3%	161.6	3%
MINEBEA MITSUMI	6479	116.9	150.0	155.0	1.5	109.4	-6%	124.4	6%
MABUCHI MOTOR	6592	27.2	150.0	150.0	0.3	25.7	-6%	28.7	6%
TDK	6762	309.8	150.0	150.0	2.0	299.8	-3%	319.8	3%
ALPS ALPINE	6770	50.8	150.0	150.0	1.2	44.8	-12%	56.8	12%
NIHON DEMPACO KOGYO	6779	3.8	150.0	150.0	0.1	3.4	-11%	4.2	11%
MEIKO ELECTRONICS	6787	33.6	150.0	150.0	0.1	33.1	-1%	34.1	1%
HIROSE ELECTRIC	6806	46.6	150.0	150.0	0.3	45.0	-4%	48.3	4%
JAE	6807	10.3	150.0	153.0	0.3	8.8	-15%	11.8	15%
CMK	6958	2.9	150.0	150.0	0.0	2.9	0%	2.9	0%
DAISHINKU	6962	1.3	150.0	150.0	0.1	1.1	-19%	1.6	19%
HAMAMATSU PHOTONICS	6965	14.9	150.0	151.7	0.3	13.4	-10%	16.4	10%
KYOCERA	6971	121.5	150.0	150.0	1.0	116.5	-4%	126.5	4%
TAIYO YUDEN	6976	31.8	150.0	149.8	0.9	27.3	-14%	36.3	14%
MURATA MANUFACTURING	6981	372.8	150.0	149.1	4.5	350.3	-6%	395.3	6%
NICHICON	6996	8.4	150.0	145.0	0.1	8.0	-5%	8.9	5%
NIPPON CHEMI-CON	6997	5.6	150.0	153.0	0.3	4.3	-23%	6.9	23%
KOA	6999	5.2	150.0	157.0	0.1	4.6	-12%	5.8	12%
Total		1,396.9			14.8	1,322.8	-5%	1,471.1	5%

Changes in the competitiveness of core products under a weak-yen environment									
IBIDEN	4062	↑	Flip-chip package costs rising vs. peers in S. Korea, Taiwan, Austria; profits highly likely to grow						
NITERRA	5334	→	Expecting profit growth from weak yen as core parts for automotive plugs/sensors made in Japan, but likely no change to competitiveness given large number of Japanese peers						
MINEBEA MITSUMI	6479	→	Potential sales decline with over 70% of sales \$-denominated, but expecting THB/USD, CNY/USD rates to have big impact on profits						
MABUCHI MOTOR	6592	→	Forecasting profit growth, but expecting no major shift in competitiveness for mainstay products given high overseas production weighting						
TDK	6762	↑	Cost competitiveness for passive components may improve versus foreign peers, but limited impact for rechargeable batteries, HDD heads given lack of ¥-denominated costs						
ALPS ALPINE	6770	→	Forecasting profit growth, but expecting no major shift in product competitiveness given limited rivalry with foreign firms for mainstay products						
NIHON DEMPACO KOGYO	6779	↑	Expecting increase in competitiveness on high foreign sales exposure, limited use of imported materials						
MEIKO ELECTRONICS	6787	↑	Expecting no major change in product competitiveness given high foreign production exposure, some use of imported materials						
HIROSE ELECTRIC	6806	→	Expecting no major change in product competitiveness as US micro connector peers also undertaking production development/mass production in Japan						
JAE	6807	→	Expecting no major change in product competitiveness as US micro connector peers also undertaking production development/mass production in Japan						
CMK	6958	→	Expecting no major shift in competitiveness given high sales exposure to domestic customers						
DAISHINKU	6962	↑	Expecting increase in competitiveness on high foreign sales exposure, limited use of imported materials						
HAMAMATSU PHOTONICS	6965	↑	Yen weakness positive for earnings as 95% of production in Japan & 75% of sales abroad, although still certain level of ¥-denominated sales overseas						
KYOCERA	6971	→	Forecasting growth in ¥-converted profits, but expecting no major shift in competitiveness of mainstay products						
TAIYO YUDEN	6976	↑	Cost competitiveness of mainstay MLCCs rising vs. peers in S. Korea & Taiwan; potential growth in ¥-converted profits on high foreign sales exposure						
MURATA MANUFACTURING	6981	↑	Cost competitiveness of mainstay MLCCs rising vs. peers in S. Korea & Taiwan; potential growth in ¥-converted profits on high foreign sales exposure						
NICHICON	6996	→	Positive on aluminum electrolytic capacitor earnings, but worries over pressure on NECST product earnings given rising costs of imported parts/materials						
NIPPON CHEMI-CON	6997	↑	Cost competitiveness of mainstay aluminum electrolytic capacitors rising vs. peers in China; potential growth in ¥-converted profits on high foreign sales exposure						
KOA	6999	→	Expecting substantial profit growth but no major change in product competitiveness as domestic peers with high market share also getting same benefits						

Note: e = Morgan Stanley Research estimates

Source: Company data, Morgan Stanley Research

Sales and Production by Region

■ Sales by region (FY24)

Code	Company Name	Japan	US/America	Europe	China	Others
4062	IBIDEN	29.3%	12.9%	7.3%	-	50.5%
5334	NITERRA	18.9%	25.3%	19.3%	10.5%	26.0%
6479	MINEBEA MITSUMI	24.4%	27.2%	11.3%	17.8%	19.4%
6592	MABUCHI MOTOR	10.3%	18.1%	22.9%	0.0%	48.7%
6594	NIDEC	15.4%	22.8%	22.8%	21.1%	17.8%
6762	TDK	7.9%	6.4%	7.9%	54.1%	23.7%
6770	ALPS ALPINE	11.2%	18.0%	-	22.1%	48.7%
6779	NIHON DEMP A KOGYO	15.4%	13.7%	8.4%	35.3%	27.2%
6787	MEIKO ELECTRONICS	36.1%	0.0%	1.8%	16.7%	33.3%
6806	HIROSE ELECTRIC	15.7%	-	-	39.6%	44.7%
6807	JAE	36.6%	1.0%	-	37.0%	15.1%
6908	IRISO ELECTRONICS	16.0%	9.1%	15.5%	36.7%	22.6%
6958	CMK	58.2%	-	11.2%	18.2%	12.4%
6962	DAISHINKU	11.7%	5.3%	11.1%	32.5%	39.5%
6965	HAMAMATSU PHOTONICS	22.6%	24.2%	23.4%	19.1%	10.7%
6971	KYOCERA	29.0%	21.0%	20.2%	11.8%	18.0%
6976	TAIYO YUDEN	7.0%	-	-	32.2%	47.4%
6981	MURATA MANUFACTURING	7.4%	13.9%	9.2%	47.7%	21.7%
6996	NICHICON	50.6%	7.9%	5.7%	26.0%	9.9%
6997	NIPPON CHEMI-CON	21.4%	11.4%	10.7%	32.2%	24.4%
6999	KOA	31.3%	17.1%	18.9%	-	32.7%
Total		17.7%	16.9%	13.0%	27.6%	24.9%

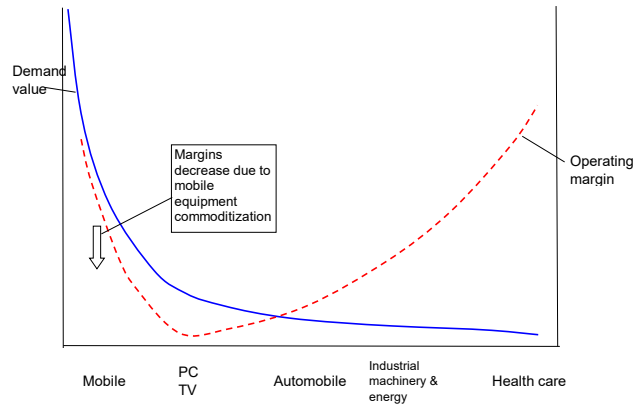
■ Production by region (FY25e)

Code	Company Name	Japan	US/America	Europe	China	Others
4062	IBIDEN	75.0%	4.0%	3.0%	2.0%	16.0%
5334	NITERRA	60.0%	10.0%	8.0%	8.0%	14.0%
6479	MINEBEA MITSUMI	22.0%	26.0%	11.0%	19.0%	22.0%
6592	MABUCHI MOTOR	0.0%	7.0%	3.0%	35.0%	55.0%
6762	TDK	12.5%	5.0%	10.0%	60.0%	12.5%
6770	ALPS ALPINE	30.0%	10.0%	5.0%	25.0%	30.0%
6779	NIHON DEMP A KOGYO	70.0%	0.0%	0.0%	20.0%	10.0%
6787	MEIKO ELECTRONICS	10.0%	0.0%	0.0%	45.0%	45.0%
6806	HIROSE ELECTRIC	50.0%	0.0%	0.0%	10.0%	40.0%
6807	JAE	60.0%	5.0%	0.0%	20.0%	15.0%
6908	IRISO ELECTRONICS	13.0%	0.0%	0.0%	49.0%	38.0%
6958	CMK	20.0%	0.0%	0.0%	35.0%	45.0%
6962	DAISHINKU	60.0%	0.0%	0.0%	30.0%	10.0%
6965	HAMAMATSU PHOTONICS	80.0%	5.0%	10.0%	5.0%	0.0%
6971	KYOCERA	60.0%	10.0%	10.0%	10.0%	10.0%
6976	TAIYO YUDEN	35.0%	0.0%	0.0%	20.0%	45.0%
6981	MURATA MANUFACTURING	70.0%	3.0%	3.0%	15.0%	9.0%
6996	NICHICON	75.0%	0.0%	0.0%	15.0%	10.0%
6997	NIPPON CHEMI-CON	65.0%	5.0%	0.0%	15.0%	15.0%
6999	KOA	70.0%	0.0%	0.0%	5.0%	25.0%
Total		40.9%	8.7%	7.9%	24.2%	18.3%

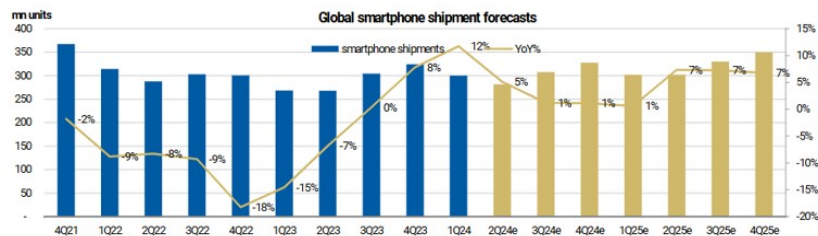
Note: Production data based on Morgan Stanley Research estimation
Source: Company data, Morgan Stanley Research

Preparing for Smartphone Price Compression and Slowdown in Technological Innovation

■ Potential for smartphone commoditization to knock down margins, but also greater earnings opportunities from computerization of autos and other areas



■ Global smartphone shipments



Note: e = Morgan Stanley Research estimates
Source: Company data, Morgan Stanley Research

■ Electronic component firms: Sales breakdown by applications (F24)

Sales in F24							
	(JPY bn)	Smartphone	Computers	AV/Office	Automotive	Industrial	Others
Ibiden	369.4	10.3	186.9	0.0	73.0	14.6	84.6
Nittera	653.0	2.4	0.0	0.0	538.9	32.6	79.0
Minebea Mitsumi	1,522.7	270.9	132.8	345.4	550.0	91.4	132.2
Mabuchi Motor	196.2	0.0	0.0	17.3	152.5	8.6	17.8
Nidec	2,607.1	0.0	178.5	315.6	664.6	1,051.0	397.4
TDK	2,204.8	866.9	400.1	66.7	453.5	360.8	56.7
Alps Alpine	990.4	298.0	43.8	53.0	537.2	22.0	36.3
NDK	53.1	10.5	0.4	0.4	29.2	5.0	7.6
Meiko Electronics	206.8	22.5	13.8	0.0	96.7	5.2	68.6
Hirose Electric	189.4	46.1	10.4	9.8	50.5	46.4	26.2
JAE	221.6	66.9	1.5	1.5	112.8	28.7	10.4
IRISO	56.3	0.0	0.6	4.1	48.2	3.2	0.4
CMK	95.5	0.0	0.5	0.6	81.7	3.0	9.7
Daishinku	38.6	10.2	5.3	0.0	13.8	4.0	5.3
Hamamatsu Photonics	204.0	0.0	0.0	1.0	5.6	66.3	131.0
Kyocera	2,014.5	310.0	130.6	472.1	288.2	398.1	415.4
Taiyo Yuden	341.4	84.6	58.9	27.3	102.4	68.3	0.0
Murata Mfg.	1,743.4	674.2	281.9	150.4	453.1	183.7	0.0
Nichicon	175.8	0.0	15.4	18.8	42.7	58.1	40.9
Nippon Chemi-Con	122.7	0.0	31.7	8.5	45.0	23.1	14.3
KOA	64.1	0.0	8.0	3.5	35.3	7.6	9.8
Total	14,070.8	2,673.5	1,501.1	1,495.8	4,375.0	2,481.7	1,543.6

Sales in F24							
	Smartphone	Computers	AV/Office	Automotive	Industrial	Others	
Ibiden	100%	3%	51%	0%	20%	4%	23%
Nittera	100%	0%	0%	0%	83%	5%	12%
Minebea Mitsumi	100%	18%	9%	23%	36%	6%	9%
Mabuchi Motor	100%	0%	0%	9%	78%	4%	9%
Nidec	100%	0%	7%	12%	25%	40%	15%
TDK	100%	39%	18%	3%	21%	16%	3%
Alps Alpine	100%	30%	4%	5%	54%	2%	4%
NDK	100%	20%	1%	1%	55%	9%	14%
Meiko Electronics	100%	11%	7%	0%	47%	3%	33%
Hirose Electric	100%	24%	5%	5%	27%	25%	14%
JAE	100%	30%	1%	1%	51%	13%	5%
IRISO	100%	0%	1%	7%	86%	6%	1%
CMK	100%	0%	1%	1%	86%	3%	10%
Daishinku	100%	26%	14%	0%	36%	10%	14%
Hamamatsu Photonics	100%	0%	0%	1%	3%	33%	64%
Kyocera	100%	15%	6%	23%	14%	20%	21%
Taiyo Yuden	100%	25%	17%	8%	30%	20%	0%
Murata Mfg.	100%	39%	16%	9%	26%	11%	0%
Nichicon	100%	0%	9%	11%	24%	33%	23%
Nippon Chemi-Con	100%	0%	26%	7%	37%	19%	12%
KOA	100%	0%	12%	5%	55%	12%	15%
Total	100%	19%	11%	11%	31%	18%	11%

Memory Price Outlook

	Mar-25 1Q25	Jun-25 2Q25	Sep-25 3Q25	Dec-25 4Q25	Mar-25 1Q26E	Jan-26 2Q26E	Mar-25 2Q26E	Mar-26 3Q26E	Mar-25 4Q26E
PC DRAM	DDR4: down 13-18% DDR5: down 10-15% Blended ASP: down 10-15%	DDR4: up 13-18% DDR5: up 3-8% Blended ASP: up 3-8%	DDR4: up 38-43% DDR5: up 3-8% Blended ASP: up 8-13%	DDR4: up 43-48% DDR5: up 38-43% Blended ASP: up 38-43%	DDR4: up 120-125% DDR5: up 110-115% Blended ASP: up 110-115%	DDR4: up 28-33% DDR5: up 20-25% Blended ASP: up 20-25%	DDR4: up 28-33% DDR5: up 20-25% Blended ASP: up 20-25%	up 3-8%	up 0-5%
Server DRAM	DDR4: down 10-15% DDR5: down 3-8% Blended ASP: down 5-10%	DDR4: up 18-23% DDR5: up 3-8% Blended ASP: up 3-8%	DDR4: up 28-33% DDR5: up 3-8% Blended ASP: up 3-8%	DDR4: up 60-65% DDR5: up 53-58% Blended ASP: up 53-58%	DDR4: up 95-100% DDR5: up 93-98% Blended ASP: up 93-98%	DDR4: up 20-25% DDR5: up 20-25% Blended ASP: up 20-25%	DDR4: up 28-33% DDR5: up 28-33% Blended ASP: up 28-33%	up 3-8%	up 0-5%
Mobile DRAM	LPDDR4X: down 8-13% LPDDR5(x): down 3-8%	LPDDR4X: up 0-5% LPDDR5(x): up 3-8%	LPDDR4X: up 38-43% LPDDR5(x): up 10-15%	LPDDR4X: up 48-53% LPDDR5(x): up 43-48%	LPDDR4X: up 88-93% LPDDR5(x): 88-93%	LPDDR4X: up 20-25% LPDDR5(x): up 20-25%	LPDDR4X: up 20-25% LPDDR5(x): up 20-25%	LPDDR4X: mostly flat LPDDR5(x): up 8-13%	LPDDR4X: mostly flat LPDDR5(x): up 3-8%
Graphics DRAM	GDDR6: down 8-13% GDDR7: down 0-5%	GDDR6: mostly flat GDDR7: down 0-5%	GDDR6: up 28-33% GDDR7: 5-10%	GDDR6: up 25-30% GDDR7: up 30-35%	GDDR6: up 95-100% GDDR7: up 95-100%	GDDR6: up 20-25% GDDR7: up 20-25%	GDDR6: up 20-25% GDDR7: up 20-25%	GDDR6: up 0-5% GDDR7: up 3-8%	GDDR6: mostly flat GDDR7: up 0-5%
Consumer DRAM	DDR3: down 3-8% DDR4: down 10-15%	DDR3: mostly flat DDR4: up 18-23%	DDR3: up 50-60% DDR4: up 85-90%	DDR3: up 55-60% DDR4: up 45-50%	DDR3: up 75-80% DDR4: up 75-80%	DDR3: up 20-25% DDR4: up 20-25%	DDR3: up 20-25% DDR4: up 20-25%	DDR3: up 0-5% DDR4: up 0-5%	DDR3: mostly flat DDR4: mostly flat
Total DRAM	Conventional DRAM: down 8-13% HBM Blended: up 0-5% (HBM Penetration: 8%)	Conventional DRAM: up 5-10% HBM Blended: up 5-10% (HBM Penetration: 9%)	Conventional DRAM: up 10-15% HBM Blended: up 15-20% (HBM penetration: 10%)	Conventional DRAM: up 45-50% HBM Blended: up 50-55% (HBM penetration: 11%)	Conventional DRAM: up 93-98% HBM Blended: up 83-88%	Conventional DRAM: up 20-25% HBM Blended: up 18-23%	Conventional DRAM: up 23-28% HBM Blended: up 20-25%	Conventional DRAM: up 3-8%	Conventional DRAM: up 0-5%
	1Q25	2Q25	3Q25	4Q25	1Q26E	2Q26E	2Q26E	3Q26E	4Q26E
eMMC									
UFS	down 15-20%	up 5-10%	Mostly flat	up 20-25%	up 95-100%	up 20-25%	up 30-35%	up 5-10%	mostly flat
Enterprise SSD	down 18-23%	mostly flat	up 3-8%	up 25-30%	up 75-80%	up 15-20%	up 33-38%	up 5-10%	up 0-5%
Client SSD	down 18-23%	up 5-10%	up 5-10%	up 20-25%	up 105-110%	up 20-25%	up 30-35%	up 5-10%	mostly flat
3D NAND Wafers (TLC & QLC)	down 8-13%	up 15-20%	up 8-13%	up 145-150%	up 75-80%	up 15-20%	up 25-30%	up 5-10%	up 0-5%
Total NAND Flash	down 15-20%	up 3-8%	up 3-8%	up 33-38%	up 85-90%	up 18-23%	up 30-35%	up 5-10%	up 0-5%

Source: TrendForce, Morgan Stanley Research

End-product Market and Major Components Market (Calendar Year)

	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026e	2027e
(mn units)																
Server	8.7	9.0	9.2	9.7	9.6	10.2	11.8	11.8	12.7	13.5	14.9	12.4	14.5	16.4	16.7	16.8
YoY % Change	0%	3%	3%	5%	-2%	7%	16%	0%	7%	7%	10%	-17%	17%	13%	1%	1%
PC	346.1	311.8	304.7	272.1	256.2	261.3	260.2	267.5	306.6	353.6	293.8	253.2	261.6	288.5	250.2	241.5
YoY % Change	-4%	-10%	-2%	-11%	-6%	2%	0%	3%	15%	15%	-17%	-14%	3%	10%	-13%	-3%
Desktop	146.0	134.2	131.3	110.7	100.8	100.8	96.0	95.9	85.8	95.8	85.7	72.2	73.0	83.6	73.2	71.5
YoY % Change	-4%	-8%	-2%	-16%	-9%	0%	-5%	0%	-11%	12%	-11%	-16%	1%	15%	-12%	-2%
Notebook	200.1	177.5	173.4	161.4	155.4	160.5	164.1	171.6	220.8	257.7	208.1	181.0	188.6	204.9	176.9	170.0
YoY % Change	-4%	-11%	-2%	-7%	-4%	3%	2%	5%	29%	17%	-19%	-13%	4%	9%	-14%	-4%
Smartphone	726.7	1,018.7	1,301.5	1,437.6	1,469.5	1,465.3	1,402.5	1,372.6	1,281.2	1,359.8	1,206.2	1,165.5	1,237.5	1,264.0	1,094.0	1,132.6
YoY % Change	47%	40%	28%	10%	2%	0%	-4%	-2%	-7%	6%	-11%	-3%	6%	2%	-13%	4%
5G smartphone	0.0	0.0	0.0	0.0	0.0	0.0	0.0	16.1	255.3	559.5	636.8	703.4	814.6	893.1	829.6	901.3
YoY % Change	-	-	-	-	-	-	-	-	1485%	119%	14%	10%	16%	10%	-5%	9%
iPhone	135.9	153.5	192.7	231.5	215.4	215.8	204.9	199.3	203.4	235.8	226.7	235.6	234.3	251.7	252.8	267.0
YoY % Change	46%	13%	26%	20%	-7%	0%	-5%	-3%	2%	16%	-4%	4%	-1%	7%	0%	6%
HDD	578.0	551.9	564.1	468.8	423.9	403.1	375.4	317.7	260.0	259.3	172.2	122.3	123.9	123.8	126.8	125.7
YoY % Change	-7%	-5%	2%	-17%	-10%	-5%	-7%	-15%	-18%	0%	-34%	-29%	1%	0%	2%	-1%
Enterprise	33.2	31.8	32.4	32.1	24.8	22.6	21.5	17.4	12.9	13.4	8.6	5.8	4.4	3.2	2.3	1.5
NL	28.1	35.5	36.9	37.2	41.0	42.4	51.3	52.7	59.1	73.5	61.5	39.4	58.6	67.6	75.5	79.6
3.5"ATA	216.2	203.2	202.6	161.0	142.9	135.4	120.5	103.9	87.0	89.9	55.0	45.2	35.6	30.9	28.8	27.6
2.5"	298.7	280.7	291.7	238.6	215.1	202.7	182.0	143.6	101.0	82.5	46.3	32.0	25.3	22.0	20.3	17.1
Global Light vehicle sales	75.0	78.2	81.4	83.5	92.3	94.4	93.9	89.9	77.2	80.3	79.0	86.7	88.7	91.1	91.4	93.2
YoY % Change	5%	4%	4%	3%	11%	2%	-1%	-4%	-14%	4%	-2%	10%	2%	3%	0%	2%
ICE				83.5	91.8	93.7	92.6	88.5	75.1	75.2	71.0	76.2	77.0	76.2	75.4	75.0
YoY % Change					10%	2%	-1%	-4%	-15%	0%	-6%	7%	1%	-1%	-1%	-1%
EV				-	0.5	0.7	1.3	1.4	2.1	5.1	8.0	10.5	11.7	14.9	16.0	18.2
YoY % Change						40%	86%	8%	50%	143%	57%	31%	11%	27%	7%	14%

Note: e = Morgan Stanley Research estimates

Source: Company data, Morgan Stanley Research

End-product Market and Major Components Market (Quarterly)

	CY2021			CY2022			CY2023			CY2024			CY2025			CY2026					
(mn units)	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q	1Q
Server	2.9	3.4	3.5	3.7	3.4	3.8	3.9	3.8	3.0	3.1	3.1	3.2	3.2	3.6	3.7	4.0	3.9	4.2	4.2	4.4	3.9
YoY % Change	9%	3%	8%	8%	14%	13%	12%	1%	-10%	-19%	-22%	-16%	7%	17%	19%	25%	23%	16%	15%	11%	0%
Sequential % Change	-15%	15%	4%	7%	-10%	14%	3%	-4%	-20%	2%	0%	3%	2%	13%	1%	8%	0%	7%	0%	4%	-10%
PC	85.7	85.7	88.3	93.8	80.6	71.6	74.4	67.2	57.3	61.4	68.7	65.9	59.5	63.4	69.1	69.6	64.0	69.7	76.5	78.3	65.6
YoY % Change	58%	14%	5%	1%	-6%	-16%	-16%	-28%	-29%	-14%	-8%	-2%	4%	3%	1%	6%	8%	10%	11%	12%	3%
Sequential % Change	-8%	0%	3%	6%	-14%	-11%	4%	-10%	-15%	7%	12%	-4%	-10%	7%	9%	1%	-8%	9%	10%	2%	-16%
Desktop	22.3	23.6	24.3	25.6	22.7	21.5	21.8	19.7	17.5	17.4	18.6	18.7	17.3	17.3	18.8	19.6	18.5	19.5	22.3	23.4	18.2
YoY % Change	14%	19%	10%	5%	2%	-9%	-10%	-23%	-23%	-19%	-15%	-5%	-1%	0%	1%	5%	7%	12%	19%	19%	-2%
Sequential % Change	-9%	6%	3%	5%	-11%	-5%	2%	-10%	-11%	-1%	7%	0%	-7%	0%	8%	4%	-6%	5%	15%	5%	-22%
Notebook (incl. Netbook)	63.5	62.1	64.0	68.1	57.9	50.1	52.6	47.5	39.8	44.0	50.0	47.2	42.2	46.0	50.3	50.0	45.5	50.3	54.2	54.9	47.4
YoY % Change	83%	12%	3%	0%	-9%	-19%	-18%	-30%	-31%	-12%	-5%	-1%	6%	5%	1%	6%	8%	9%	8%	10%	4%
Sequential % Change	-7%	-2%	3%	6%	-15%	-13%	5%	-10%	-16%	11%	14%	-6%	-11%	9%	9%	-1%	-9%	10%	8%	1%	-14%
Smartphone	344.7	313.4	334.2	367.6	314.5	287.6	303.2	300.6	268.4	267.9	303.7	324.0	300.4	292.3	314.7	328.8	302.6	297.4	326.5	337.5	293.8
YoY % Change	25%	13%	-6%	-2%	-9%	-8%	-9%	-18%	-15%	-7%	0%	8%	12%	9%	4%	1%	1%	2%	4%	3%	-3%
Sequential % Change	-8%	-9%	7%	10%	-14%	-9%	5%	-1%	-11%	0%	13%	7%	-7%	-3%	8%	4%	-8%	-2%	10%	3%	-13%
5G smartphone	125.1	118.1	136.5	179.8	155.5	146.3	161.6	173.4	160.5	149.3	182.6	211.1	187.9	186.0	211.4	229.3	213.6	203.9	228.0	247.6	219.0
YoY % Change	491%	139%	110%	50%	24%	24%	18%	-4%	3%	2%	13%	22%	17%	25%	16%	9%	14%	10%	8%	8%	3%
Sequential % Change	4%	-6%	16%	32%	-14%	-6%	10%	7%	-7%	-7%	22%	16%	-11%	-1%	14%	8%	-7%	-5%	12%	9%	-12%
iPhone	55.3	44.4	51.1	85.0	56.5	45.4	52.3	72.5	55.8	44.8	54.4	80.6	53.0	46.1	57.4	77.9	59.1	48.4	59.6	84.7	61.8
YoY % Change	50%	18%	23%	-3%	2%	2%	2%	-15%	-1%	-1%	4%	11%	-5%	3%	6%	-3%	12%	5%	4%	9%	4%
Sequential % Change	-37%	-20%	15%	66%	-34%	-20%	15%	39%	-23%	-20%	21%	48%	-34%	-13%	25%	36%	-24%	-18%	23%	42%	-27%
HDD	64.1	67.6	66.3	61.2	52.8	44.6	38.5	36.3	33.5	31.3	28.6	28.9	29.7	30.3	32.1	31.7	28.7	30.4	31.6	33.0	33.0
YoY % Change	-5%	15%	4%	-13%	-18%	-34%	-42%	-41%	-37%	-30%	-26%	-20%	-11%	-3%	12%	10%	-3%	0%	-2%	4%	15%
Sequential % Change	-9%	5%	-2%	-8%	-14%	-15%	-14%	-6%	-8%	-7%	-9%	1%	3%	2%	6%	-1%	-10%	6%	4%	5%	0%
Enterprise	3.2	3.5	3.5	3.3	2.7	2.4	1.6	1.7	1.4	1.6	1.4	1.4	1.5	1.0	1.0	0.9	0.8	0.8	0.7	0.8	0.8
NL	16.0	19.3	19.7	18.5	18.7	18.0	14.2	10.6	11.4	9.2	9.0	9.9	12.3	13.8	15.9	16.5	14.6	16.5	17.3	19.2	18.7
3.5"ATA	22.2	23.6	23.2	20.8	16.5	14.1	13.0	12.8	11.2	12.1	11.2	10.7	9.3	9.4	9.0	7.9	7.6	7.9	8.0	7.6	8.3
2.5"	22.7	21.2	19.9	18.7	14.9	10.0	9.7	11.3	9.5	8.4	7.1	6.9	6.6	6.1	6.2	6.4	5.7	5.3	5.6	5.5	5.3
HDD/PC	0.7	0.8	0.8	0.7	0.7	0.6	0.5	0.5	0.6	0.5	0.4	0.4	0.5	0.5	0.5	0.5	0.4	0.4	0.4	0.4	0.5

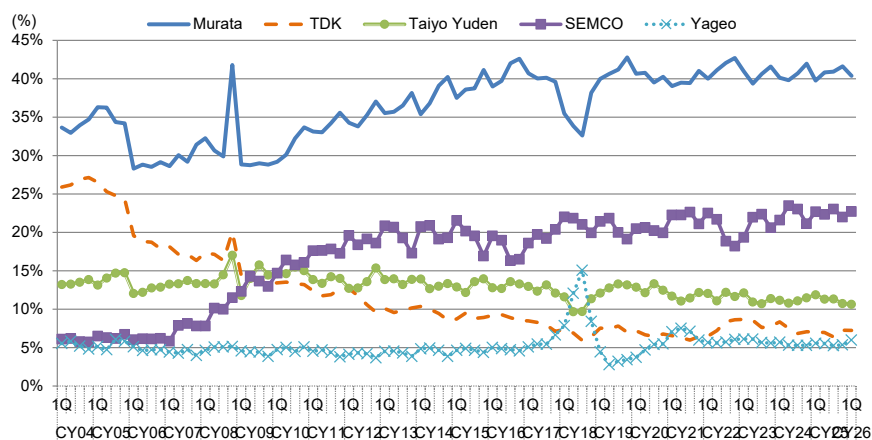
Source: Company data, Morgan Stanley Research

MLCC Market Data

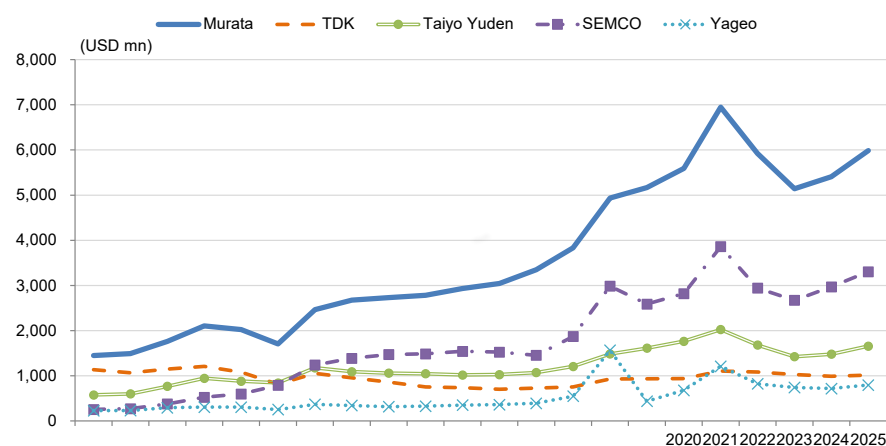
■ MLCC market forecasts

Base Case	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026e	2027e	2028e
Total worldwide shipment quantity (Bil pcs)	1,504	1,480	1,640	2,050	2,029	2,167	2,293	2,605	2,735	3,080	3,480	3,828	3,254	3,695	4,446	3,572	3,187	3,386	3,773	4,037	4,360	4,709
YoY (%)	14.9%	-1.6%	10.8%	24.9%	-1.0%	6.8%	5.8%	13.6%	5.0%	12.6%	13.0%	10.0%	-15.0%	13.5%	20.3%	-19.7%	-10.8%	6.2%	11.4%	7.0%	8.0%	8.0%
CAGR (1991~)	14.9%	13.8%	13.7%	14.2%	13.4%	13.1%	12.8%	12.8%	12.5%	12.5%	12.5%	12.4%	11.3%	11.4%	11.6%	10.5%	9.7%	9.6%	9.7%	9.6%	9.6%	9.5%
ASP in worldwide shipment (Cent)	0.47	0.42	0.36	0.38	0.39	0.36	0.33	0.30	0.29	0.26	0.27	0.37	0.39	0.38	0.39	0.40	0.40	0.39	0.39	0.41	0.42	0.43
YoY (%)	-0.2%	-10.1%	-14.0%	6.8%	1.2%	-7.7%	-7.2%	-10.6%	-4.1%	-7.2%	3.8%	34.3%	4.6%	-2.8%	4.7%	2.0%	-1.0%	-1.0%	-1.0%	5.0%	3.0%	2.5%
CAGR (1991~)	-8.6%	-8.7%	-9.0%	-8.2%	-7.8%	-7.8%	-7.7%	-7.9%	-7.7%	-7.7%	-7.3%	-6.0%	-5.6%	-5.5%	-5.2%	-5.0%	-4.9%	-4.8%	-4.6%	-4.4%	-4.2%	-4.0%
Total worldwide shipment value (Bil \$)	7.00	6.19	5.90	7.87	7.89	7.77	7.63	7.75	7.80	8.15	9.56	14.13	12.56	13.86	17.47	14.32	12.64	13.30	14.67	16.48	18.34	20.30
YoY (%)	14.6%	-11.6%	-4.7%	33.4%	0.2%	-1.5%	-1.8%	1.6%	0.7%	4.5%	17.3%	47.7%	-11.1%	10.3%	26.0%	-18.1%	-11.7%	5.2%	10.3%	12.4%	11.2%	10.7%
CAGR (1991~)	5.0%	4.0%	3.5%	4.8%	4.6%	4.3%	4.0%	3.9%	3.8%	3.8%	4.3%	5.7%	5.0%	5.2%	5.8%	5.0%	4.4%	4.4%	4.6%	4.8%	5.0%	5.1%

■ Global share of the top 5 ceramic capacitor firms



■ MLCC sales



Note: e = Morgan Stanley Research estimates

Source: Company data, Morgan Stanley Research

Capacitor Market: 016008 MLCCs are 75% Smaller than 0201 MLCCs

■ MLCC size comparison

	4532	3216	2012	1608	1005	0603	0402	0201	016008
Length	4.5mm	3.2mm	2.0mm	1.6mm	1.0mm	0.6mm	0.4mm	0.25mm	0.16mm
Width	3.2mm	1.6mm	1.25mm	0.8mm	0.5mm	0.3mm	0.2mm	0.125mm	0.08mm
Thickness	2.0mm	0.85mm	0.6mm	0.8mm	0.5mm	0.3mm	0.2mm	0.125mm	0.08mm
Volume	28.8mm ³	4.35mm ³	1.5mm ³	1.024mm ³	0.25mm ³	0.054mm ³	0.016mm ³	0.004mm ³	0.001mm ³
Volume ratio	7,373	1,114	384	262	64.0	13.8	4.1	1.0	0.26

■ Number of MLCCs per vehicle

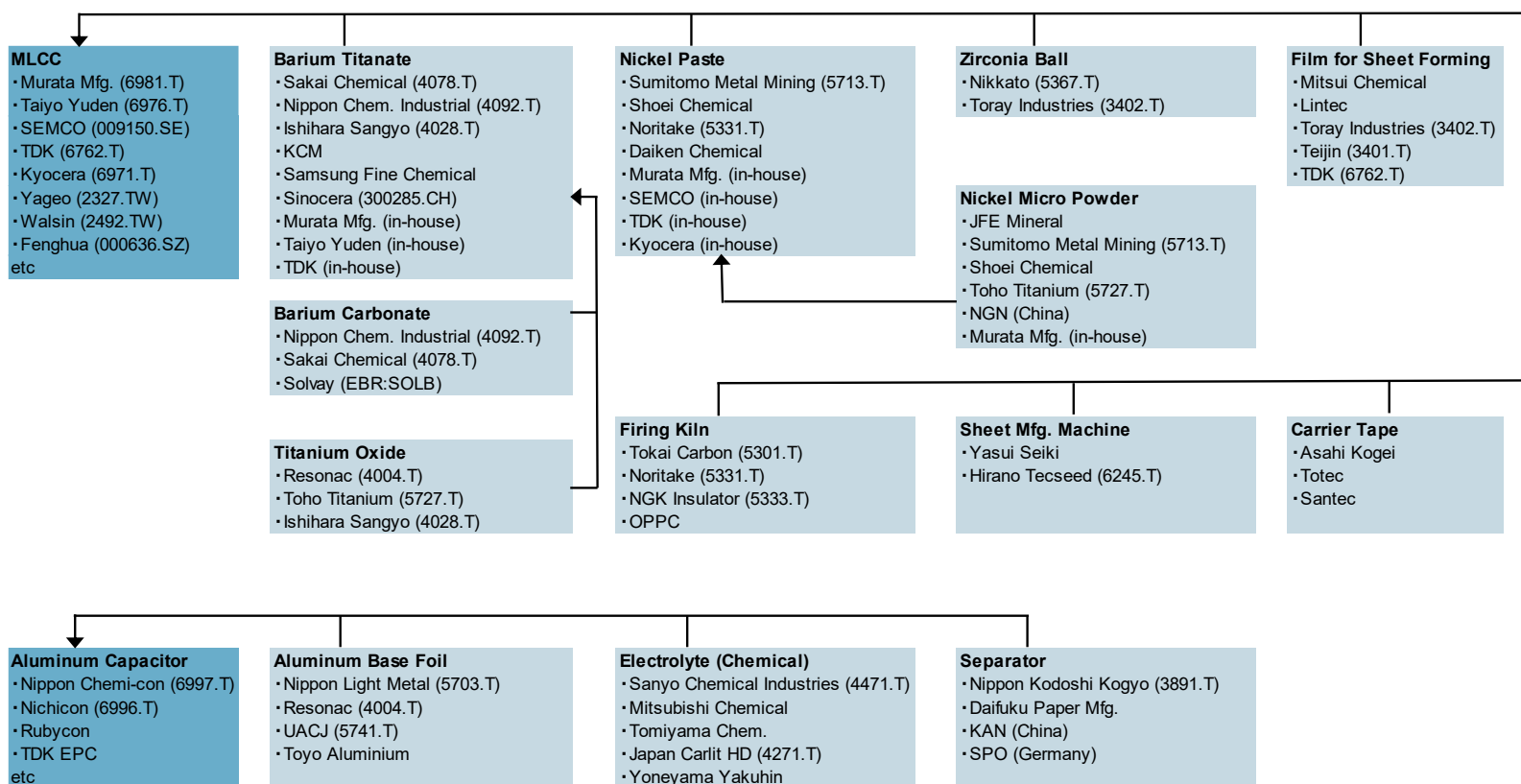
	GAS	HV	BEV	PHEV
(units: pcs)				
Power Train	300-500	1,000-1,600	2,000-2,500	2,500-3,000
ADAS(Lv2-5)	1,500-5,000			
Safety	300-1,000			
Infotainment	500-2,500			
Other	500-2,500			

Source: Company data, Morgan Stanley Research

MLCC Sales, Market Share, OP

	CY23				CY24				CY25				CY26				2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	
	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q																		
Total (USD mn)	2,940	3,131	3,354	3,221	3,171	3,276	3,518	3,334	3,321	3,682	3,937	3,731	3,825	5,917	7,868	7,885	7,769	7,626	7,731	7,802	8,167	9,564	14,129	12,562	13,861	17,311	14,316	12,645	13,299	14,672				
Major 5	2,567	2,724	2,918	2,802	2,759	2,850	3,061	2,900	2,889	3,203	3,426	3,246	3,328	4,411	6,308	6,462	6,444	6,405	6,586	6,660	7,000	8,215	11,903	10,744	11,798	15,013	12,467	11,010	11,571	12,764				
Murata	1,205	1,233	1,362	1,340	1,273	1,305	1,432	1,399	1,320	1,504	1,611	1,553	1,545	1,708	2,467	2,677	2,730	2,784	2,933	3,043	3,349	3,834	4,934	5,168	5,591	6,945	5,921	5,140	5,409	5,988				
TDK	256	268	256	251	265	247	241	236	234	256	253	271	277	816	1,053	957	864	757	738	701	730	753	931	934	939	1,105	1,095	1,031	988	1,015				
Taiyo Yuden	357	343	359	367	354	354	390	383	395	416	447	401	408	843	1,177	1,093	1,061	1,047	1,020	1,028	1,073	1,209	1,484	1,613	1,767	2,026	1,683	1,425	1,481	1,659				
SEMCO	569	689	750	664	685	770	811	705	754	823	907	821	869	791	1,241	1,389	1,471	1,488	1,542	1,525	1,455	1,871	2,987	2,590	2,820	3,860	2,943	2,672	2,972	3,305				
Yageo	180	191	190	180	181	175	187	177	185	204	207	200	229	254	370	345	318	329	353	363	393	549	1,567	439	681	1,077	825	742	720	797				
Other	373	407	436	419	412	426	457	433	432	479	512	485	497	1,505	1,548	1,423	1,326	1,222	1,145	1,142	1,162	1,438	2,400	1,815	2,063	2,298	1,848	1,635	1,729	1,907				
Market Share																																		
Major 5	87.3%	87.0%	87.0%	87.0%	87.0%	87.0%	87.0%	87.0%	87.0%	87.0%	87.0%	87.0%	87.0%	74.6%	80.2%	81.9%	82.9%	84.0%	85.2%	85.4%	85.7%	85.9%	84.2%	85.5%	85.1%	86.7%	87.1%	87.1%	87.0%	87.0%				
Murata	41.0%	39.4%	40.6%	41.7%	40.2%	39.8%	40.7%	42.0%	39.8%	40.8%	40.9%	41.6%	40.4%	28.9%	31.4%	33.9%	35.1%	36.5%	37.9%	39.0%	41.0%	40.1%	34.9%	41.1%	40.3%	40.1%	41.4%	40.6%	40.7%	40.8%				
TDK	8.7%	8.6%	7.6%	7.8%	8.4%	7.5%	6.8%	7.1%	7.1%	7.0%	6.4%	7.3%	7.2%	13.8%	13.4%	12.1%	11.1%	9.9%	9.5%	9.0%	8.9%	7.9%	6.6%	7.4%	6.8%	6.4%	7.6%	8.1%	7.4%	6.9%				
Taiyo Yuden	12.1%	10.9%	10.7%	11.4%	11.2%	10.8%	11.1%	11.5%	11.9%	11.3%	11.4%	10.7%	10.7%	14.2%	15.0%	13.9%	13.7%	13.7%	13.2%	13.2%	13.1%	12.6%	10.5%	12.8%	12.7%	11.7%	11.8%	11.3%	11.1%	11.3%				
SEMCO	19.4%	22.0%	22.4%	20.7%	21.6%	23.5%	23.1%	21.2%	22.7%	22.3%	23.0%	22.0%	22.7%	13.4%	15.8%	17.6%	18.9%	19.5%	19.9%	19.5%	17.8%	19.6%	21.1%	20.6%	20.3%	22.3%	20.6%	21.1%	22.3%	22.5%				
Yageo	6.1%	6.1%	5.7%	5.3%	5.7%	5.3%	5.3%	5.3%	5.6%	5.5%	5.3%	5.4%	6.0%	4.3%	4.7%	4.4%	4.1%	4.3%	4.6%	4.7%	4.8%	5.7%	11.1%	3.5%	4.9%	6.2%	5.8%	5.9%	5.4%	5.4%				
Other	12.7%	13.0%	13.0%	13.0%	13.0%	13.0%	13.0%	13.0%	13.0%	13.0%	13.0%	13.0%	13.0%	25.4%	19.7%	18.1%	17.1%	16.0%	14.8%	14.6%	14.2%	15.0%	17.0%	14.4%	14.9%	13.3%	12.9%	12.9%	13.0%	13.0%				
Market Share in Major 5																																		
Murata	47.0%	45.3%	46.7%	47.8%	46.2%	45.8%	46.8%	48.2%	45.7%	46.9%	47.0%	47.8%	46.4%	38.7%	39.1%	41.4%	42.4%	43.5%	44.5%	45.7%	47.8%	46.7%	41.5%	48.1%	47.4%	46.3%	47.5%	46.7%	46.8%	46.9%				
TDK	10.0%	9.9%	8.8%	9.0%	9.6%	8.7%	7.9%	8.1%	8.1%	8.0%	7.4%	8.4%	8.3%	18.5%	16.7%	14.8%	13.4%	11.8%	11.2%	10.5%	10.4%	9.2%	7.8%	8.7%	8.0%	7.4%	8.8%	9.4%	8.5%	7.9%				
Taiyo Yuden	13.9%	12.6%	12.3%	13.1%	12.8%	12.4%	12.7%	13.2%	13.7%	13.0%	13.1%	12.3%	12.3%	19.1%	18.7%	16.9%	16.5%	16.4%	15.5%	15.4%	15.3%	14.7%	12.5%	15.0%	13.5%	13.5%	12.9%	12.8%	13.0%					
SEMCO	22.2%	25.3%	25.7%	23.7%	24.8%	27.0%	26.5%	24.3%	26.1%	25.7%	26.5%	25.3%	26.1%	17.9%	19.7%	21.5%	22.8%	23.2%	23.4%	22.9%	20.8%	22.8%	25.1%	24.1%	23.9%	25.7%	23.6%	24.3%	25.7%	25.9%				
Yageo	7.0%	7.0%	6.5%	6.4%	6.6%	6.1%	6.1%	6.1%	6.4%	6.4%	6.1%	6.2%	6.9%	5.7%	5.9%	5.3%	4.9%	5.1%	5.4%	5.5%	6.7%	6.7%	13.2%	4.1%	5.8%	7.2%	6.6%	6.7%	6.2%	6.2%				
YoY Growth	-29.4%	-17.1%	0.3%	6.2%	7.9%	4.7%	4.9%	3.5%	4.7%	12.4%	11.8%	11.9%	15.2%																					
Major 5	-28.9%	-17.2%	-0.1%	5.9%	7.5%	4.7%	4.9%	3.5%	4.7%	12.4%	11.9%	11.9%	15.2%																					
Murata	-27.6%	-20.8%	-3.0%	3.6%	5.6%	5.9%	5.1%	4.4%	3.7%	15.2%	12.5%	11.0%	17.0%																					
TDK	-5.6%	-3.1%	-9.9%	-4.9%	3.7%	-8.1%	-5.8%	-6.0%	-11.5%	4.0%	4.9%	14.9%	18.1%																					
Taiyo Yuden	-28.9%	-18.5%	-11.9%	3.8%	-0.7%	3.3%	8.6%	4.5%	11.6%	17.6%	14.6%	4.6%	3.2%																					
SEMCO	-39.3%	-16.3%	19.0%	20.3%	20.4%	11.9%	8.1%	6.2%	10.0%	6.8%	11.9%	16.4%	15.3%																					
Yageo	-23.9%	-9.9%	-1.2%	-1.9%	0.7%	-8.7%	-1.8%	-1.8%	2.2%	16.7%	11.2%	13.1%	23.7%																					
Other	-32.0%	-16.8%	2.7%	8.7%	10.4%	4.7%	4.9%	3.5%	4.7%	12.4%	11.9%	11.9%	15.2%																					
QoQ Growth	-3.0%	6.5%	7.1%	-4.0%	-1.5%	3.3%	7.4%	-5.2%	-0.4%	10.9%	6.9%	-5.2%	2.5%																					
Major 5	-3.0%	6.1%	7.1%	-4.0%	-1.5%	3.3%	7.4%	-5.2%	-0.4%	10.9%	6.9%	-5.2%	2.5%																					
Murata	-6.8%	2.3%	10.5%	-1.7%	-5.0%	2.5%	9.8%	-2.3%	-5.6%	13.9%	7.2%	-3.6%	-0.5%																					
TDK	-3.2%	5.0%	-4.7%	-1.7%	5.5%	-6.9%	-2.4%	-1.9%	-0.7%	9.4%	-1.5%	7.4%	2.1%																					
Taiyo Yuden	1.1%	-3.9%	4.8%	2.0%	-3.4%	0.0%	10.1%	-1.8%	3.2%	5.4%	7.3%	-10.4%	1.8%																					
SEMCO	3.1%	21.0%	9.0%	-11.5%	3.2%	12.4%	5.3%	-13.0%	6.9%	9.2%	10.3%	-9.5%	5.8%																					
Yageo	-2.1%	6.3%	-0.8%	-5.1%	0.5%	-3.6%	6.7%	-5.0%	4.6%	10.0%	1.7%	-3.4%	14.4%																					
Other	-3.0%	9.0%	7.1%	-4.0%	-1.5%	3.3%	7.4%	-5.2%	-0.4%	10.9%	6.9%	-5.2%	2.5%																					
OP																																		
Murata	35.8	42.8	52.0	50.8	47.4	63.7	64.8	58.8	47.9	61.2	73.0	66.4	70.1	-8.8	31.9	22.3	19.3	50.6	72.2	115.6	73.3	105.8	176.1	190.5	196.6	312.1	270.5	181.4	234.7	248.4				
TDK	7.9	9.7	8.2	10.7	8.8	8.1	6.4	4.7	2.4	1.8	3.0	6.6	6.0	-20.3	-3.9	-9.4	-15.4	-0.2	3.4	5.9	3.3	7.4	19.0	20.4	16.1	27.6	43.3	36.5	27.9	13.8				
Taiyo Yuden	-2.9	0.0	0.8	2.0	1.3	1.6	2.4	0.5	2.9	3.5	4.5	5.8	3.1	-0.2	14.7	-0.6	0.4	9.1	6.7	15.9	6.9	15.4	33.0	41.8	37.5	55.4	43.4	-0.1	5.8	16.7				
SEMCO	7.5	12.3	10.8	7.9	8.9	14.5	17.0	14.0	11.5	14.3	18.1	15.2	22.6	12.2	19.1	12.3	8.6	11.1	8.1	20.5	11.9	26.1	46.1	44.5	48.7	96.4	58.9	38.5	54.4	59.0				
- SEMCO (KRW bn)	72.6	117.8	98.6	70.2	80.1	127.9	154.2	128.0	109.5	138.3	169.7	142.5	211.0	160.2	252.5	168.8	146.9	126.3	78.0	190.4	126.7	261.2	459.7	470.3	536.0	1,005.3	585.8	359.2	490.2	560.0				
OP (USD mn)																																		
- Murata	270.8	311.1	359.4	343.5	319.5	408.5	434.0	385.9	313.7	423.2	495.1	431.1	446.6	-90.8	367.2	277.5	241.4	515.0	672.0	953.7	680.8	942.8	1,587.4	1,746.5	1,847.2	2,836.6	2,075.8	1,28						

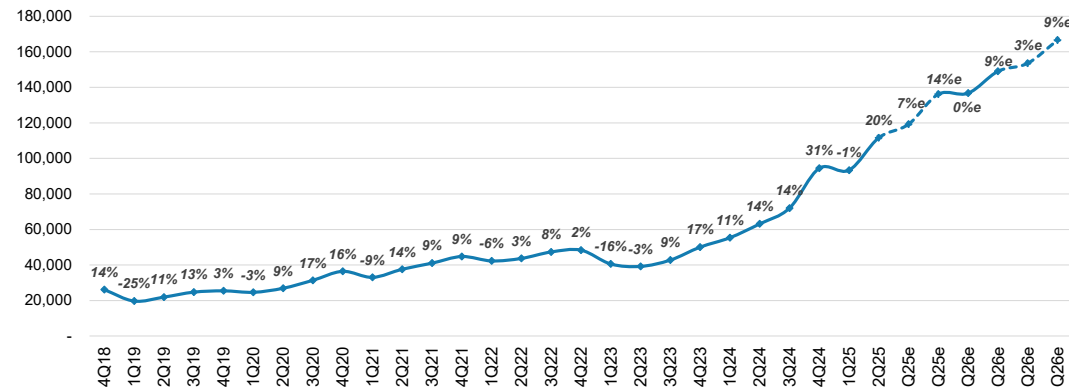
Supply Chains of MLCC and Aluminium Capacitors



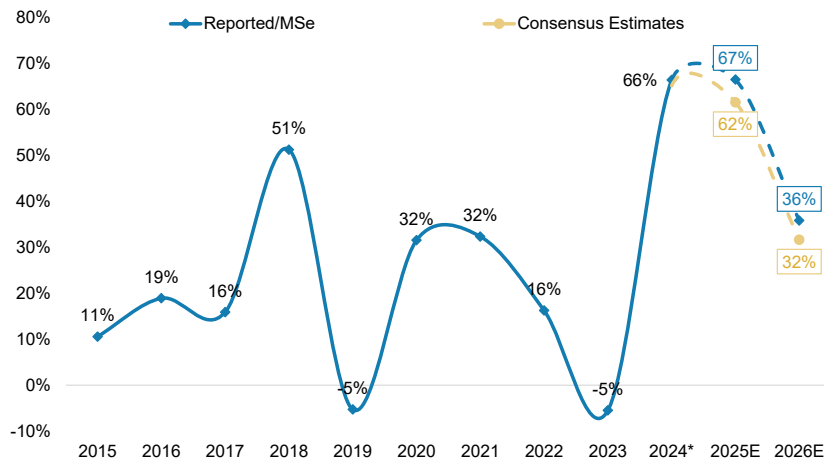
Source: Company data, Morgan Stanley Research

Capex by Hyperscalers Has Increased Significantly

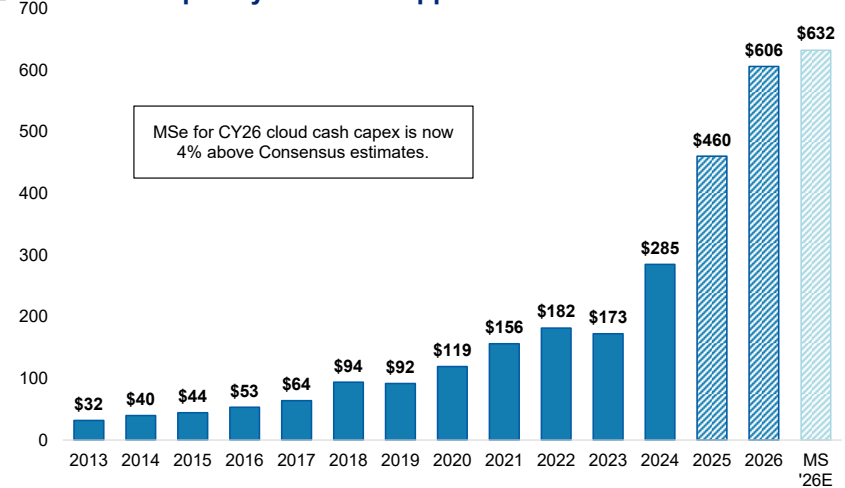
■ Total cash capex by 11 cloud suppliers and QoQ change



■ Total cash capex by 11 cloud suppliers and YoY change



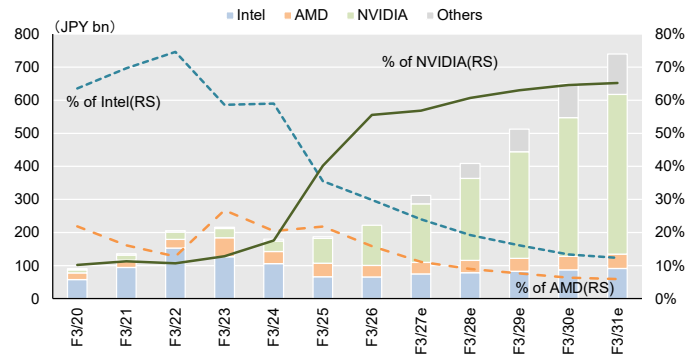
■ Total cash capex by 11 cloud suppliers



Note: E = FactSet consensus, MS'26E and e = Morgan Stanley Research estimates
Source: Company data, Morgan Stanley Research

We Expect Ibiden Will Maintain Nearly 100% Market Share in FC Packages for NVIDIA

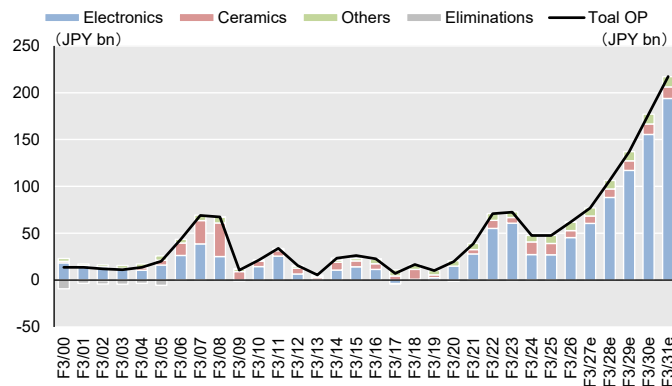
■ ABF package substrate sales by customers*



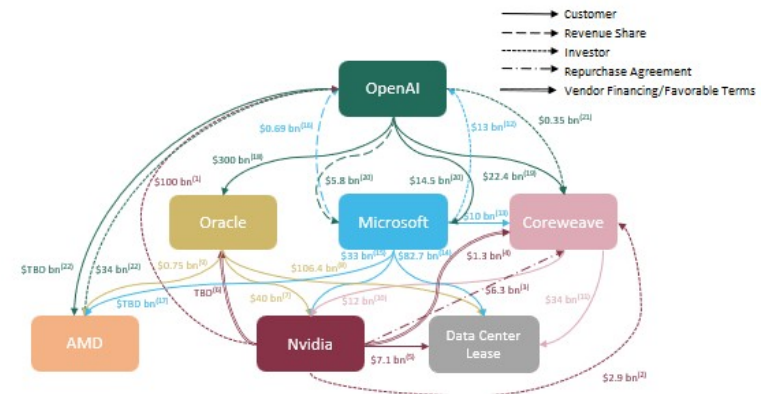
■ Size and # of layers for NVIDIA ABF packages

	Package Size	Surface Area	# of BU Layers	SAP Demand Size
Hopper	55 x 60mm	3,300	10	100
Blackwell	70 x 80mm	5,600	12	204
Rubin	80 x 100mm	8,000	16	388

■ OP by segment and total OP



■ AI ecosystem capital flows



Note: *Morgan Stanley Research assumption based on company data, e = Morgan Stanley Research estimates
Source: Company data, Morgan Stanley Research

Ibiden: Medium-term Results and Forecasts

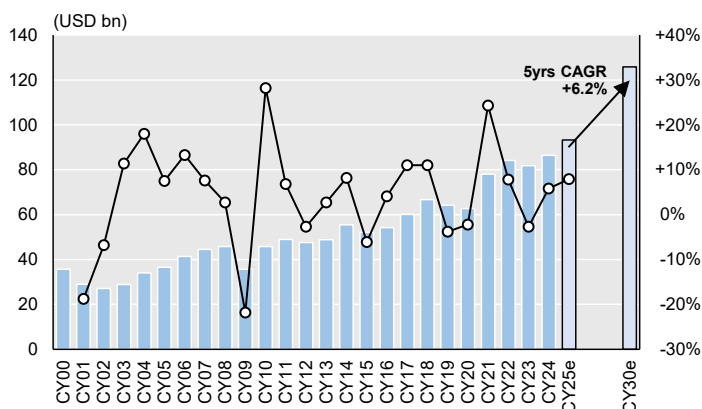
(bn JPY, %)	F3/07	F3/19	F3/20	F3/21	F3/22	F3/23	F3/24	F3/25	F3/26	F3/27 MSe	F3/28 MSe	F3/29 MSe	F3/30 MSe	F3/31 MSe	CAGR F3/25- F3/31e
Sales	398.6	291.1	296.0	323.5	401.1	417.5	370.5	369.4	416.2	499.7	602.5	711.4	854.5	950.8	17.1%
Electronics	205.0	116.0	132.2	166.1	237.0	250.7	190.7	197.2	243.3	322.1	419.1	522.4	659.7	749.9	24.9%
Ceramics	96.2	102.5	88.4	87.4	90.7	89.9	96.5	84.1	82.6	83.8	86.3	88.9	91.8	94.8	2.0%
Others	97.4	72.7	75.4	70.0	73.5	76.9	83.3	88.1	90.3	93.9	97.1	100.1	103.1	106.1	3.1%
OP	69.0	10.1	19.7	38.6	70.8	72.4	47.6	47.6	62.0	77.0	106.4	137.0	177.1	217.2	28.8%
Electronics	38.6	2.5	14.9	27.8	55.1	60.6	27.3	26.8	45.2	60.5	88.2	117.2	155.5	193.8	39.0%
Ceramics	24.9	3.0	-1.0	4.6	8.7	6.1	13.4	12.2	7.6	7.9	9.0	10.0	11.2	12.4	0.2%
Others	5.9	4.7	5.8	6.6	7.1	6.1	7.1	8.7	9.0	8.5	9.2	9.8	10.4	11.0	3.6%
Eliminations	-0.4	-0.0	-0.1	-0.4	-0.1	-0.6	-0.1	-0.2	0.2	0.0	0.0	0.0	0.0	0.0	
OPM	17.3%	3.5%	6.7%	11.9%	17.7%	17.3%	12.8%	12.9%	14.9%	15.4%	17.7%	19.3%	20.7%	22.8%	
ROE	19.0%	1.2%	4.2%	8.8%	12.1%	13.3%	6.9%	6.8%	12.2%	9.5%	12.3%	14.6%	16.5%	17.3%	
CapEx	70.4	22.9	57.1	78.2	60.6	131.3	146.6	157.3	64.3	160.0	160.0	210.0	110.0	110.0	
Electronics	54.1	14.2	48.7	70.4	54.6	121.5	136.9	146.8	45.2	150.0	150.0	200.0	100.0	100.0	
Ceramics	9.3	5.4	6.1	3.1	3.0	3.1	5.2	7.1	7.6	5.0	5.0	5.0	5.0	5.0	
Others	6.9	3.4	2.3	4.7	3.0	6.6	4.5	3.4	9.0	5.0	5.0	5.0	5.0	5.0	
Depreciation	31.5	25.1	24.2	35.4	52.7	54.9	46.0	54.2	66.2	79.5	95.0	94.0	94.0	94.0	
Electronics	16.6	13.8	12.3	25.5	43.1	45.4	36.7	44.2	55.8	71.0	86.0	85.0	85.0	85.0	
Ceramics	12.5	9.6	8.4	6.8	6.1	5.6	5.0	5.6	6.0	4.5	5.0	5.0	5.0	5.0	
Others	2.4	3.0	3.5	3.1	3.5	3.9	4.3	4.4	4.4	4.0	4.0	4.0	4.0	4.0	

Note: MSe = Morgan Stanley Research estimates

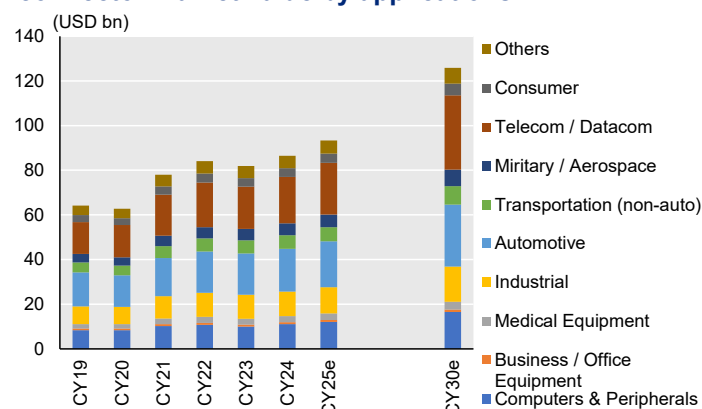
Source: Company data, Morgan Stanley Research

Connector Market

Global connector market value



Global connector market value by applications



Global connector market value CAGR by applications

	CY04-09	CY09-14	CY14-19	CY19-24	CY24-30e
Computers & Peripherals	3.0%	5.9%	-0.3%	5.7%	6.9%
Business / Office Equipment	4.1%	5.1%	-0.7%	3.1%	3.5%
Medical Equipment	8.2%	6.4%	2.6%	6.5%	4.4%
Industrial	-0.6%	9.0%	4.6%	7.0%	6.1%
Automotive	-5.7%	12.0%	4.6%	4.7%	6.4%
Transportation (non-auto)	1.4%	9.4%	7.4%	6.1%	5.4%
Military / Aerospace	4.4%	2.1%	3.4%	6.9%	5.9%
Telecom / Datacom	5.1%	10.2%	5.8%	7.9%	8.0%
Consumer	3.6%	6.4%	3.0%	4.6%	5.1%
Others	2.3%	15.5%	-6.9%	5.3%	4.3%
Market total	1.0%	9.2%	3.0%	6.1%	6.5%

Global connector sales by major suppliers

(USD bn)	CY13		(USD bn)	CY23	
	Sales	Share		Sales	Share
1 TE Connectivity	8.7	17.8%	1 TE Connectivity	12.7	15.6%
2 Amphenol	4.3	8.8%	2 Amphenol	7.0	8.5%
3 Molex	3.6	7.4%	3 Aptiv(Delphi)	4.7	5.8%
4 Aptiv(Delphi)	3.0	6.0%	4 Molex	4.6	5.7%
5 Foxconn	2.7	5.5%	5 Foxconn	2.5	3.1%
6 Yazaki	2.4	4.9%	6 Luxshare Precision	2.3	2.9%
7 JST	1.5	3.1%	7 Yazaki	2.2	2.7%
8 JAE	1.3	2.7%	8 JAE	1.7	2.1%
9 Sumitomo Wiring Systems	1.0	2.1%	9 Rosenberger	1.4	1.7%
10 Hirose Electric	1.0	2.0%	10 Hirose Electric	1.1	1.4%
Top 10	29.5	60.4%	Top 10	40.3	49.3%
Total	48.9	100.0%	Total	81.9	100.0%

Note: e = TTI estimates

Source: TTI, Company data, Morgan Stanley Research

Connector Coverage Companies: Medium-term Results and Forecasts

■ Hirose Electric (6806)

(JPY bn, %)	F3/20	F3/21	F3/22	F3/23	F3/24	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Sales	121.8	133.5	163.7	183.2	165.5	189.4	211.3	224.2	238.6	254.2
General Industrial	37.3	40.4	56.4	69.1	50.5	46.9	62.0	68.4	73.9	79.8
Smartphone	30.0	31.7	33.9	34.0	35.5	46.4	44.3	43.8	45.2	46.5
Consumer/Mobile equipment	19.3	26.3	31.3	30.0	25.3	34.7	34.7	33.3	34.0	34.7
Automotive/Mobility related	23.1	23.8	29.2	36.0	42.9	49.2	54.3	60.1	66.1	72.7
Telecommunications infrastructure equipment	5.4	5.2	6.2	6.0	4.5	5.4	6.7	9.3	10.3	11.3
OA equipment and Others	6.7	6.1	6.7	8.1	6.8	6.8	9.3	9.2	9.2	9.2
OP	20.4	27.9	40.8	46.8	34.0	42.7	43.0	46.6	49.6	53.5
OPM	16.7%	20.9%	24.9%	25.5%	20.6%	22.5%	20.4%	20.8%	20.8%	21.0%
ROE	5.0%	6.3%	9.4%	10.0%	7.4%	9.0%	8.9%	9.5%	10.0%	10.5%

■ JAE (6807)

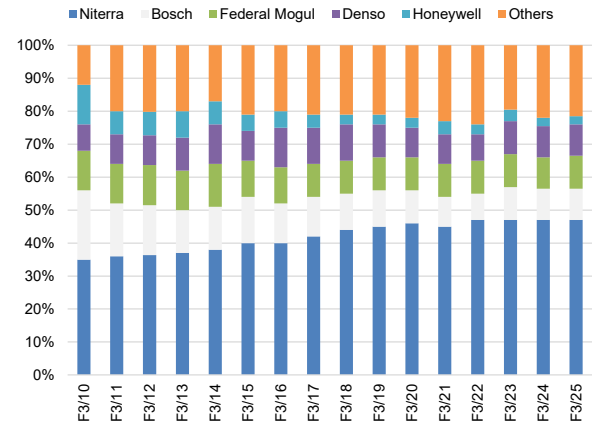
(JPY bn, %)	F3/20	F3/21	F3/22	F3/23	F3/24	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Sales	208.1	209.7	225.1	235.9	225.8	221.6	227.9	231.4	241.4	252.0
Connector	183.6	185.7	198.3	204.2	194.8	192.8	199.2	203.0	211.1	219.8
Handhelds	100.2	99.7	88.9	86.8	75.5	66.9	65.2	65.2	66.6	68.1
Automotive	63.7	64.0	76.8	85.6	100.2	107.1	111.0	113.8	119.4	125.4
Industrial & Infrastructure	14.9	16.9	25.7	26.3	14.8	14.5	17.6	18.7	19.9	21.0
Others	4.8	5.1	6.9	5.5	4.4	4.3	5.4	5.3	5.3	5.3
Aerospace	14.9	13.3	15.3	19.1	20.1	19.3	20.5	20.5	22.2	24.1
Automotive	0.8	1.3	1.1	1.1	1.0	0.2	0.1	0.1	0.1	0.1
Industrial & Infrastructure	9.6	8.0	9.1	13.6	13.7	10.7	9.3	9.5	10.1	10.7
Aviation & Space (Defence)	4.5	4.0	5.0	4.4	5.4	8.4	11.1	10.9	12.0	13.2
UIS	9.0	10.1	10.8	11.7	10.1	9.0	7.7	7.4	7.6	7.7
Automotive	5.9	6.5	6.3	5.4	5.1	5.5	4.6	4.5	4.5	4.5
Industrial & Infrastructure	3.1	3.6	4.5	6.3	5.0	3.5	3.0	2.9	3.0	3.1
Others	0.7	0.6	0.8	0.9	0.7	0.5	0.5	0.5	0.5	0.5
OP	14.2	8.7	18.0	17.6	14.4	15.6	8.9	10.3	11.5	12.4
Connector	17.4	12.2	20.4	17.6	15.4	17.7	11.9	14.2	15.2	16.0
Aerospace	1.4	0.3	1.6	4.1	3.6	2.6	2.2	1.4	1.6	1.7
UIS	-0.4	0.2	0.4	0.5	0.4	0.3	0.1	0.0	0.1	0.1
Others	0.2	0.3	0.3	0.3	0.1	0.1	0.1	0.1	0.1	0.1
Adjustment	-4.4	-4.2	-4.6	-4.8	-5.1	-5.0	-5.4	-5.5	-5.5	-5.5
OPM	6.8%	4.2%	8.0%	7.4%	6.4%	7.0%	3.9%	4.5%	4.8%	4.9%
ROE	8.3%	4.1%	9.6%	8.9%	8.2%	8.9%	5.1%	5.2%	5.7%	6.0%

Note: e = Morgan Stanley Research estimates

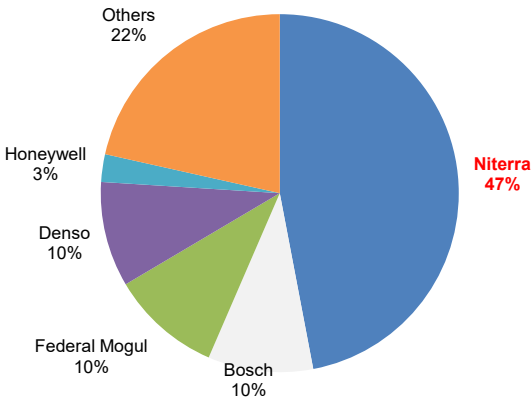
Source: Company data, Morgan Stanley Research

Global Market Share Trend and Split for Spark Plugs / Exhaust Gas Oxygen Sensors

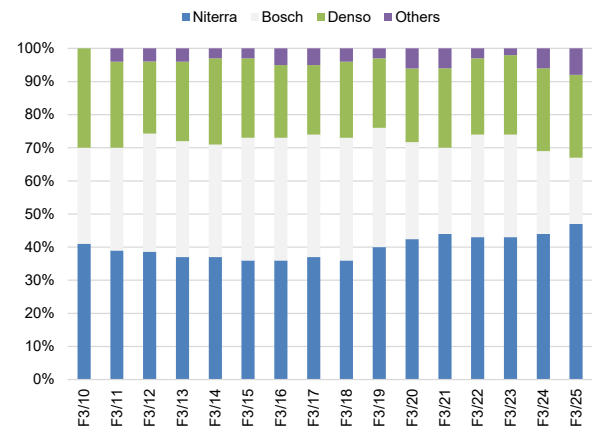
Spark plug global market share trend



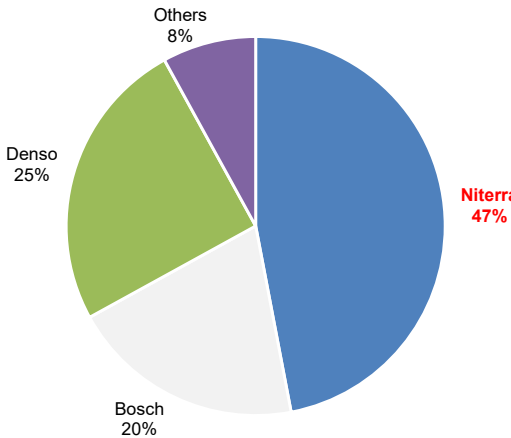
Spark plug global market share split in F3/25



Exhaust gas oxygen sensor global market share



Exhaust gas oxygen sensor global market share split in F3/25



Source: Company data, Morgan Stanley Research

Niterra: Medium-term Results and Forecasts

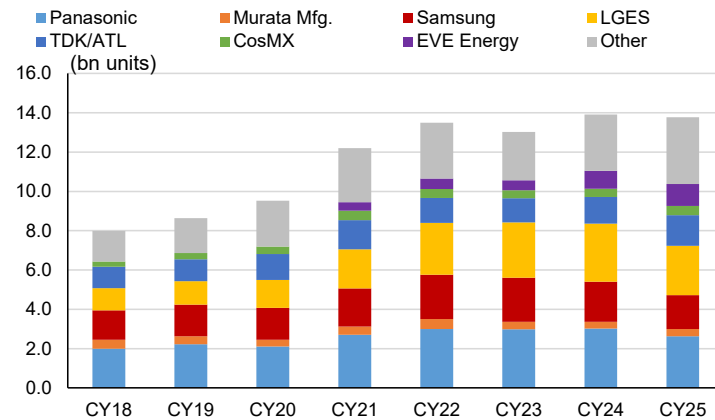
(bn JPY, %)	F3/20	F3/21	F3/22	F3/23	F3/24	F3/25	F3/26	F3/27 MSe	F3/28 MSe	F3/29 MSe	F3/30 MSe	CAGR F3/22-F3/28MSe
Sales	562.6	427.5	491.7	562.6	614.5	653.0	731.2	754.6	788.6	848.9	914.9	8.2%
Spark & Glow Plugs	203.5	205.3	250.0	285.1	335.3	362.6	390.7	414.4	438.0	479.6	525.4	9.8%
Automotive Sensors	140.9	133.4	137.8	158.9	170.0	180.2	196.8	197.0	200.9	206.9	213.1	6.5%
Component Solutions	36.3	80.0	95.5	110.8	96.5	103.1	130.7	135.3	143.7	156.4	170.4	7.1%
New Business	0.0	4.5	4.6	4.5	5.1	0.0	0.0	0.0	0.0	0.0	0.0	-
Others/Elimination	4.1	4.4	3.9	3.3	7.4	7.1	13.0	8.0	6.0	6.0	6.0	7.5%
OP	48.4	47.4	75.5	89.2	107.6	129.7	138.2	156.6	171.7	196.6	222.0	14.7%
Spark & Glow Plugs	36.2	40.4	55.0	67.4	95.8	109.8	109.2	122.4	132.7	148.1	165.0	15.8%
Automotive Sensors	16.6	16.2	19.3	25.9	25.4	24.4	26.1	27.1	28.5	30.6	32.8	6.7%
Component Solutions	-1.1	-0.2	14.7	11.0	-0.6	-6.3	-4.6	6.6	10.1	17.5	23.8	
New Business	0.0	-9.4	-13.6	-17.1	-13.2	0.0	0.0	0.0	0.0	0.0	0.0	
Others/Elimination	-0.6	0.4	0.1	2.0	0.2	1.8	7.4	0.4	0.4	0.4	0.4	
OPM	8.6%	11.1%	15.4%	15.9%	17.5%	19.9%	18.9%	20.7%	21.8%	23.2%	24.3%	
ROE	8.5%	9.1%	12.5%	12.3%	13.8%	14.1%	15.7%	15.2%	15.6%	16.4%	16.8%	

Note: MSe = Morgan Stanley Research estimates

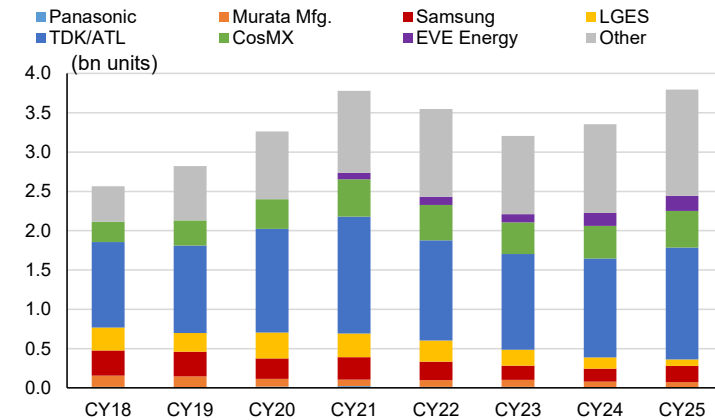
Source: Company data, Morgan Stanley Research

Small Size LiB Market Overview

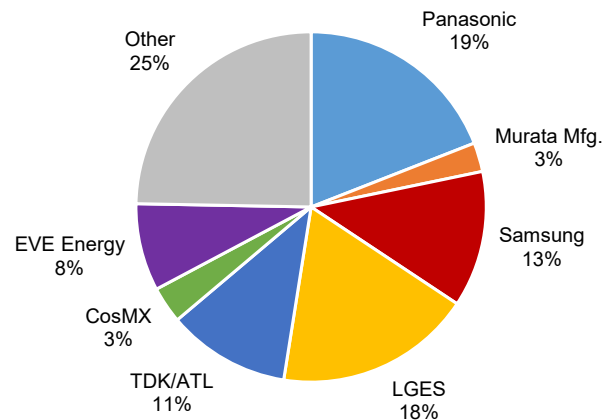
■ Small size LiB shipment volume by supplier



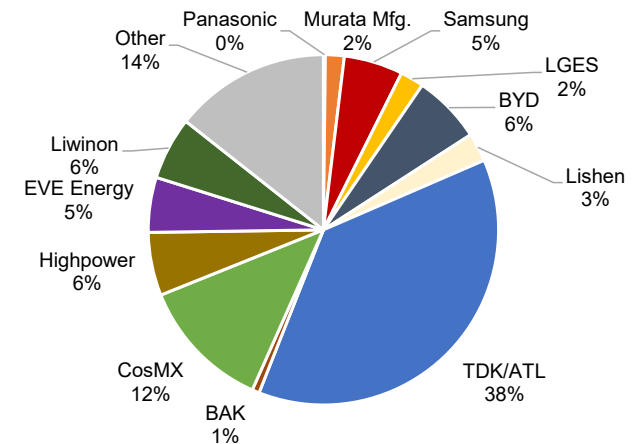
■ Laminated small size LiB shipment volume by supplier



■ Small size LiB market share by supplier (CY25)



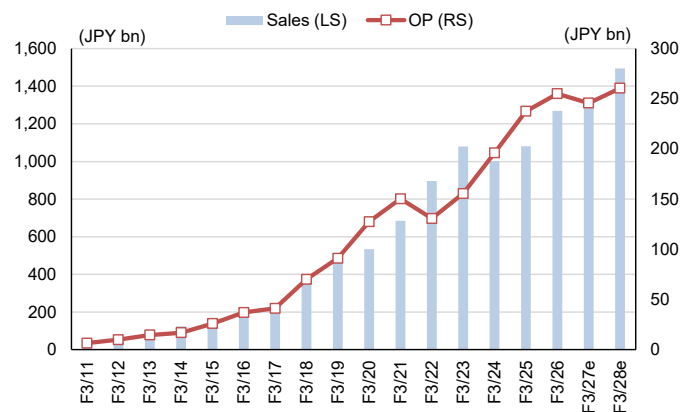
■ Laminated small size LiB market share by supplier (CY25)



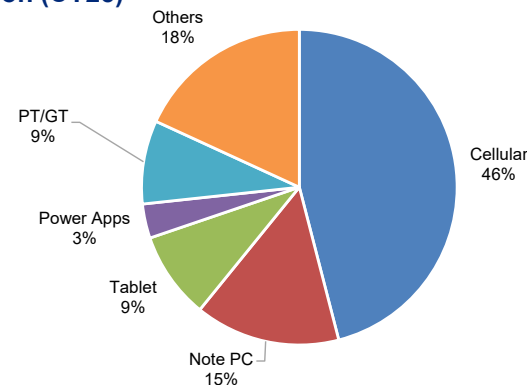
Source: TSR, Morgan Stanley Research

TDK: LiB Earnings Expansion, Major Acquisition and Transfer of Business

TDK: Sales and OP for rechargeable battery business



TDK/ATL: Small size battery shipment volume composition by application (CY25)



TDK: Major acquisition and transfer of business

	Acquisition and Transfer of Business	Business	Acquisition/Sales value (JPY mn)
Aug-1986	Acquisition of SAE Magnetics	HDD Heads	unknown
May-1989	Acquisition of Silicon Systems	Mixed-signal Semiconductors	28,000
Jun-1996	Transfer of Silicon Systems to Texas Instruments	Mixed-signal Semiconductors	62,700
Mar-2000	Acquisition of Headway Technologies	HDD Heads	26,410
May-2005	Acquisition of Amperex Technology(ATL)	Lithium Polymer Batteries	8,700
Oct-2005	Acquisition of Lambda Power Group	Power	25,500
Aug-2007	Transfer of Recording Media business to Imation	Recording Media	31,400
Sep-2007	Acquisition of HDD head business from Alps	HDD Heads	34,400
Nov-2007	Acquisition of Magnecomp International	HDD Head Suspensions	20,900
Oct-2008	Acquisition of Epcos	Passive Components	170,600
Mar-2016	Acquisition of Micronas	MEMS Magnetic Sensors	22,500
Oct-2016	Acquisition of Hutchinson Technology	HDD Head Suspensions	14,262
Dec-2016	Acquisition of Tronics	MEMS Inertial Sensors	4,107
Feb-2017	Transfer of Epcos High Frequency Components business to Qualcomm	High Frequency Components	339,000
Mar-2017	Acquisition of ICsense	Gyroscope Sensors	2,396
May-2017	Acquisition of InvenSense	MEMS Inertial Sensors	142,758
Feb-2018	Acquisition of Chirp Microsystems	Ultrasonic 3D Sensors	3,015
May-2018	Acquisition of Faraday Semi LLC	Power Management IC	unknown
Apr-2025	Transfer of Micro Actuator Solution Business to QT Investment	Micro Actuator	unknown
Jun-2025	Acquisition of SoftEye	Smart glasses	unknown
Jun-2025	Acquisition of RF power business from QEI Corporation	RF power business	unknown

Note: e = Morgan Stanley Research estimates

Source: TSR, Company data, Morgan Stanley Research

TDK: Actions for Business Portfolio Improvement

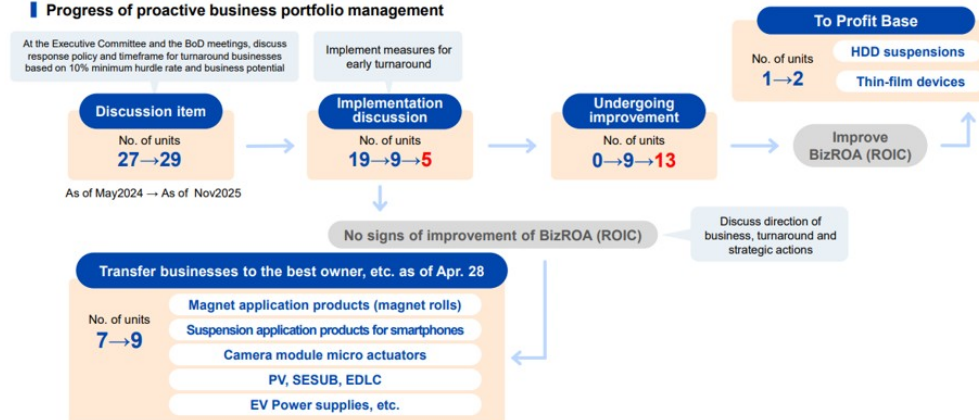
Chart for growth and capital efficiency by segment



Net sales, BizROA by segment

	(¥bn)	FY3/25	FY3/26
Net sales		2,204.8	2,504.8
Passive Components		559.6	593.2
Sensor Application Products		189.5	224.7
Magnetic Application Products		223.6	262.9
Energy Application Products		1,176.5	1,370.3
OP margin		10.2%	10.9%
BizROA (ROIC)		6.7%	7.5%
Passive Components		3.3%	4.8%
Sensor Application Products		0.2%	4.7%
Magnetic Application Products		1.0%	8.4%
Energy Application Products		27.3%	25.7%

Progress of proactive business portfolio management



Source: Company data

TDK: Medium-term Results and Forecasts

(bn JPY, %)	F3/18	F3/19	F3/20	F3/21	F3/22	F3/23	F3/24	F3/25	F3/26	F3/27 MSe	F3/28 MSe	F3/29 MSe	CAGR F3/24-F3/27e MSe
Sales	1,271.7	1,381.8	1,363.0	1,479.0	1,902.1	2,180.8	2,103.9	2,204.8	2,504.8	2,648.5	2,869.4	3,116.3	8.0%
Passive Components	417.6	433.4	395.5	407.1	507.8	578.8	565.6	559.6	593.2	625.3	650.7	677.6	3.4%
Sensor Application Products	77.4	76.5	77.9	81.3	130.8	169.5	180.5	189.5	224.6	236.7	252.7	269.9	9.4%
Magnetic Application Products	277.5	272.8	219.7	199.3	248.4	200.6	184.2	223.6	262.9	301.9	311.5	321.6	17.9%
Energy Application Products	442.8	537.5	597.7	740.2	965.3	1,173.4	1,121.7	1,176.5	1,370.3	1,432.7	1,604.6	1,797.2	8.5%
Other	56.4	61.6	72.3	51.1	49.7	58.6	51.8	55.6	53.8	52.0	50.0	50.0	0.1%
OP	85.6	107.8	97.9	111.5	166.8	168.8	172.9	224.2	272.4	309.8	348.1	383.6	21.5%
Passive Components	46.0	58.4	39.1	40.2	76.8	95.5	53.9	34.1	41.8	52.9	63.9	72.1	-0.6%
Sensor Application Products	-19.4	-22.1	-25.0	-24.9	-0.3	10.7	6.0	5.0	20.7	24.3	30.4	33.6	
Magnetic Application Products	16.1	17.0	0.4	-2.4	4.6	-56.4	-35.6	3.4	27.0	42.9	49.2	56.0	
Energy Application Products	72.9	91.0	124.1	147.4	123.2	147.4	195.7	234.4	246.7	243.7	258.5	275.9	7.6%
Other	0.6	-6.7	-8.6	-16.1	-1.4	-0.4	-1.8	-4.4	-10.2	-4.0	-4.0	-4.0	
Intersegment Elimination	-30.5	-29.8	-32.2	-32.7	-36.1	-28.0	-45.3	-48.3	-53.6	-50.0	-50.0	-50.0	
OPM	6.7%	7.8%	7.2%	7.5%	8.8%	7.7%	8.2%	10.2%	10.9%	11.7%	12.1%	12.3%	
ROE	7.8%	9.7%	6.7%	8.6%	11.4%	8.3%	7.9%	9.5%	9.7%	11.5%	11.9%	12.0%	

(JPY bn)	F3/18	F3/19	F3/20	F3/21	F3/22	F3/23	F3/24	F3/25	F3/26	F3/27 MSe	F3/28 MSe	F3/29 MSe	CAGR F3/18-F3/25	CAGR F3/25-F3/28 MSe
Total Sales	1,271.7	1,381.8	1,363.0	1,479.0	1,902.1	2,180.8	2,103.9	2,204.8	2,504.8	2,648.5	2,869.4	3,116.3	8.2%	9.2%
Group A Sales	633.2	727.5	752.1	889.5	1,165.4	1,309.7	1,237.6	1,375.7	1,613.6	1,718.6	1,904.3	2,111.5	11.7%	11.4%
HDD Head	175.5	181.4	136.4	116.5	148.5	104.1	100.1	134.5	142.5	168.9	171.4	174.0		
HDD Suspension	60.5	56.0	54.3	57.5	66.3	49.7	43.7	56.2	81.0	100.6	107.6	115.2		
Magnetic Sensor	26.2	28.1	27.5	30.5	54.6	76.6	91.9	104.3	120.3	124.1	131.5	139.4		
LiB	371.0	461.9	533.8	685.0	896.0	1,079.3	1,001.9	1,080.7	1,269.9	1,325.0	1,493.7	1,682.9		
Group B Sales	638.6	654.3	611.0	589.5	736.7	871.1	866.3	829.1	891.2	930.0	965.1	1,004.7	3.8%	5.2%
Capacitors	157.0	173.3	153.9	158.2	198.1	242.5	245.0	234.3	257.5	276.1	292.6	310.2		
Inductive Devices	157.5	150.0	137.6	140.0	180.2	198.5	192.0	204.3	216.2	226.2	235.3	244.7		
Other Passive Components	103.1	110.1	104.0	109.0	129.4	137.8	128.7	121.1	119.5	123.0	122.8	122.8		
Temperature/Pressure Sensor	24.9	26.4	26.8	27.2	33.0	36.9	40.4	41.6	42.3	45.1	46.9	48.7		
MEMS Sensor	25.5	22.0	23.7	23.6	43.2	56.0	48.3	43.5	62.0	67.5	74.3	81.7		
Magnet	41.5	35.4	28.9	25.2	33.6	46.8	40.7	32.9	39.5	32.4	32.4	32.4		
Power Supply	71.8	75.6	63.9	55.2	69.3	94.0	119.7	95.8	100.4	107.7	110.9	114.2		
Others	57.2	61.6	72.3	51.1	49.8	58.6	51.6	55.6	53.8	52.0	50.0	50.0		
Total OP	85.6	107.8	97.9	111.5	166.8	168.8	172.9	224.2	272.4	309.8	348.1	383.6		
Group A OP	86.2	118.1	145.1	154.3	153.4	126.7	193.7	273.0	309.9	318.3	342.4	369.7		
Group B OP	30.0	20.0	-15.1	-10.0	53.3	70.1	24.5	-0.6	16.1	41.5	55.7	63.9		
Intersegment elimination	-30.5	-30.3	-32.2	-32.7	-39.8	-28.0	-45.3	-48.3	-53.6	-50.0	-50.0	-50.0		
OPM	6.7%	7.8%	7.2%	7.5%	8.8%	7.7%	8.2%	10.2%	10.9%	11.7%	12.1%	12.3%		
ROE	7.8%	9.7%	6.7%	8.6%	11.4%	8.3%	7.9%	9.5%	9.7%	11.5%	11.9%	12.0%		

Note: MSe = Morgan Stanley Research estimates

Source: Company data, Morgan Stanley Research

iPhone Sell-in Forecasts / iPhone Camera Actuators by Model

mn units	1Q22	2Q22	3Q22	4Q22	2022	1Q23	2Q23	3Q23	4Q23	2023	1Q24	2Q24	3Q24	4Q24	2024	1Q25	2Q25	3Q25	4Q25e	2025	1Q26e	2Q26e
iPhone Sell-in																						
iPhone SE3 / SE 5G / 16e / 17e	3.0	13.0	6.0	2.0	24.0	1.0	2.5	5.0	3.0	11.5	1.0	2.0	5.0	3.0	11.0	4.0	9.0	4.0	1.0	18.0	2.0	10.0
iPhone XR	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
iPhone 11	1.0	1.0	0.5	0.5	3.0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
iPhone 12 mini (5.4")	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
iPhone 12 (6.1")	2.0	1.0	1.0	0.5	4.5	0.5	0.5	-	-	1.0	-	-	-	-	-	-	-	-	-	-	-	-
iPhone 12 Pro (6.1")	1.0	0.5	0.5	0.5	2.5	0.5	-	-	-	0.5	-	-	-	-	-	-	-	-	-	-	-	-
iPhone 12 Pro Max (6.7")	1.0	0.5	0.5	0.5	2.5	0.5	-	-	-	0.5	-	-	-	-	-	-	-	-	-	-	-	-
iPhone 13 mini (5.4")	5.0	2.0	2.0	1.0	10.0	1.0	0.5	-	-	1.5	-	-	-	-	-	-	-	-	-	-	-	-
iPhone 13 (6.1")	16.0	11.5	8.0	3.0	38.5	1.0	0.5	-	-	1.5	-	-	-	-	-	-	-	-	-	-	-	-
iPhone 13 Pro (6.1")	13.0	8.0	6.0	3.5	30.5	1.0	2.0	1.0	-	4.0	0.5	0.5	-	-	1.0	-	-	-	-	-	-	-
iPhone 13 Pro Max (6.7")	12.0	7.5	5.0	4.0	28.5	2.0	2.0	1.0	-	5.0	0.5	0.5	-	-	1.0	-	-	-	-	-	-	-
iPhone 14 (6.1")	-	-	6.5	13.0	19.5	7.0	4.0	5.0	2.0	18.0	1.0	0.5	-	-	1.5	-	-	-	-	-	-	-
iPhone 14 Plus (6.7")	-	-	2.0	13.5	15.5	5.0	3.0	1.0	1.0	10.0	1.0	0.5	-	-	1.5	-	-	-	-	-	-	-
iPhone 14 Pro (6.1")	-	-	6.0	16.0	22.0	15.0	12.0	7.0	3.0	37.0	2.0	2.0	1.0	1.0	6.0	1.0	1.0	1.0	-	3.0	-	-
iPhone 14 Pro Max (6.7")	-	-	6.0	19.0	25.0	19.5	14.0	9.0	4.0	46.5	2.0	2.0	1.0	1.0	6.0	1.0	1.0	1.0	-	3.0	-	-
iPhone 15 (6.1")	-	-	-	-	-	-	-	6.0	12.0	18.0	6.0	5.0	4.0	1.0	16.0	1.0	0.5	0.5	0.5	2.5	-	-
iPhone 15 Plus (6.7")	-	-	-	-	-	-	-	5.0	10.0	15.0	4.0	3.0	1.0	1.0	9.0	1.0	0.5	0.5	0.5	2.5	-	-
iPhone 15 Pro (6.1")	-	-	-	-	-	-	-	8.0	16.0	24.0	12.0	10.0	8.5	2.0	32.5	2.0	2.0	2.0	1.0	7.0	1.0	1.0
iPhone 15 Pro Max (6.7")	-	-	-	-	-	-	-	2.0	24.0	26.0	18.0	13.0	10.5	3.0	44.5	3.0	3.0	2.0	1.0	9.0	1.0	1.0
iPhone 16 (6.1")	-	-	-	-	-	-	-	-	-	-	-	-	6.0	12.0	18.0	5.0	4.0	4.0	1.0	14.0	2.0	2.0
iPhone 16 Plus (6.7")	-	-	-	-	-	-	-	-	-	-	-	-	5.0	6.0	11.0	4.0	3.0	1.0	-	8.0	-	-
iPhone 16 Pro (6.3")	-	-	-	-	-	-	-	-	-	-	-	-	8.0	17.0	25.0	11.5	9.5	4.0	2.0	27.0	2.0	2.0
iPhone 16 Pro Max (6.9")	-	-	-	-	-	-	-	-	-	-	-	-	4.0	26.0	30.0	16.5	13.0	10.0	4.0	43.5	5.0	3.0
iPhone 17 (6.1")	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	6.0	16.0	22.0	10.0	8.0
iPhone Air (6.7")	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	4.0	4.0	8.0	-	-
iPhone 17 Pro (6.3")	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	8.0	22.0	30.0	18.0	13.0
iPhone 17 Pro Max (6.9")	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	7.0	23.0	30.0	15.0	12.0
Total	54.0	45.0	50.0	77.0	226.0	54.0	41.0	50.0	75.0	220.0	48.0	39.0	54.0	73.0	214.0	50.0	46.5	55.0	76.0	227.5	56.0	52.0
YoY	7%	1%	0%	-9%	-2%	0%	-9%	0%	-3%	-3%	-11%	-5%	8%	-3%	-3%	-4%	19%	2%	4%	6%	12%	12%
QoQ	-36%	-17%	11%	54%		-30%	-24%	22%	50%		-36%	-19%	38%	35%		-32%	-7%	18%	38%		-26%	-7%

	Type	Rear cam									Front cam	
		FF	OIS	VCM+OIS	OIS+FF	OIS x 2	SS+FF	OIS x 2 + FF	SS + OIS + FF	SS + periscope + FF	F Camera VCM	
	number of cameras	Single	Single	Dual	Dual	Dual	Dual	Triple	Triple	Triple		Single
2022	iPhone 14 (6.1")						○					○
	iPhone 14 Plus (6.7")						○					○
	iPhone 14 Pro (6.1")								○			○
	iPhone 14 Pro Max(6.7")								○			○
2023	iPhone 15 (6.1")						○					○
	iPhone 15 Plus (6.7")						○					○
	iPhone 15 Pro (6.1")								○			○
	iPhone 15 Pro Max(6.7")									○		○
2024/25	iPhone 16 (6.1")						○					○
	iPhone 16 Plus (6.7")						○					○
	iPhone 16 Pro (6.3")									○		○
	iPhone 16 Pro Max(6.9")									○		○
	iPhone 16e (6.1")			○								○
2025/26	iPhone 17 (6.3")						○					○
	iPhone Air (6.5")		○									○
	iPhone 17 Pro (6.3")									○		○
	iPhone 17 Pro Max (6.9")									○		○
	iPhone 17e (6.1")			○								○

Note: e = Morgan Stanley Research estimates

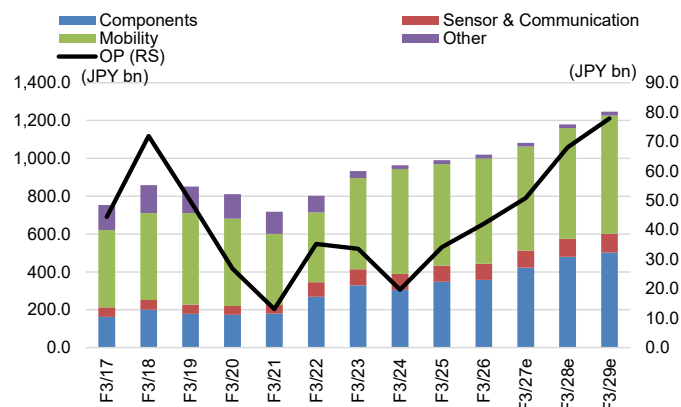
Source: Morgan Stanley Research

Alps Alpine: We Expect New Smartphone Camera Actuators to Contribute to Earnings

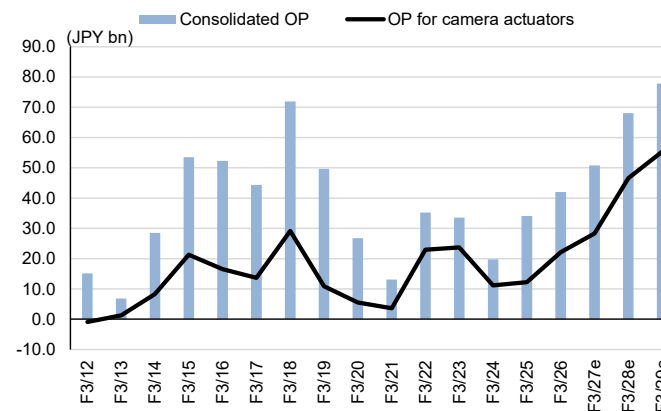
■ Alps Alpine: Medium-term results and forecasts

(bn JPY, %)	F3/17	F3/18	F3/19	F3/20	F3/21	F3/22	F3/23	F3/24	F3/25	F3/26	F3/27 MSe	F3/28 MSe	F3/29 MSe	F3/27 Ce	F3/28 Ce
Sales	753.3	858.3	851.3	810.6	718.0	802.9	933.1	964.1	990.4	1,019.5	1,082.0	1,179.9	1,246.6	1,045.0	1,075.0
Components	162.8	200.3	178.9	173.6	180.1	268.5	329.0	305.3	348.0	358.4	422.6	479.9	502.1	398.0	350.0
Sensor & Communication	50.1	52.1	48.1	46.7	44.3	76.7	85.5	84.1	84.2	85.3	90.5	95.0	99.8	87.0	105.0
Mobility	408.3	458.6	484.4	461.1	376.3	369.4	481.4	554.4	537.2	555.1	548.9	584.9	624.6	540.0	600.0
Other	132.1	147.4	139.9	129.2	117.3	88.2	37.2	20.3	21.0	20.8	20.0	20.0	20.0	20.0	20.0
OP	44.4	71.9	49.6	26.8	13.1	35.2	33.6	19.7	34.1	42.0	50.8	68.1	77.8	48.5	71.0
Components	20.9	41.5	19.5	14.8	10.3	32.6	38.3	20.5	30.4	30.2	41.8	56.1	64.8	28.0	35.0
Sensor & Communication	9.6	7.9	6.0	3.0	1.7	3.0	-1.6	-1.5	-3.4	-3.5	-2.0	0.0	1.0	5.0	5.0
Mobility	7.9	15.9	18.0	3.7	-4.6	-8.0	-6.6	-1.1	5.6	14.2	10.0	11.0	11.0	19.0	30.0
Other/Elimination	6.0	6.6	6.2	5.3	5.8	7.6	3.5	1.9	1.5	1.3	1.0	1.0	1.0	1.0	1.0
OPM	5.9%	8.4%	5.8%	3.3%	1.8%	4.4%	3.6%	2.0%	3.4%	4.1%	4.7%	5.8%	6.2%	4.6%	6.6%
ROE	14.4%	17.0%	6.6%	-1.2%	-1.1%	6.2%	2.9%	-7.6%	9.4%	6.2%	8.7%	11.0%	11.6%	-	10.0%

■ Alps Alpine: Sales by segment and consolidated OP



■ Alps Alpine: Consolidated OP and OP for camera actuators

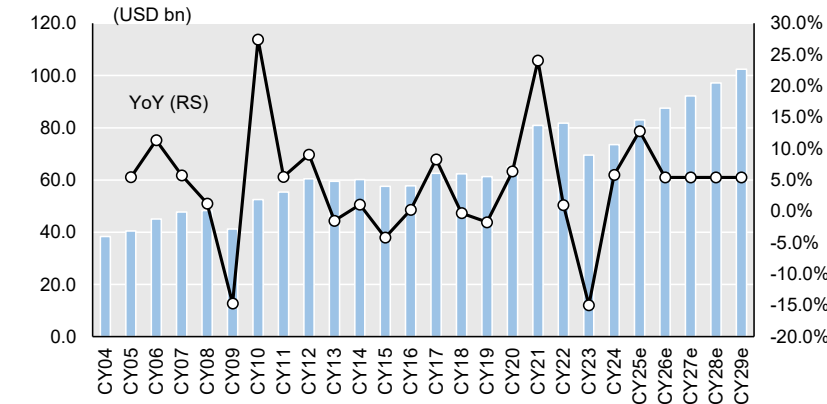


Note: MSe/e = Morgan Stanley Research estimates, Ce = Company estimates

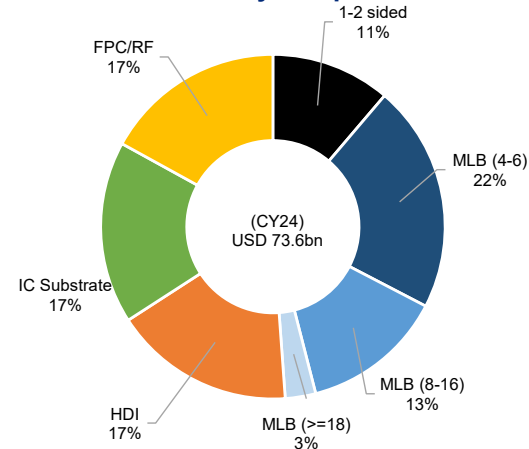
Source: Company data, Morgan Stanley Research

PWB Market

Global PWB market value

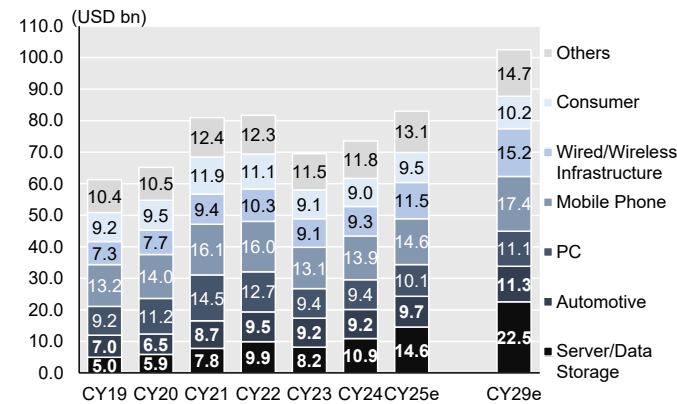


Global PWB market value by composition



Note: e = Morgan Stanley Research estimates
Source: Prismark, Morgan Stanley Research

Global PWB market value by applications



Major composition by application

Usage/Product	1-2 sided	MLB	HDI	Flexible/ Rigid-flex	IC Substrate
Server/Computer	○	○ (4-16)	○	○	○
Automotive	○	○ (4-8)	○	○	○
Mobile	○	○ (4-12)	○	○	○
Consumer electronics	○	○ (4-6)	○	○	○
5G	○	○ (20-26)	○	○	○
Industrial	○	○ (4-16)	○	○	○
Medical	○	○ (-20)	○	○	○
Aerospace/Military	○	○ (4-6)	○	○	○
IC Packaging	○	○	○	○	○

Meiko Electronics: Medium-term Results and Forecasts

(JPY bn)	F3/21	F3/22	F3/23	F3/24	F3/25	F3/26	F3/27e	(F3/27Ce)	F3/28e	F3/29e
Sales	119.3	151.3	167.3	179.5	206.8	240.6	305.1	320.0	368.7	429.5
Automotive	52.3	73.2	84.5	94.7	90.4	97.5	99.8	97.5	102.9	106.2
Smartphone/Tablet	28.8	33.2	26.6	23.6	25.5	29.4	30.6	37.0	54.3	70.6
Satellite communications	-	-	-	7.6	16.6	28.2	57.9	55.0	72.4	82.9
Module/AI server*	5.0	6.8	9.9	8.5	13.4	16.5	22.0	30.0	25.2	28.9
Amusement/Smart appliance/Others	-	-	-	14.1	23.7	24.4	30.2	29.0	44.6	66.7
Electronic equipment	11.6	12.6	20.3	30.3	36.2	42.6	60.1	66.0	63.7	67.5
Semi Package	0.0	0.0	0.4	0.7	1.0	2.0	3.0	3.0	4.0	5.0
OP	6.7	13.3	9.6	11.7	19.1	24.6	33.6	38.0	45.5	57.4
OPM	5.6%	8.8%	5.7%	6.5%	9.2%	10.2%	11.0%	11.9%	12.3%	13.4%
ROE	12.8%	23.2%	13.0%	12.9%	14.5%	16.2%	18.2%	-	21.0%	22.1%

Note: Ce=Company guidance, e=Morgan Stanley Research estimates; *SSD/IoT segment until F3/23
Source: Company data, Morgan Stanley Research

HDD Market by Company (Quarterly, Fiscal Year)

(mn units)	CY23				CY24				CY25				CY26				CY14	CY15	CY16	CY17	CY18	CY19	CY20	CY21	CY22	CY23	CY24	CY25	CY26e	CY27e	CY28e
	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q															
HDD Shipment (by Size)																															
Total	33.5	31.3	28.6	28.9	29.7	30.3	32.1	31.7	28.7	30.4	31.6	33.0	33.0				564.1	468.8	423.9	403.1	375.4	317.7	260.0	259.3	172.2	122.3	123.9	123.8	126.8	125.7	125.3
Enterprise	1.4	1.6	1.4	1.4	1.5	1.0	1.0	0.9	0.8	0.8	0.7	0.8	0.8				32.4	32.1	24.8	22.6	21.5	17.4	12.9	13.4	8.4	5.8	4.4	3.2	2.3	1.5	0.8
NL	11.4	9.2	9.0	9.9	12.3	13.8	15.9	16.5	14.6	16.5	17.3	19.20	18.7				36.9	37.2	41.0	42.4	51.3	52.7	59.1	73.5	61.5	39.4	58.6	67.6	75.5	79.6	81.3
3.5"ATA	11.2	12.1	11.2	10.7	9.3	9.4	9.0	7.9	7.6	7.9	8.0	7.6	8.3				202.6	161.0	142.9	135.4	120.5	103.9	87.0	89.9	55.0	45.2	35.6	30.9	28.8	27.6	26.8
2.5"	9.5	8.4	7.1	6.9	6.6	6.1	6.2	6.4	5.7	5.3	5.6	5.5	5.3				291.7	238.6	215.1	202.7	182.0	143.6	101.0	82.5	46.3	32.0	25.3	22.0	20.3	17.1	16.4
1.8"	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0				0.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
YoY % Change	-37%	-30%	-26%	-20%	-11%	-3%	12%	10%	-3%	0%	-2%	4%	15%				2%	-17%	-10%	-5%	-7%	-15%	-18%	0%	-33.6%	-29%	1%	0%	2%	-1%	0%
Enterprise	-48%	-34%	-15%	-16%	6%	-36%	-30%	-33%	-45%	-21%	-23%	-16%	-7%				2%	-1%	-23%	-9%	-5%	-19%	-26%	3%	-37%	-32%	-23%	-29%	-28%	-36%	-47%
NL	-39%	-49%	-37%	-7%	8%	51%	78%	67%	19%	19%	9%	16%	28%				4%	1%	10%	3%	21%	3%	12%	24%	-16.3%	-36%	49%	15.5%	11.6%	5.4%	2.2%
3.5"ATA	-33%	-4%	-14%	-16%	-16%	-22%	-19%	-26%	-19%	-17%	-12%	-4%	9%				0%	-21%	-11%	-5%	-11%	-14%	-16%	3%	-39%	-18%	-21%	-13%	-7%	-4%	-3%
2.5"	-36%	-20%	-26%	-39%	-30%	-28%	-13%	-7%	-14%	-13%	-10%	-14%	-7%				4%	-18%	-10%	-6%	-10%	-21%	-30%	-18%	-44%	-31%	-21%	-13%	-8%	-16%	-4%
1.8"	-	-	-	-	-	-	-	-	-	-	-	-	-				-6%	-98%	-	-	-	-	-	-	-	-	-	-	-	-	-
Sequential % Change	-8%	-7%	-9%	1%	3%	2%	6%	-1%	-10%	6%	4%	5%	0%				-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Enterprise	-16%	13%	-16%	4%	7%	-32%	-7%	-1%	-13%	-1%	-10%	8%	-4%				-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
NL	8%	-20%	-2%	11%	24%	12%	15%	4%	-12%	13%	5%	11%	-3%				-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
3.5"ATA	-13%	9%	-7%	-4%	-13%	1%	-4%	-13%	-4%	4%	2%	-5%	10%				-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
2.5"	-16%	-11%	-16%	-3%	-4%	-8%	2%	3%	-11%	-8%	6%	-1%	-3%				-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
1.8"	-	-	-	-	-	-	-	-	-	-	-	-	-				-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Enterprise	4%	5%	5%	5%	5%	3%	3%	3%	3%	3%	2%	2%	2%				6%	7%	6%	6%	6%	5%	5%	5%	5%	4%	3%	2%	1%	1%	1%
NL	34%	29%	31%	34%	41%	46%	50%	52%	51%	54%	55%	58%	57%				7%	8%	10%	11%	14%	17%	23%	28%	36%	32%	47%	55%	60%	63%	65%
3.5"ATA	33%	39%	39%	37%	31%	31%	28%	25%	26%	26%	25%	23%	25%				36%	34%	34%	34%	32%	33%	33%	35%	32%	37%	29%	25%	23%	22%	21%
2.5"	28%	27%	25%	24%	22%	20%	19%	20%	20%	17%	18%	17%	16%				52%	51%	51%	50%	48%	45%	39%	32%	27%	26%	20%	18%	16%	14%	13%
1.8"	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%				0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%
HDD Shipment (by Company)																															
Total	33.5	31.3	28.6	28.9	29.7	30.3	32.2	31.7	28.7	30.4	31.6	33.0					564.1	468.8	423.9	403.1	375.4	317.7	260.0	259.3	172.2	122.3	123.9	123.8	126.8	125.7	125.3
Seagate	14.5	14.3	12.0	11.6	11.4	12.3	13.0	13.0	11.4	12.5	12.5	13.1					224.1	188.6	155.2	148.2	150.0	129.1	126.7	110.3	74.1	52.4	49.7	49.4	-	-	-
Western Digital	12.6	11.8	10.4	10.8	11.7	12.1	13.2	13.5	12.1	12.9	13.9	14.3					249.3	215.6	191.4	161.8	139.7	114.2	110.8	94.3	63.9	45.6	50.5	53.2	-	-	-
Toshiba	6.4	5.2	6.2	6.5	6.7	5.9	6.0	5.2	5.2	5.0	5.2	5.7					90.7	75.8	93.5	93.0	85.7	74.4	70.0	54.7	34.3	24.3	23.8	21.1	-	-	-
Market Share																															
Seagate	43%	46%	42%	40%	38%	41%	40%	41%	40%	41%	39%	39%					40%	40%	37%	37%	40%	41%	49%	43%	43%	43%	40%	40%	-	-	-
Western Digital	38%	38%	36%	37%	39%	40%	41%	43%	42%	42%	44%	43%					44%	46%	45%	40%	37%	36%	43%	36%	37%	37%	41%	43%	-	-	-
Toshiba	19%	17%	22%	22%	23%	20%	19%	17%	18%	16%	17%	17%					16%	16%	22%	23%	23%	23%	27%	21%	20%	20%	19%	17%	-	-	-
YoY % Change	-37%	-30%	-26%	-20%	-11%	-3%	12%	10%	-3%	0%	-2%	4%					2%	-17%	-10%	-5%	-7%	-15%	-18%	0%	-34%	-29%	1%	0%	2%	-1%	0%
Seagate	-37%	-28%	-24%	-25%	-21%	-14%	8%	12%	0%	2%	-4%	0%					1%	-16%	-18%	-4%	1%	-14%	-2%	-13%	-33%	-29%	-5%	0%	-	-	-
Western Digital	-36%	-28%	-29%	-16%	-7%	3%	27%	25%	3%	7%	5%	6%					1%	-14%	-11%	-15%	-14%	-18%	-3%	-15%	-32%	-29%	11%	5%	-	-	-
Toshiba	-37%	-37%	-22%	-19%	5%	14%	-4%	-19%	-22%	-16%	-12%	8%					8%	-16%	23%	-1%	-8%	-13%	-6%	-22%	-37%	-29%	-2%	-11%	-	-	-
Sequential % Change	-8%	-7%	-9%	1%	3%	2%	6%	-1%	-10%	6%	4%	5%					-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Seagate	-6%	-1%	-16%	-3%	-2%	8%	5%	0%	-12%	10%	0%	5%					-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Western Digital	-2%	-6%	-12%	4%	8%	3%	9%	2%	-10%	7%	8%	3%					-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Toshiba	-21%	-19%	20%	5%	3%	-11%	0%	-12%	-1%	-4%	5%	8%					-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
HDD Shipment Capacity (EB, by Company)																															
Total	236.0	189.8	196.1	214.3	264.6	289.4	340.3	362.3	346.6	391.8	432.5	457.0	477.2				534.9	535.8	622.6	720.7	855.3	902.5	1,019.1	1,344.8	1,163.5	836.2	1,256.6	1,627.9	2,004.5	2,463.5	3,027.8
Seagate	118.7	91.2	89.6	95.1	99.1	114.2	137.5	150.8	143.6	162.5	181.7	190.1	199.0				221.6	222.7	252.2	287.2	366.5	367.0	480.6	614.2	539.5	394.6	501.6	677.9	-	-	-
Western Digital	96.5	78.8	79.1	90.3	127.3	142.6	163.8	172.5	162.0	187.0	204.1	215.0	222.0				240.1	250.1	286.1	342.7	390.0	422.7	422.3	544.0	476.3	344.7	606.2	768.1	-	-	-
Toshiba	20.8	19.8	27.4	28.9	38.2	32.6	39.2	39.0	41.0	42.3	46.7	51.9	56.1				73.2	63.0	84.3	90.8	98.8	112.9	116.2	187.5	147.7	96.9	149.0	181.9	-	-	-
YoY % Change	-28%	-41%	-31%	-6%	12%	52%	74%	69%	31%	35%	27%	26%	38%				16%	0%	16%	16%	19%	6%	13%	32%	-13%	-28%	50%	30%	23%	23%	23%
Seagate	-23%	-41%	-24%	-15%	-17%	25%	53%	59%	45%	42%	32%	26%	39%				14%	0%	13%	14%	28%	0%	31%	28%	-12%	-27%	27%	35%	-	-	-
Western Digital	-27%	-39%	-39%	8%	32%	81%	107%	91%	27%	31%	25%	25%	37%				19%	4%	14%	20%	14%	8%	0%	29%	-12%	-28%	76%	27%	-	-	-
Toshiba	-51%	-45%	-26%	-10%	84%	65%	43%	35%	7%	30%	19%	33%	37%				14%	-14%	34%	8%	9%	14%	3%	61%	-21%	-34%	54%	22%	-	-	-
Sequential % Change	3%	-20%	3%	9%	23%	9%	18%	6%	-4%	13%	10%	6%	4%				-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Seagate	6%	-23%	-2%	6%	4%	15%	20%	10%	-5%	13%	12%	5%	5%				-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Western Digital	15%	-18%	0%	14%	41%	12%	15%	5%	-6%	15%	9%	5%	3%				-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Toshiba	-35%	-5%	38%	5%	32%	-15%	20%	-1%	5%	3%	10%	11%	8%				-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

HDD Head Market

(mn units)	CY23				CY24				CY25				CY26															
	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q	CY14	CY15	CY16	CY17	CY18	CY19	CY20	CY21	CY22	CY23	CY24	CY25
HDD Head Shipment																												
Total	262.2	202.9	217.1	237.7	262.5	297.1	327.3	332.4	330.0	348.7	377.6	397.8	401.9				1,931.6	1,756.1	1,702.8	1,714.4	1,715.4	1,496.2	1,517.9	1,747.9	1,349.0	919.9	1,219.3	1,454.1
TDK	39.6	26.4	35.0	40.3	43.3	47.4	49.3	43.4	47.9	45.2	49.7	51.0	52.8				471.6	376.0	375.0	361.9	330.2	299.6	254.4	338.2	230.8	141.3	183.4	193.8
Seagate	118.8	92.1	86.9	101.7	103.3	116.1	132.8	140.7	128.3	140.8	155.1	165.2	165.4				634.0	616.2	617.1	636.6	683.5	594.1	652.2	755.2	579.1	399.5	492.9	589.4
Western Digital	103.8	84.4	95.2	95.7	115.9	133.6	145.2	148.3	153.8	162.7	172.8	181.6	183.7				826.0	763.9	710.7	715.9	701.7	602.5	611.3	654.5	539.1	379.1	543.0	670.9
Market Share																												
TDK	15%	13%	16%	17%	16%	16%	15%	13%	15%	13%	13%	13%	13%				24%	21%	22%	21%	19%	20%	17%	19%	17%	15%	15%	13%
Seagate	45%	45%	40%	43%	39%	39%	41%	42%	39%	40%	41%	42%	41%				33%	35%	36%	37%	40%	40%	43%	43%	43%	43%	40%	41%
Western Digital	40%	42%	44%	40%	44%	45%	44%	45%	47%	47%	46%	46%	46%				43%	43%	42%	42%	41%	40%	40%	37%	40%	41%	45%	46%
YoY % Change																												
TDK	-38%	-47%	-33%	9%	0%	46%	51%	40%	26%	17%	15%	20%	22%				0%	-9%	-3%	1%	0%	-13%	1%	15%	-23%	-32%	33%	19%
Seagate	-45%	-53%	-41%	-5%	9%	80%	41%	8%	11%	-5%	1%	18%	10%				-3%	-20%	0%	-3%	-9%	-9%	-15%	33%	-32%	-39%	30%	6%
Western Digital	-40%	-47%	-32%	26%	-13%	26%	53%	38%	24%	21%	17%	17%	29%				1%	-3%	0%	3%	7%	-13%	10%	16%	-23%	-31%	23%	20%
Western Digital	-33%	-44%	-31%	2%	12%	58%	53%	55%	33%	22%	19%	22%	19%				2%	-8%	-7%	1%	-2%	-14%	1%	7%	-18%	-30%	43%	24%
HDD Head/HDD																												
Total	7.84	6.49	7.59	8.22	8.83	9.80	10.20	10.47	11.50	11.47	11.95	12.04	12.18				3.42	3.75	4.02	4.25	4.57	4.71	5.84	6.74	7.83	7.52	9.84	11.75
TDK's Head Shipment by Customer																												
Total	39.6	26.4	35.0	40.3	43.3	47.4	49.3	43.4	47.9	45.2	49.7	51.0	52.8				471.6	376.0	375.0	361.9	330.2	299.6	254.4	338.2	230.8	141.3	183.4	193.8
Seagate	4.3	3.7	1.0	0.3	0.1	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.2				17.4	14.0	25.7	24.0	41.3	51.7	43.1	45.5	28.1	9.3	0.3	0.0
Maxtor	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0				0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
HGST	2.9	2.0	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.7				13.1	16.0	11.9	21.5	22.6	1.4	0.0	4.8	7.6	5.1	0.0	0.0
Western Digital	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0				43.7	17.5	16.2	12.2	5.7	1.4	0.0	0.0	0.0	0.0	0.0	0.0
Samsung	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0				130.2	86.0	21.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Toshiba	32.4	20.7	33.8	40.0	43.2	47.3	49.2	43.4	47.9	45.2	49.7	51.0	51.9				267.2	242.5	299.8	304.2	260.6	245.1	211.3	287.9	195.1	126.9	183.1	193.8
Fujitsu	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0				0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Others	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0				0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
TDK's Shipment Share by Customer																												
Seagate	4%	4%	1%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%				3%	2%	4%	4%	6%	8%	6%	6%	5%	2%	0%	0%
Maxtor	-	-	-	-	-	-	-	-	-	-	-	-	-				-	-	-	-	-	-	-	-	-	-	-	-
HGST	3%	3%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%				4%	5%	3%	6%	6%	0%	0%	1%	2%	2%	0%	0%
Western Digital	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%				8%	4%	4%	3%	2%	1%	0%	0%	0%	0%	0%	0%
Samsung	-	-	-	-	-	-	-	-	-	-	-	-	-				-	-	-	-	-	-	-	-	-	-	-	-
Toshiba	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%				100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%
Fujitsu	-	-	-	-	-	-	-	-	-	-	-	-	-				-	-	-	-	-	-	-	-	-	-	-	-
Others	-	-	-	-	-	-	-	-	-	-	-	-	-				-	-	-	-	-	-	-	-	-	-	-	-
YoY % Change																												
Seagate	-45%	-53%	-41%	-5%	9%	80%	41%	8%	11%	-5%	1%	18%	10%				-3%	-20%	0%	-3%	-9%	-9%	-15%	33%	-32%	-39%	30%	6%
Sequential % Change	-7%	-33%	33%	15%	7%	9%	4%	-12%	10%	-6%	10%	3%	4%				-	-	-	-	-	-	-	-	-	-	-	-

Note: e = Morgan Stanley Research estimates

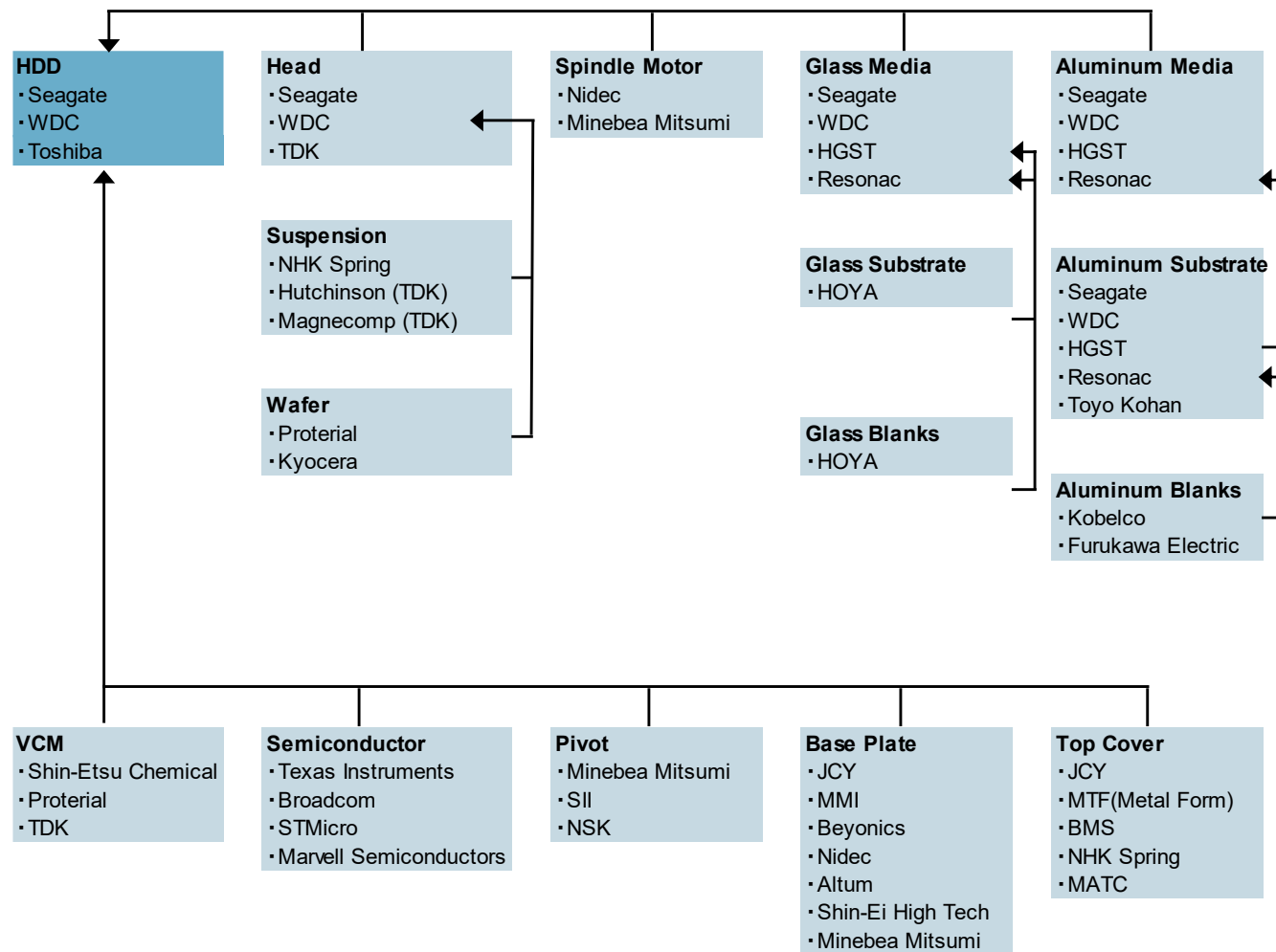
Source: TSR, Morgan Stanley Research

HDD Motor Market

CY23														CY24				CY25				CY25				CY14	CY15	CY16	CY17	CY18	CY19	CY20	CY21	CY22	CY23	CY24	CY25
(mn units)	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q	1Q	CY14	CY15	CY16	CY17	CY18	CY19	CY20	CY21	CY22	CY23	CY24	CY25												
HDD Motor Shipment																																					
Total	34.2	27.2	31.6	29.5	30.5	32.7	33.2	32.6	30.8	31.7	32.5	34.3	34.0	575.2	476.5	431.7	407.3	381.6	307.7	271.8	248.8	146.1	122.5	129.0													
Nidec	19.2	13.2	19.1	17.4	18.6	18.9	19.3	19.2	18.8	18.8	20.0	20.2	20.4	408.6	380.3	362.6	341.9	317.6	250.1	210.9	133.7	82.1	68.9	76.0													
Minebea Mitsumi	15.0	14.0	12.5	12.1	11.9	13.8	13.9	13.4	12.0	12.9	12.5	14.1	13.6	70.9	67.7	69.1	65.4	63.8	57.5	60.9	115.1	64.0	53.6	53.0													
Market Share																																					
Nidec	56%	49%	60%	59%	61%	58%	58%	59%	61%	59%	62%	59%	60%	71%	80%	84%	84%	83%	81%	78%	54%	56%	56%	59%													
Minebea Mitsumi	44%	51%	40%	41%	39%	42%	42%	41%	39%	41%	38%	41%	40%	12%	14%	16%	16%	17%	19%	22%	46%	44%	44%	41%													
YoY % Change																																					
Nidec	-32%	-27%	2%	8%	-11%	20%	5%	11%	1%	-3%	-2%	5%	10%	3%	-17%	-9%	-6%	-6%	-19%	-12%	-8%	-41%	-16%	5%													
Minebea Mitsumi	-25%	-30%	3%	-9%	-3%	43%	1%	10%	1%	-1%	4%	5%	9%	4%	-7%	-5%	-6%	-7%	-21%	-16%	-37%	-39%	-16%	10%													
	-40%	-25%	1%	49%	-21%	-1%	11%	11%	1%	-7%	-10%	5%	13%	-4%	-5%	2%	-5%	-2%	-10%	6%	89%	-44%	-16%	-1%													
Sequential % Change																																					
Nidec	25%	-20%	16%	-7%	3%	7%	2%	-2%	-6%	3%	3%	6%	-1%	-	-	-	-	-	-	-	-	-	-	-													
Minebea Mitsumi	0%	-31%	45%	-9%	7%	2%	2%	-1%	-2%	0%	6%	1%	1%	-	-	-	-	-	-	-	-	-	-	-													
	85%	-7%	-11%	-3%	-2%	16%	1%	-4%	-10%	8%	-3%	13%	-4%	-	-	-	-	-	-	-	-	-	-	-													
HDD Motor/HDD														1.02	1.02	1.02	1.01	1.02	0.97	1.05	0.96	0.85	1.00	1.04	1.04												
Nidec's Motor Shipment by Size																																					
Total	19.2	13.2	19.1	17.4	18.6	18.9	19.3	19.2	18.8	18.8	20.0	20.2	20.4	408.6	380.3	362.6	341.9	317.6	250.1	210.9	133.7	82.1	68.9	76.0													
Enterprise	0.6	0.5	0.7	0.8	0.8	0.4	0.3	0.4	0.4	0.4	0.4	0.4	0.4	23.1	22.4	15.0	14.9	13.0	7.3	4.6	3.4	2.9	2.6	1.9													
NL	7.0	4.0	7.1	6.3	8.6	9.5	10.4	10.9	11.1	11.4	11.3	11.8	12.1	28.6	31.9	32.5	39.2	50.0	48.6	59.7	41.8	31.0	24.4	39.4													
3.5"ATA	5.9	4.6	6.0	5.4	5.2	4.7	4.3	3.7	3.4	3.6	4.3	4.3	4.5	146.4	143.1	134.2	124.6	108.2	87.6	76.7	45.2	26.2	21.9	17.9													
2.5"	5.7	4.1	5.3	4.9	4.0	4.3	4.2	3.9	3.4	4.0	3.7	3.5	3.5	210.2	209.2	180.9	163.2	146.5	106.9	69.9	43.5	22.2	20.0	16.8													
1.8"	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.6	0.4	-	-	-	-	-	-	-	-	-													
1"	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	-	-	-	-	-	-	-	-	-													
As % of Total																																					
Enterprise	3%	4%	4%	5%	4%	2%	2%	2%	2%	2%	2%	2%	2%	6%	6%	4%	4%	4%	3%	2%	3%	4%	4%	3%													
NL	36%	30%	37%	36%	46%	50%	54%	57%	59%	61%	57%	58%	59%	7%	8%	9%	11%	16%	19%	28%	31%	38%	35%	52%													
3.5"ATA	31%	35%	31%	31%	28%	25%	22%	19%	18%	19%	22%	21%	22%	36%	38%	37%	36%	34%	35%	36%	34%	32%	32%	24%													
2.5"	30%	31%	28%	28%	22%	23%	22%	22%	21%	18%	20%	18%	17%	51%	55%	50%	48%	46%	43%	33%	33%	27%	29%	22%													
1.8"	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	-	-	-	-	-	-	-	-	-													
1"	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	-	-	-	-	-	-	-	-	-													
YoY % Change																																					
Enterprise	-33%	0%	40%	-20%	33%	-20%	-57%	-50%	-50%	0%	33%	0%	0%	-3%	-7%	-5%	-6%	-7%	-21%	-16%	-37%	-39%	-16%	10%													
NL	-24%	-57%	6%	9%	23%	138%	46%	73%	29%	20%	9%	8%	9%	20%	12%	2%	21%	28%	-3%	23%	-30%	-26%	-21%	61%													
3.5"ATA	-24%	-57%	6%	9%	23%	138%	46%	73%	29%	20%	9%	8%	9%	0%	-2%	-6%	-7%	-13%	-19%	-12%	-41%	-42%	-16%	-18%													
2.5"	-34%	-10%	3%	-14%	-12%	2%	-28%	-31%	-35%	-23%	0%	16%	32%	6%	0%	-14%	-10%	-10%	-27%	-35%	-38%	-49%	-10%	-16%													
Sequential % Change																																					
Enterprise	-40%	-17%	40%	14%	0%	-50%	-25%	33%	0%	0%	0%	0%	0%	-	-	-	-	-	-	-	-	-	-	-													
NL	21%	-43%	78%	-11%	37%	10%	9%	5%	2%	3%	-1%	4%	3%	-	-	-	-	-	-	-	-	-	-	-													
3.5"ATA	21%	-43%	78%	-11%	37%	10%	9%	5%	2%	3%	-1%	4%	3%	-	-	-	-	-	-	-	-	-	-	-													
2.5"	-6%	-22%	30%	-10%	-4%	-10%	-9%	-14%	-8%	6%	19%	0%	5%	-	-	-	-	-	-	-	-	-	-	-													

Note: e = Morgan Stanley Research estimates
Source: TSR, Morgan Stanley Research

Supply Chain of HDDs



Source: TSR, Morgan Stanley Research

Ibiden: Key Financial Data and Risk Reward

Income Statement					
(Y bn)	F3/24	F3/25	F3/26e	F3/27e	F3/28e
Net Sales	370.5	369.4	418.1	499.7	602.5
COGS	268.0	256.1	288.1	343.9	409.6
Gross profit	102.5	113.3	130.0	155.8	192.9
SG&A	54.9	65.7	68.2	78.8	86.5
Operating Profit	47.6	47.6	61.8	77.0	106.4
OP Incl. depreciation of inactive noncurrent assets	47.6	45.9	57.9	73.0	102.4
Non-operating Income/Loss	3.6	0.3	-2.1	-3.3	-3.3
Recurring Profit	51.1	47.9	59.7	73.7	103.1
Extraordinary Income/Loss	-3.7	3.6	-0.9	-2.0	-2.0
Pretax Profit	47.4	51.5	58.8	71.7	101.1
Income Tax	15.7	17.4	16.1	19.4	27.3
(Effective Tax Rate)	33%	34%	27%	27%	27%
Minority Interest	0.3	0.3	0.4	0.2	0.2
Net Profit	31.5	33.7	42.3	52.1	73.6
Capex	146.6	157.3	67.0	160.0	160.0
Depreciation and Amortization	46.0	54.2	65.5	79.5	95.0
R&D Expenses	20.0	27.5	29.6	34.5	35.0
EBITDA	93.6	101.8	127.3	156.5	201.4
ROE	6.9%	6.8%	8.3%	9.5%	12.3%
Average Share Outstanding (Diluted)	279.4	279.4	289.5	289.5	289.5
EPS (Diluted, Yen)	112.7	120.7	146.2	180.0	254.3
BPS (Yen)	1,786.1	1,780.0	1,840.0	1,888.7	2,206.8
DPS (Yen)	20.0	20.0	30.0	35.0	40.0
USD/JPY: Average Exchange Rate	142.0	152.0	147.5	150.0	150.0
EUR/JPY: Average Exchange Rate	154.0	162.0	167.5	170.0	170.0
YoY % Change					
Net Sales	-11%	0%	13%	20%	21%
Operating Profit	-34%	0%	30%	25%	38%
Net Profit	-40%	7%	26%	23%	41%
As % of Sales					
Gross profit	27.7%	30.7%	31.1%	31.2%	32.0%
Operating Profit	12.8%	12.9%	14.8%	15.4%	17.7%
Net Profit	8.5%	9.1%	10.1%	10.4%	12.2%
Capex	39.6%	42.6%	16.0%	32.0%	26.6%
Depreciation and Amortization	12.4%	14.7%	15.7%	15.9%	15.8%
Balance Sheet					
(Y bn)	F3/24	F3/25	F3/26e	F3/27e	F3/28e
Total Assets	1,130.0	1,081.7	1,125.4	1,177.6	1,251.5
Total Current Assets	600.8	549.6	591.8	563.5	572.4
Cash and Cash Equivalents	443.6	390.7	427.4	373.6	351.7
Marketable Securities	0.0	0.0	0.0	0.0	0.0
Account Receivables	65.4	68.2	73.8	88.2	106.4
Inventory	65.3	67.2	67.1	78.2	90.8
Other Current Assets	26.5	23.5	23.5	23.5	23.5
Total Long Term Assets	529.2	532.1	533.6	614.1	679.1
Net PP&E	408.8	460.1	461.6	542.1	607.1
Investments in Securities	110.2	58.8	58.8	58.8	58.8
Other Assets	10.2	13.3	13.3	13.3	13.3
Total Current Liabilities	358.4	327.7	336.1	345.3	356.0
ST Debt & Current Portion of LT Debt	90.0	90.0	90.0	90.0	90.0
Accounts Payable	43.9	38.8	47.2	56.3	67.0
Other Current Liabilities	144.4	106.9	106.9	106.9	106.9
Total Long Term Liabilities	269.8	256.7	256.7	256.7	256.7
Long Term Debt	253.5	253.0	253.0	253.0	253.0
Other Long Term Liabilities	16.4	-3.7	-3.7	-3.7	-3.7
Net Assets	501.8	497.3	532.6	575.7	638.8
Minority Interests	6.9	6.8	6.8	6.8	6.8
Stock Acquisition Rights	0.0	0.0	0.0	0.0	0.0
Net Debt	-210.3	-106.5	-143.2	-89.4	-67.5

Cash Flow Statement					
(Y bn)	F3/24	F3/25	F3/26e	F3/27e	F3/28e
Net Cash from Operations	145.2	118.9	110.7	115.2	148.5
Net Income	31.5	33.7	42.3	52.1	73.6
Depreciation and Amortization	46.0	54.2	65.5	79.5	95.0
Changes in Working Capital	33.5	-8.8	2.9	-16.4	-20.1
Other-net	34.2	40.7	0.0	0.0	0.0
Net Cash from Investing Activities	-77.3	-164.2	-67.0	-160.0	-160.0
Purchase of PP&E	-85.3	-107.5	-67.0	-160.0	-160.0
Sale/Purchase of Investments	0.1	34.6	0.0	0.0	0.0
Other-net	7.9	-1.3	0.0	0.0	0.0
Free Cash Flow	68.0	-45.3	43.7	-44.8	-11.5
Net Cash from Financing Activities	67.5	-7.1	-7.0	-9.1	-10.5
Change in Interest-bearing Debt	73.5	0.0	0.0	0.0	0.0
Change in Treasury Stock	0.1	-0.5	0.0	0.0	0.0
Payment of Cash Dividends	-5.6	-5.6	-7.0	-9.1	-10.5
Other-net	-0.5	-1.0	0.0	0.0	0.0
Net Effect of Exchange Rate Changes	5.7	-0.5	0.0	0.0	0.0
Net Change in Cash	141.2	-52.9	36.8	-53.8	-21.9
Cash at End of Period	443.6	390.7	427.4	373.6	351.7
Segment Information					
(Y bn)	F3/24	F3/25	F3/26e	F3/27e	F3/28e
Net Sales	370.5	369.4	418.1	499.7	602.5
Electronics	190.7	197.2	245.9	322.1	419.1
FC Package	179.5	186.9	235.9	312.1	409.1
Print Circuit Board	11.2	10.3	10.0	10.0	10.0
FC CSP	0.0	0.0	0.0	0.0	0.0
Ceramics	96.5	84.1	81.1	83.8	86.3
DPF	43.6	34.6	31.0	29.7	29.7
Sealing Materials	32.8	30.4	29.3	30.4	31.3
Graphite Specialty Products	16.3	16.2	14.1	15.1	15.9
Others	3.8	3.0	6.7	8.5	9.4
Others	83.3	88.1	91.1	93.9	97.1
Total Operating Profit	47.6	47.6	61.8	77.0	106.4
Electronics	27.3	26.8	45.8	60.5	88.2
Ceramics	13.4	12.2	7.9	7.9	9.0
Others	6.9	8.6	8.1	8.5	9.2
Operating Profit Margin	12.8%	12.9%	14.8%	15.4%	17.7%
Electronics	14.3%	13.6%	18.6%	18.8%	21.1%
Ceramics	13.8%	14.5%	9.7%	9.5%	10.4%
Others	8.3%	9.7%	8.9%	9.1%	9.5%

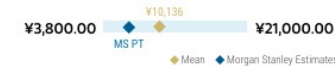
Risk Reward – Ibiden (4062.T)

Market's excessively high expectations likely to correct

PRICE TARGET ¥6,500

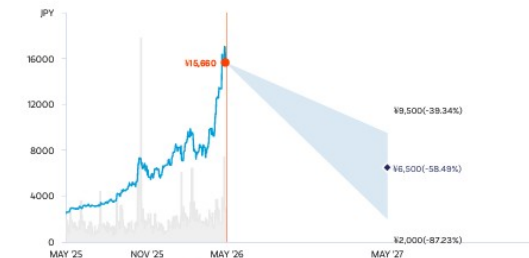
Derived from the base case. DCF assumptions: 2.2% risk-free rate, 1.31 beta, 3.7% risk premium, for 6.5% WACC. Terminal growth of 0% from F3/36.

Consensus Price Target Distribution



Source: Refinitiv, Morgan Stanley Research

RISK REWARD CHART



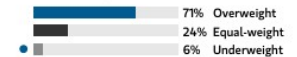
Key: — Historical Stock Performance • Current Stock Price ♦ Price Target

Source: Refinitiv, Morgan Stanley Research

UNDERWEIGHT THESIS

- We expect ABF packages for AI servers sales to increase and earnings growth to continue.
- We see Ibiden sustaining a competitive advantage in ABF package technology and keeping dominant market share in GPU ABF packages for AI servers.
- We expect earnings growth to continue due to customer and product mix improvement, even though costs increase due mainly to new lines being activated in F3/26.
- Glass cloth shortages seem likely to continue until 3Q F3/27 and F3/27 earnings to fall short of the market's overly high expectations.
- We see a high risk that market consensus will come down and the stock price will fall.

Consensus Rating Distribution



• MS Rating

Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Contrarian: Positive
New Data Era: Positive

View descriptions of Risk Rewards Themes [here](#)

BULL CASE ¥9,500

Derived from DCF

Earnings ahead of the base case on stronger sales of value-added ABF packages for AI servers: OP of ¥73.0bn in F3/26, ¥103.4bn in F3/27, ¥142.9bn in F3/28 and ¥277.0bn in F3/31.

Same FX and DCF assumptions as the base case. Fair value equates to 38.5x F3/27 diluted EPS of ¥246.7 and 27.5x F3/28 diluted EPS of ¥346.3.

BASE CASE ¥6,500

Derived from DCF

Earnings expansion to continue due to value-added ABF package sales growth: OP ¥61.8bn in F3/26, ¥77.0bn in F3/27, ¥106.4bn in F3/28, ¥217.2bn in F3/31.

FX assumptions ¥150/\$, ¥170/€. Fair value is based on DCF as explained in the valuation methodology section. Base case fair value equates to 36.1x our F3/27e diluted EPS of ¥180.0, 25.6x our F3/28e diluted EPS of ¥217.2.

BEAR CASE ¥2,000

F3/26 BPS ¥1,821 x P/B 1.1

Limited earnings improvement from F3/26: OP of ¥54.3bn in F3/26, ¥62.4bn in F3/27, ¥62.5bn in F3/28, and ¥67.9bn in F3/31.

Same FX assumptions as the base case. ROE improves from 6.8% in F3/25 to 7.5% in F3/26 and 8.0% in F3/27, higher than 7.1% cost of capital. However, this is below the market consensus.

Note: e = Morgan Stanley Research estimates; Stock price as of the May 15 close for Risk Reward Chart
Source: Company data, Morgan Stanley Research

Niterra: Key Financial Data and Risk Reward

Income Statement					
(Y bn)	F3/24	F3/25	F3/26e	F3/27e	F3/28e
Sales	614.5	653.0	694.3	754.6	788.6
COGS	388.3	394.9	422.2	439.3	455.7
Gross Profit	226.2	258.1	272.0	315.3	332.9
SG&A	118.6	128.4	133.0	158.8	161.2
Operating Profit	107.6	129.7	139.1	156.6	171.7
Non-operating Income/Loss	9.6	3.7	7.3	7.3	7.3
Recurring Profit	117.2	133.3	146.3	163.8	178.9
Extraordinary Income/Loss	0.0	0.0	0.0	0.0	0.0
Pretax Profit	117.2	133.3	146.3	163.8	178.9
Income Tax	34.8	40.5	39.2	45.9	50.1
(Effective Tax Rate)	30%	30%	27%	28%	28%
Minority Interest	-0.3	0.2	3.4	3.4	3.4
Net Profit	82.6	92.6	103.8	114.6	125.5
Capex	41.2	37.5	37.6	40.0	45.0
Depreciation and Amortization	39.8	39.7	41.4	40.0	43.0
R&D	27.8	0.0	6.5	30.0	31.0
EBITDA	147.4	169.4	180.5	196.6	214.7
ROE	13.8%	14.1%	14.8%	15.2%	15.6%
Average Share Outstanding (Diluted)	201.8	198.6	198.4	193.4	188.4
EPS (Diluted, Yen)	409.5	466.3	523.1	592.4	665.8
BPS (Yen)	3,175.8	3,399.4	3,668.8	4,013.6	4,435.2
DPS (Yen)	164.0	179.0	180.0	186.0	192.0
USD/JPY: Average Exchange Rate	145.0	153.0	148.0	150.0	150.0
EUR/JPY: Average Exchange Rate	157.0	164.0	169.0	170.0	170.0
YoY % Change					
Sales	9%	6%	6%	9%	5%
Operating Profit	21%	21%	7%	13%	10%
Net Profit	25%	12%	12%	10%	9%
As % of Sales					
Gross Profit	36.8%	39.5%	39.2%	41.8%	42.2%
Operating Profit	17.5%	19.9%	20.0%	20.7%	21.8%
Net Profit	13.4%	14.2%	14.9%	15.2%	15.9%
Capex	6.7%	5.7%	5.4%	5.3%	5.7%
Depreciation and Amortization	6.5%	6.1%	6.0%	5.3%	5.5%
Balance Sheet					
(Y bn)	F3/24	F3/25	F3/26e	F3/27e	F3/28e
Total Assets	975.7	991.0	1,178.0	1,199.3	1,231.0
Total Current Assets	564.2	588.5	629.4	650.6	680.4
Cash and Cash Equivalents	180.7	208.2	238.6	250.7	278.5
Marketable Securities	27.2	0.0	0.0	0.0	0.0
Account Receivables	141.4	147.6	156.9	170.5	178.2
Inventory	189.6	183.9	185.1	180.5	174.8
Other Current Assets	25.2	48.8	48.8	48.8	48.8
Total Long Term Assets	411.6	402.4	548.7	548.7	550.7
Net PP&E	224.3	220.0	226.2	226.2	228.2
Investments in Securities	96.0	93.1	93.1	93.1	93.1
Other Assets	91.3	89.4	229.4	229.4	229.4
Total Current Liabilities	172.9	153.6	157.6	160.1	162.5
ST Debt & Current Portion of LT Debt	42.4	16.7	16.7	16.7	16.7
Accounts Payable	58.8	57.7	61.7	64.1	66.5
Other Current Liabilities	71.8	79.3	79.3	79.3	79.3
Total Long Term Liabilities	164.5	162.6	292.6	292.6	232.6
Long Term Debt	135.7	140.0	270.0	240.0	210.0
Other Long Term Liabilities	28.8	22.6	22.6	22.6	22.6
Net Assets	638.3	674.7	727.8	776.5	835.9
Minority Interests	0.8	0.1	0.1	0.1	0.1
Net Debt	-125.8	-144.6	-45.0	-87.1	-144.9

Cash Flow Statement					
(Y bn)	F3/24	F3/25	F3/26e	F3/27e	F3/28e
Net Cash from Operations	118.2	132.9	138.7	148.0	168.9
Net Income	82.6	92.6	103.8	114.6	125.5
Depreciation and Amortization	39.8	39.7	41.4	40.0	43.0
Changes in Working Capital	11.7	-6.2	-6.5	-6.6	0.5
Other-net	-15.9	6.8	0.0	0.0	0.0
Net Cash from Investing Activities	-92.2	-34.2	-187.6	-40.0	-45.0
Purchase of PP&E	-31.1	-27.8	-37.6	-40.0	-45.0
Sale/Purchase of Investments	-51.3	-3.4	-150.0	0.0	0.0
Other-net	-9.7	-3.0	0.0	0.0	0.0
Free Cash Flow	26.0	98.7	-48.9	108.0	123.9
Net Cash from Financing Activities	-57.5	-71.0	79.3	-95.8	-96.1
Change in Interest-bearing Debt	-14.9	-25.9	130.0	-30.0	-30.0
Change in Treasury Stock	-10.0	-10.0	-15.0	-30.0	-30.0
Payment of Cash Dividends	-33.1	-34.4	-35.7	-35.8	-36.1
Other-net	0.6	-0.7	0.0	0.0	0.0
Net Effect of Exchange Rate Changes	10.5	-0.2	0.0	0.0	0.0
Net Change in Cash	-20.9	27.5	30.4	12.2	27.8
Cash at End of Period	180.7	208.2	238.6	250.7	278.5
Segment Information					
(Y bn)	F3/24	F3/25	F3/26e	F3/27e	F3/28e
Net sales	614.5	653.0	694.3	754.6	788.6
Automotive Components	505.4	542.8	556.7	611.4	638.8
Spark & Glow Plugs	335.3	362.6	364.9	414.4	438.0
Automotive Sensors	170.0	180.2	191.8	197.0	200.9
Ceramics	96.5	103.1	127.0	135.3	143.7
New Business	5.1	0.0	0.0	0.0	0.0
Others/Elimination	7.4	7.1	11.2	8.0	6.0
Total Operating Profit	107.6	129.7	139.1	156.6	171.7
Automotive Components	121.2	134.2	135.8	149.6	161.2
Ceramics	-0.6	-6.3	1.8	6.6	10.1
New Business	-13.2	0.0	0.0	0.0	0.0
Others/Elimination	0.2	1.8	1.4	0.4	0.4
Operating Profit Margin	17.5%	19.9%	20.0%	20.7%	21.8%
Automotive Components	24.0%	24.7%	24.4%	24.5%	25.2%
Ceramics	-0.6%	-6.1%	1.5%	4.9%	7.0%

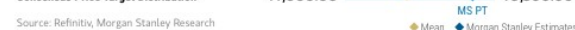
Risk Reward – Niterra (5334.T)

Expect earnings growth in auto components with volume & unit price going up

PRICE TARGET ¥9,100

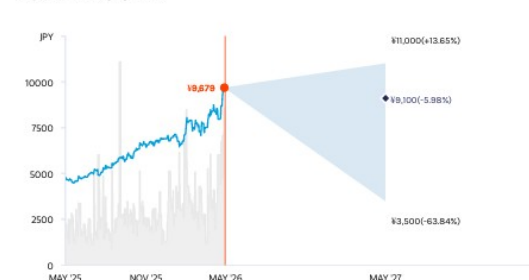
Calculated from our DCF model derived from our base case. Our DCF model assumes a 2.2% risk-free rate, 1.08 equity beta, and 3.7% risk premium, yielding a 5.4% WACC. We assume a 0% perpetual growth rate from F3/36.

Consensus Price Target Distribution



Source: Refinitiv, Morgan Stanley Research

RISK REWARD CHART



Key: — Historical Stock Performance • Current Stock Price ♦ Price Target

Source: Refinitiv, Morgan Stanley Research

OVERWEIGHT THESIS

- We expect the firm to maintain competitiveness in automotive plugs and exhaust gas sensors, and market share to keep gradually expanding.
- Although demand for plugs is unlikely to grow substantially (Niterra boasts the top global share at 47%), we expect Niterra's market share to increase gradually and the firm to book large profits on aftermarket products.
- In automotive sensors, we expect Niterra to gradually increase its 44% global market share, and earnings to keep growing until F3/28 alongside the tightening of environmental regulations.
- In non-auto business, we believe earnings from electrostatic chucks for SPE will continue growing having bottomed in F3/24.

Consensus Rating Distribution



MS Rating

Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Contrarian: Positive

View descriptions of Risk Rewards Themes [here](#)

BULL CASE	¥11,000	BASE CASE	¥9,100	BEAR CASE	¥3,500
Derived from DCF		Derived from DCF			
Automotive plug & exhaust gas sensor sales exceed the base case by 4% and earnings continue to grow further. This scenario assumes OP of ¥150.6bn in F3/26, ¥179.3bn in F3/27, ¥196.5bn in F3/28, ¥225.6bn in F3/29, and uses the same FX & DCF assumptions as the base case. Bull case fair value equates to P/E of 16.7x on F3/27e diluted EPS of ¥659.4 and 15.2x on F3/28e diluted EPS of ¥721.6.		Expect profit to continue to reach new heights, driven by increased sales of automobile plugs and SPE electro static chuck. We forecast OP of ¥139.1bn in F3/26, ¥155.6bn in F3/27, ¥171.7bn in F3/28, ¥196.6bn in F3/29, assuming FX rates of ¥150/US\$, ¥170/€. Our PT is calculated from our DCF model, the conditions for which are outlined in our valuation methodology. Our base case PT equates to P/E of 15.4x on F3/27e diluted EPS of ¥592.4 and 13.7x on F3/28e diluted EPS of ¥655.8.		F3/27e diluted BPS ¥3,713 x P/B 0.9	
				Lack of growth in autos-related volumes, limited improvement in SPE-related earnings: This scenario assumes OP of ¥123.9bn in F3/26, ¥107.7bn in F3/27, ¥97.3bn in F3/28, ¥89.0bn in F3/29, and uses the same FX assumptions as the base case. In this case, the market consensus moves lower and the share price drops to P/B 0.9x.	

Note: e = Morgan Stanley Research estimates; Stock price as of the May 15 close for Risk Reward Chart
Source: Company data, Morgan Stanley Research

Minebea Mitsumi: Key Financial Data and Risk Reward

Income Statement					
(¥ bil)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Net Sales	1,522.7	1,664.4	1,651.1	1,703.3	1,763.6
Cost of Goods Sold	1,252.3	1,371.4	1,348.7	1,387.6	1,438.4
Gross Profit	270.4	293.0	302.4	315.7	325.2
SG&A Expenses	175.9	189.0	185.5	188.2	191.2
Operating Profit	94.5	104.0	116.9	127.5	134.0
Interest/Dividend Income	6.9	40.9	1.8	1.8	1.8
Interest Expense	18.8	11.1	2.5	2.5	2.5
Other Non-Operating Income/Loss	0.0	0.0	0.0	0.0	0.0
Recurring Profit	82.6	133.8	116.2	126.8	133.3
Extraordinary Income/Loss	0.0	0.0	0.0	0.0	0.0
Pre-tax profit	82.6	133.8	116.2	126.8	133.3
Income Tax	22.8	34.5	30.2	33.0	34.7
(Effective Tax Rate)	28%	26%	26%	26%	26%
Minority Interest	0.4	0.2	0.0	0.0	0.0
Net Profit	59.5	99.0	86.0	93.8	98.7
Capex	94.8	96.5	99.1	102.2	105.8
Depreciation and Amortization	66.2	70.4	69.9	72.1	74.6
R&D	45.5	49.8	49.5	51.1	52.9
EBITDA	160.7	174.4	186.7	199.6	208.7
ROE	8.2%	12.5%	10.4%	10.8%	10.1%
Average Share Outstanding (mil, Diluted)	406.3	401.6	401.6	401.6	401.6
EPS (Diluted, Yen)	147.7	246.6	214.0	233.7	245.7
BPS (Yen)	1,830.0	2,102.8	2,143.7	2,326.4	2,518.0
DPS (Yen)	45.0	50.0	48.0	51.0	54.0
USD/JPY: Average Exchange Rate	152.6	150.0	150.0	150.0	150.0
YoY % Change					
Net Sales	0%	9%	2%	3%	4%
Operating Profit	28%	10%	14%	9%	5%
Net Profit	10%	67%	25%	9%	5%
As % of Sales					
Gross Profit	17.8%	17.6%	18.3%	18.5%	18.4%
Operating Profit	6.2%	6.2%	7.1%	7.5%	7.6%
Net Profit	3.9%	6.0%	5.2%	5.5%	5.6%
Balance Sheet					
(¥ bil)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Total Assets	1,584.8	1,782.3	1,724.1	1,806.7	1,895.6
Total Current Assets	904.1	1,041.2	992.9	1,045.3	1,103.1
Cash and Cash Equivalents	214.3	220.0	202.1	232.1	262.1
Marketable Securities	0.0	0.0	0.0	0.0	0.0
Account Receivables	293.3	344.6	367.2	378.8	392.2
Inventory	350.9	411.7	377.9	388.8	403.0
Other Current Assets	170.3	197.4	170.3	170.3	170.3
Total Long Term Assets	680.7	741.1	731.2	761.3	792.5
Net PP&E	529.0	576.6	579.5	609.6	640.8
Investments in Securities	27.1	32.0	27.1	27.1	27.1
Other assets	0.0	0.0	0.0	0.0	0.0
Total Current Liabilities	477.2	618.1	499.4	508.6	520.5
ST Debt & Current Portion of LT Debt	175.1	271.6	175.1	175.1	175.1
Accounts Payable	213.6	251.5	218.5	222.7	230.9
Other Current Liabilities	88.5	94.9	107.9	110.8	114.6
Total Long Term Liabilities	353.0	308.2	353.0	353.0	353.0
Long Term Debt	288.5	235.3	288.5	288.5	288.5
Other Long Term Liabilities	64.4	73.0	64.4	64.4	64.4
Accrued pension cost	0.0	0.0	0.0	0.0	0.0
Minority Interests	11.2	11.5	10.9	10.9	10.9
Total Equity	754.6	856.0	871.7	945.1	1,022.1
Net Debt	249.3	286.9	261.5	231.5	201.4

Cash Flow Statement					
(¥ bil)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Net Cash from Operations	133.7	94.9	146.1	149.7	153.8
Net Income After Tax	59.5	99.0	86.0	93.8	98.7
Depreciation and Amortization	66.2	70.4	69.9	72.1	74.6
Changes in Working Capital:	9.7	-53.9	-9.7	-16.3	-19.5
Other-net	-1.7	-20.7	0.0	0.0	0.0
Net Cash from Investing Activities	-125.8	-82.8	-97.7	-99.3	-102.0
Purchase of PP&E	-77.4	-77.1	-99.1	-102.2	-105.8
Sale/Purchase of Investments	-0.4	-0.2	0.0	0.0	0.0
Other-net	-48.0	-5.5	1.3	2.9	3.8
Free cash flow	7.9	12.1	48.3	50.4	51.8
Net Cash from Financing Activities	64.0	-16.1	-19.3	-20.5	-21.7
Change in Interest-bearing Debt	96.4	-139.0	0.0	0.0	0.0
Change in Treasury Stock	-3.7	0.0	0.0	0.0	0.0
Payment of Cash Dividends	-16.1	-20.1	-19.3	-20.5	-21.7
Other-net	-7.6	-9.1	0.0	0.0	0.0
Net Effect of Exchange Rate Changes	-4.3	17.3	0.0	0.0	0.0
Cash at End of Period	189.3	222.0	177.2	207.1	237.2

Segment Information					
(¥ bil)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Sales by Segment					
Total Sales	1,522.7	1,664.4	1,651.1	1,703.3	1,763.6
Precision Technologies	255.7	281.2	287.3	298.4	315.5
Ball bearings	171.8	186.4	198.2	206.1	219.9
Rod-ends/Fasteners	59.1	66.0	64.0	67.2	70.6
Pivot assemblies	24.8	28.7	25.0	25.0	25.0
Motor, Lighting & Sensing	407.7	456.5	448.8	460.2	472.0
Motors	322.4	340.3	342.7	353.0	363.6
Electronic devices	41.1	68.0	64.6	64.6	64.6
Measuring components	36.9	38.5	37.5	36.6	36.8
Other	7.4	9.8	4.0	4.0	4.0
Semiconductors & Electronics	527.6	590.3	576.3	589.0	602.7
Access Solutions	328.1	332.2	337.8	354.7	372.5
Others	3.5	4.2	1.0	1.0	1.0
Operating Profit by Segment					
Total Operating Profit	94.5	104.0	116.9	127.5	134.0
Machined Components Business	55.7	62.2	63.6	66.0	68.6
Electronic Devices & Components Business	23.0	26.9	27.7	28.5	28.3
Semiconductors & Electronics	22.0	26.7	28.7	34.5	37.7
Access Solutions	15.9	17.1	17.9	19.5	20.5
Others	-1.2	-2.7	-1.0	-1.0	-1.0
Adjustments	-20.9	-26.3	-20.0	-20.0	-20.0
Operating Profit Margin	6.2%	6.2%	7.1%	7.5%	7.6%
Machined Components Business	21.8%	22.1%	22.1%	22.1%	21.7%
Electronic Devices & Components Business	5.8%	5.9%	6.2%	6.2%	6.0%
Semiconductors & Electronics	4.2%	4.5%	5.0%	5.9%	6.3%
Access Solutions	4.9%	5.1%	5.3%	5.5%	5.5%
Others	-33.8%	-63.2%	-100.0%	-100.0%	-100.0%

Risk Reward – Minebea Mitsumi (6479.T)

Earnings should recover gradually, but this is mostly priced into the stock

PRICE TARGET ¥3,300

Derived from the base case. DCF assumptions: 2.2% risk-free rate, 1.21 equity beta, 3.7% risk premium, yielding 5.2% WACC; zero growth from F3/36.



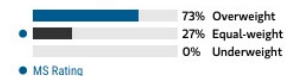
RISK REWARD CHART



EQUAL-WEIGHT THESIS

- Earnings should continue to recover having bottomed in F3/24, but this is mostly priced into the stock.
- We expect earnings growth of Minebea's more-than-60% share of the global market for miniature ball bearings to continue in mid/long term.
- We expect it will be difficult for the semiconductor & electronics segment to post a new record high in the near term.
- Access Solution earnings should improve thanks to cost cuts and market share gains.

Consensus Rating Distribution



BULL CASE

¥4,300

Derived from DCF

Semiconductor & Electronics and Precision Technology sales higher than in our base case: OP ¥112.2bn in F3/26, ¥129.7bn in F3/27, ¥142.5bn in F3/28, ¥154.9bn in F3/29. Same FX and DCF model assumptions as the base case. Fair value for the shares equates to 18.1x F3/27e diluted EPS of ¥237.7, 16.5x F3/28e diluted EPS of ¥261.2.

BASE CASE

¥3,300

Derived from DCF

Earnings continue to grow, driven by rising sales of core products, as the market expects: OP ¥102.1bn in F3/26, ¥116.9bn in F3/27, ¥127.5bn in F3/28, ¥134.0bn in F3/29. FX assumption of ¥150/\$, ¥170/€. DCF model assumptions: 2.2% risk-free rate, 1.21 equity beta, 3.7% risk premium, yielding 5.2% WACC. Base case fair value equates to 15.4x F3/27e diluted EPS of ¥214.0, 14.1x F3/28e diluted EPS of ¥233.7.

BEAR CASE

¥2,100

F3/27e BPS ¥2,078 x P/B 1.0

Earnings especially in Semiconductor & Electronics and Motor, Lighting & Sensing will be weaker than base case: OP ¥89.6bn in F3/26, ¥81.5bn in F3/27, ¥85.5bn in F3/28, ¥92.1bn in F3/29. 7.1% ROE in F3/27 is higher than cost of capital of 6.7%, but earnings are lower than market consensus and the share price declines to 1.0 x P/B.

Note: e = Morgan Stanley Research estimates; Stock price as of the May 15 close for Risk Reward Chart
Source: Company data, Morgan Stanley Research

Mabuchi Motor: Key Financial Data and Risk Reward

Income Statement					
(Y bn)	F12/24	F12/25	F12/26e	F12/27e	F12/28e
Net Sales	196.2	200.4	213.4	227.3	242.2
COGS	143.3	140.6	149.0	160.3	172.2
Gross Profit	52.9	59.8	64.3	67.0	70.0
SG&A Expenses	31.2	34.3	37.2	37.8	38.4
Operating Profit	21.6	25.5	27.2	29.3	31.6
Non-operating Income/Loss	10.8	9.6	4.4	4.4	4.4
Recurring Profit	32.4	35.1	31.6	33.7	36.0
Extraordinary Income/Loss	-8.7	-0.1	-0.2	-0.2	-0.2
Pretax Profit	23.7	35.0	31.4	33.5	35.8
Income Tax	10.9	8.7	9.1	9.7	10.4
(Effective Tax Rate)	46%	25%	29%	29%	29%
Minority Interest	0.0	0.0	0.0	0.0	0.0
Net Profit	12.8	26.3	22.3	23.8	25.4
Capex	14.1	10.3	12.6	15.0	15.0
Depreciation and Amortization	14.2	13.3	13.9	14.0	14.0
R&D Expenses	7.0	7.8	8.8	9.0	9.0
EBITDA	35.8	38.8	41.1	43.3	45.6
ROE	4.1%	8.0%	6.8%	7.5%	8.3%
Average Share Outstanding (Diluted)	254.1	248.1	234.1	234.1	234.1
EPS (Diluted, Yen)	50.5	105.9	90.7	101.5	108.6
BPS (Yen)	1,257.7	1,347.0	1,378.4	1,332.5	1,289.6
DPS (Yen)	38.0	53.0	56.0	64.0	68.0
USD/JPY: Average Exchange Rate	151.6	149.7	150.0	150.0	150.0
YoY % Change					
Net Sales	10%	2%	6%	7%	7%
Operating Profit	39%	18%	7%	8%	8%
Net Profit	-34%	105%	-15%	7%	7%
As % of Sales					
Gross Profit	26.9%	29.8%	30.2%	29.5%	28.9%
Operating Profit	11.0%	12.7%	12.7%	12.9%	13.0%
Net Profit	6.5%	13.1%	10.4%	10.4%	10.5%
Capex	7.2%	5.1%	5.9%	6.6%	6.2%
Depreciation and Amortization	7.2%	6.6%	6.5%	6.2%	5.8%
Balance Sheet					
(Y bn)	F12/24	F12/25	F12/26e	F12/27e	F12/28e
Total Assets	355.0	370.2	357.8	348.2	339.0
Total Current Assets	238.5	247.7	236.6	226.1	215.9
Cash and Cash Equivalents	134.9	143.4	131.4	118.9	107.0
Marketable Securities	0.0	0.0	0.0	0.0	0.0
Account Receivables	37.3	38.7	37.1	38.9	40.8
Inventory	59.8	57.8	60.4	60.6	60.4
Other Current Assets	6.4	7.8	7.8	7.8	7.8
Total Long Term Assets	116.5	122.4	121.1	122.1	123.1
Net PP&E	95.5	95.9	94.6	95.6	96.6
Investments in Securities	12.4	16.4	16.4	16.4	16.4
Other Assets	8.6	10.2	10.2	10.2	10.2
Total Current Liabilities	28.8	28.3	27.9	28.7	29.5
ST Debt and Current Portion of LT Debt	0.0	0.0	0.0	0.0	0.0
Accounts Payable	7.7	10.4	10.0	10.7	11.5
Other Current Liabilities	21.1	17.9	17.9	17.9	17.9
Total Long Term Liabilities	6.5	7.6	7.6	7.6	7.6
Long Term Debt	0.5	1.3	1.3	1.3	1.3
Other Long Term Liabilities	6.0	6.3	6.3	6.3	6.3
Net Assets	319.6	334.2	322.2	311.9	301.9
Minority Interests	0.0	0.0	0.0	0.0	0.0
Net Debt	-146.8	-158.5	-146.4	-133.9	-122.0

Cash Flow Statement					
(Y bn)	F12/24	F12/25	F12/26e	F12/27e	F12/28e
Net Cash from Operations	40.1	35.4	34.9	36.5	38.5
Net Income	12.8	26.3	22.3	23.8	25.4
Depreciation and Amortization	14.2	13.5	13.9	14.0	14.0
Changes in Working Capital	2.2	8.1	-1.3	-1.2	-0.9
Other-net	10.9	-12.5	0.0	0.0	0.0
Net Cash from Investing Activities	-15.8	-10.5	-12.6	-15.0	-15.0
Purchase of PP&E	-14.0	-10.2	-12.6	-15.0	-15.0
Other-net	-0.5	0.0	0.0	0.0	0.0
Net Cash from Financing Activities	-1.2	-0.3	0.0	0.0	0.0
Free Cash Flow	24.4	24.8	22.3	21.5	23.5
Net Cash from Financing Activities	-18.2	-17.4	-34.3	-34.0	-35.4
Change in Interest-bearing Debt	0.0	0.0	0.0	0.0	0.0
Change in Treasury Stock	-5.7	-6.7	-19.0	-20.0	-20.0
Payment of Cash Dividends	-10.2	-9.7	-15.3	-14.0	-15.4
Other-net	-0.3	-1.0	0.0	0.0	0.0
Net Effect of Exchange Rate Changes	10.1	1.9	0.0	0.0	0.0
Net Change in Cash	18.3	9.4	-12.0	-12.5	-11.9
Cash at End of Period	130.6	139.9	127.9	115.4	103.5
Sales by Application					
(Y bn)	F12/24	F12/25	F12/26e	F12/27e	F12/28e
Net Sales	196.2	200.4	213.4	227.3	242.2
Automotive Products	152.5	154.5	161.1	172.5	184.6
Life & Industrial Products	43.7	45.9	52.3	54.9	57.6
YoY % Change					
Net Sales	10%	2%	6%	7%	7%
Automotive Products	9%	1%	4%	7%	7%
Life & Industrial Products	13%	5%	14%	5%	5%
As % of Sales					
Automotive Products	78%	77%	76%	76%	76%
Life & Industrial Products	22%	23%	24%	24%	24%
Shipment Volume by Application					
(mn units)	F12/24	F12/25	F12/26e	F12/27e	F12/28e
Total Sales Quantity	1,346	1,340	1,342	1,420	1,502
Automotive Products	1,055	1,071	1,074	1,138	1,207
Life & Industrial Products	293	269	268	281	295
Precision & Office Equipment					
Total Sales Quantity	2%	-1%	0%	6%	6%
Automotive Products	2%	2%	0%	6%	6%
Life & Industrial Products	-1%	-8%	0%	5%	5%
As % of Sales					
Automotive Products	78%	80%	80%	80%	80%
Life & Industrial Products	22%	20%	20%	20%	20%

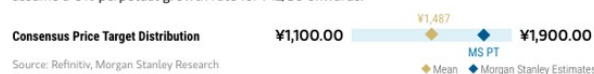
Risk Reward – Mabuchi Motor (6592.T)

Gradual profit growth on auto output recovery, headcount increases

PRICE TARGET ¥1,750

Our PT is calculated from our DCF model derived from our base case, and assumes a risk-free rate of 2.2%, equity beta of 0.81, and risk premium of 3.7%, yielding WACC of 5.2%. We assume a 0% perpetual growth rate for F12/36 onwards.

Consensus Price Target Distribution



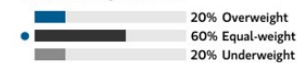
RISK REWARD CHART



EQUAL-WEIGHT THESIS

- ROE improves gradually due to moderate recovery in profit, driven by global auto production recovery and headcount increases.
- Higher automotive motor volumes drive earnings growth over the longer term.

Consensus Rating Distribution



Risk Reward Themes

Secular Growth: **Positive**
View descriptions of Risk Rewards Themes [here](#)

BULL CASE

¥2,100

Derived from DCF

Automotive motor output exceeds our base case by 8% thanks to demand growth and market share gains: This scenario assumes OP of ¥33.2bn in F12/26, ¥35.7bn in F12/27 and ¥38.5bn in F12/28 using the same FX and DCF assumptions as the base case. Our bull case fair value estimate equates to P/E of 18.5x (13.0x excluding net financial assets) on F12/26e diluted EPS of ¥113.5.

BASE CASE

¥1,750

Derived from DCF

Earnings recover gradually due mainly to global auto production recovery: We forecast OP of ¥27.2bn in F12/26, ¥29.3bn in F12/27 and ¥31.6bn in F12/28, assuming an FX rate of ¥150/\$. Our PT is derived from our DCF model, the conditions for which are outlined in our Valuation Methodology. Our PT equates to P/E of 19.3x (12.4x excluding net financial assets) on F12/26e diluted EPS of ¥90.7 and 17.2x on F12/27e diluted EPS of ¥101.5.

BEAR CASE

¥1,100

End-F12/26e diluted BPS ¥1,369 x P/B 0.8

Profit drops in F12/26 due to rising costs and automobile production decline: This scenario is for OP of ¥21.5bn in F12/26, ¥21.4bn in F12/27 and ¥19.8bn in F12/28, assuming ROE is expected to decline to 5.6% in FY12/26, remaining above the cost of capital of 5.2%; however, the market consensus has been revised down sharply, and the share price is expected to fall to a P/B of 0.8x, but end-F12/25 net financial assets/share of ¥645 and shareholder returns underpin the share price.

Note: e = Morgan Stanley Research estimates; Stock price as of the May 15 close for Risk Reward Chart

Source: Company data, Morgan Stanley Research

TDK: Key Financial Data and Risk Reward

Income Statement					
(Y bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Net Sales	2,204.8	2,504.8	2,648.5	2,869.4	3,116.3
Cost of Goods Sold	1,516.8	1,721.4	1,793.6	1,955.1	2,130.1
Gross Profit	688.0	783.4	855.0	914.3	977.2
SG&A Expenses	494.0	511.0	545.2	566.2	593.6
Others*	30.2	33.0	0.0	0.0	0.0
Operating Profit	224.2	272.4	309.8	348.1	383.6
Non-operating Income/Loss	13.6	4.4	4.5	4.5	4.5
Pretax Profit	237.8	276.8	314.3	352.6	388.1
Income Tax	67.4	77.6	78.6	88.1	97.0
(Effective Tax Rate)	28%	28%	25%	25%	25%
Minority Interest	3.2	3.5	4.0	4.0	4.0
Net Profit	167.2	195.7	231.7	260.4	287.1
Capex					
Depreciation and Amortization	225.3	298.6	290.0	290.0	300.0
R&D Expenses	196.2	204.2	210.0	210.0	220.0
EBITDA	253.6	289.7	290.0	300.0	315.0
ROE	420.4	476.6	519.8	558.1	603.6
	9.5%	9.7%	11.5%	11.9%	12.0%
Average Share Outstanding (Diluted)					
EPS (Diluted, Yen)	1,900.0	1,900.2	1,896.7	1,897.6	1,897.6
BPS (Yen)	88.0	103.0	122.1	137.2	151.3
DPS (Yen)	953.3	1,159.6	1,107.6	1,206.4	1,317.7
	30.0	36.0	36.0	40.0	40.0
USD/JPY: Average Exchange Rate					
	152.7	150.8	150.0	150.0	150.0
EUR/JPY: Average Exchange Rate					
	163.9	174.8	170.0	170.0	170.0
YoY % Change					
Net Sales	5%	14%	7%	8%	9%
Operating Profit	30%	22%	16%	12%	10%
Net Profit	34%	17%	20%	12%	10%
As % of Sales					
Gross Profit	31.2%	31.3%	32.3%	31.9%	31.4%
Operating Profit	10.2%	10.9%	11.7%	12.1%	12.3%
Net Profit	7.6%	7.8%	8.7%	9.1%	9.2%
Capex	10.2%	11.9%	10.9%	10.1%	9.6%
Depreciation and Amortization	8.9%	8.2%	7.9%	7.3%	7.1%
Balance Sheet					
(Y bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Total Assets	3,541.4	4,415.2	3,852.4	4,080.4	4,336.0
Total Current Assets					
Cash and Cash Equivalents	697.3	842.8	820.8	932.3	1,069.7
Marketable Securities	0.0	0.0	0.0	0.0	0.0
Account Receivables	583.1	700.6	576.5	601.0	627.1
Inventory	410.0	585.4	429.0	440.8	453.0
Other Current Assets	146.4	246.4	146.4	146.4	146.4
Total Long Term Assets	1,704.6	1,959.9	1,879.9	1,959.9	2,039.9
Net PP&E	1,030.1	1,225.8	1,205.4	1,285.4	1,365.4
Investments in Securities	183.8	226.8	183.8	183.8	183.8
Other Assets	278.6	269.2	276.6	276.6	276.6
Total Current Liabilities					
ST Debt and Current Portion of LT Debt	187.1	211.0	187.1	187.1	187.1
Accounts Payable	392.5	706.7	394.9	430.4	470.9
Other Current Liabilities	518.8	656.9	518.8	518.8	518.8
Total Long Term Liabilities	631.7	637.1	631.7	631.7	631.7
Long Term Debt	346.0	332.7	346.0	346.0	346.0
Other Long Term Liabilities	285.7	304.4	285.7	285.7	285.7
Minority Interests	11.2	16.3	19.1	23.1	27.1
Total Equity	1,811.3	2,293.5	2,100.8	2,289.2	2,500.3
Net Debt	-348.0	-525.9	-471.4	-583.0	-720.4

Cash Flow Statement					
(Y bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Net Cash from Operations	445.8	507.7	448.7	473.7	513.3
Net Income	167.2	195.7	231.7	260.4	287.1
Depreciation and Amortization	196.2	204.2	210.0	210.0	220.0
Changes in Working Capital	2.5	-8.9	2.9	-0.8	2.2
Other-net	80.0	116.7	4.0	4.0	4.0
Net Cash from Investing Activities	-244.8	-377.8	-290.0	-290.0	-300.0
Purchase of PP&E	-225.3	-298.6	-290.0	-290.0	-300.0
Sale/Purchase of Investments	9.9	-4.3	0.0	0.0	0.0
Other-net	-29.5	-74.9	0.0	0.0	0.0
Free Cash Flow	201.0	129.9	158.7	183.7	213.3
Net Cash from Financing Activities	-143.3	-64.7	-64.5	-72.1	-75.9
Change in Interest-bearing Debt	-81.3	9.9	0.0	0.0	0.0
Change in Treasury Stock	0.0	0.0	0.0	0.0	0.0
Payment of Cash Dividends	-48.5	-60.7	-64.5	-72.1	-75.9
Other-net	-13.5	-13.9	0.0	0.0	0.0
Net Effect of Exchange Rate Changes	-10.4	80.3	0.0	0.0	0.0
Net Change in Cash	47.3	145.5	94.2	111.6	137.4
Cash at End of Period	697.3	842.8	820.8	932.3	1,069.7
Segment Information					
(Y bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Net Sales	2,204.8	2,504.8	2,648.5	2,869.4	3,116.3
Passive Components	559.6	593.2	625.3	650.7	677.6
Capacitors	234.3	257.5	276.1	292.6	310.2
Inductive Devices	204.3	216.2	226.2	235.3	244.7
Other Passive Components	121.1	119.5	123.0	122.8	122.8
Sensor Application Products	189.5	224.6	236.7	252.7	269.9
Magnetic Application Products	223.6	262.9	301.9	311.5	321.6
Energy Application Products	1,176.5	1,370.3	1,432.7	1,604.6	1,797.2
Other	55.6	53.8	52.0	50.0	50.0
Total Operating Profit	224.2	272.4	309.8	348.1	383.6
Passive Components	34.1	41.8	52.9	63.9	72.1
Sensor Application Products	5.0	20.7	24.3	30.4	33.6
Magnetic Application Products	3.4	27.0	42.9	49.2	56.0
Energy Application Products	234.4	246.7	243.7	258.5	275.9
Other	-4.4	-10.2	-4.0	-4.0	-4.0
Corporate and Eliminations	-48.3	-53.6	-50.0	-50.0	-50.0
Profit Margin	10.2%	10.9%	11.7%	12.1%	12.3%
Passive Components	6.1%	7.1%	8.5%	9.8%	10.6%
Sensor Application Products	2.6%	9.2%	10.3%	12.0%	12.4%
Magnetic Application Products	1.5%	10.3%	14.2%	15.6%	17.4%
Energy Application Products	19.9%	18.0%	17.0%	16.1%	15.4%
Other	-8.0%	-19.0%	-7.7%	-8.0%	-8.0%

Risk Reward – TDK (6762.T)

Top Pick

We expect LiB profit growth and continued improvement beyond LiB

PRICE TARGET ¥3,200

Derived from the base case. DCF assumptions: risk-free rate 2.2%, equity beta 1.27, risk premium 3.7%, yielding WACC 6.3%. Zero growth from F3/36.



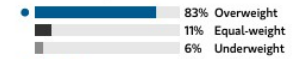
RISK REWARD CHART



OVERWEIGHT THESIS

- We expect Passive Components and Sensor Application earnings to improve from F3/26 after the bottom in F3/25.
- We envision only modest LiB profit in F3/27, but sustained LiB profit growth in the mid-long term as value-added silicon cathode sales expand.
- HDD related business earnings recovery to continue after turning to the black in F3/25.

Consensus Rating Distribution



Risk Reward Themes

New Data Era: *Positive*

View descriptions of Risk Rewards Themes [here](#)

BULL CASE

¥4,000

Derived from DCF

Earnings from rechargeable batteries, passive components, sensors, and HDD heads ahead of base case. OP ¥294.3bn in F3/26, ¥350.9bn in F3/27, ¥409.0bn in F3/28. Same currency and DCF model assumptions as base case. Fair value equates to P/E 29.0x on F3/27e diluted EPS ¥138.0, P/E 24.8x on F3/28e diluted EPS ¥161.0.

BASE CASE

¥3,200

Derived from DCF

Only moderate LiB profit growth in F3/27, but in mid-long term, profit expansion not only for LiB, but also for HDDs, sensors and passive components. OP ¥267.9bn in F3/26, ¥309.8bn in F3/27, ¥348.1bn in F3/28. Assumes ¥150/USD, ¥170/EUR; DCF model assumptions: risk-free rate 2.2%, equity beta 1.27, risk premium 3.7%, yielding WACC 6.3%. Zero growth from F3/36. Fair value equates to P/E 26.2x F3/27e diluted EPS ¥122.1, 23.3x F3/28e diluted EPS ¥137.2.

BEAR CASE

¥1,100

End-F3/27 diluted BPS ¥1,091 x P/B 1.0x

Limited profit growth due to rechargeable battery slowdown. OP ¥168.2bn in F3/26, ¥186.6bn in F3/27, ¥215.5bn in F3/28. Same currency assumptions as base case. ROE 6.4% in F3/26 and 7.0% in F3/27, almost same as the cost of capital of 6.9%, and P/B equates to 1.0x.

Note: e = Morgan Stanley Research estimates; Stock price as of the May 15 close for Risk Reward Chart
Source: Company data, Morgan Stanley Research

Alps Alpine: Key Financial Data and Risk Reward

Income Statement					
(Y bn)	F3/25	F3/26e	F3/27e	F3/28e	F3/29e
Net Sales	990.4	1,019.5	1,082.0	1,179.9	1,246.6
Cost of Goods Sold	815.2	837.6	876.5	943.1	990.5
Gross Profit	175.2	181.9	205.5	236.8	256.1
SG&A Expenses	141.1	139.8	154.7	168.7	178.3
Operating Profit	34.1	42.0	50.8	68.1	77.8
Non-Operating Income/Loss	-3.6	7.1	2.4	2.4	2.4
Pretax profit	57.9	43.9	51.2	68.4	78.2
Income Tax	19.8	11.1	14.8	19.8	22.7
(Effective Tax Rate)	34%	25%	29%	29%	29%
Minority Interest	0.2	0.4	0.0	0.0	0.0
Net Profit	37.8	26.9	36.3	48.6	55.5
Capex	51.8	61.6	70.0	60.0	60.0
Depreciation and Amortization	35.1	34.0	40.0	42.0	42.0
R&D	24.3	21.9	30.0	33.0	35.0
EBITDA	69.2	76.0	90.8	110.1	119.8
ROE	9.4%	6.2%	8.7%	11.0%	11.6%
Average Share Outstanding (Diluted)	205.7	199.5	197.0	194.2	191.3
EPS (Diluted, Yen)	184.0	134.8	184.47	250.27	290.23
BPS (Yen)	2,013.1	2,293.7	2,169.4	2,365.8	2,605.2
DPS (Yen)	60.0	62.0	60.0	60.0	60.0
USD/JPY: Average Exchange Rate	151.3	150.8	150.0	150.0	150.0
YoY % Change					
Net Sales	3%	3%	7%	9%	6%
Operating Profit	73%	23%	36%	34%	14%
Net Profit	-	-29%	41%	34%	14%
As % of Sales					
Gross Profit	17.7%	17.6%	19.0%	20.1%	20.5%
Operating Profit	3.4%	4.1%	4.7%	5.8%	6.2%
Net Profit	3.8%	2.6%	3.4%	4.1%	4.5%
Capex	5.2%	6.0%	6.5%	5.1%	4.8%
Depreciation and Amortization	3.5%	3.3%	3.7%	3.6%	3.4%
Balance Sheet					
(Y bn)	F3/25	F3/26e	F3/27e	F3/28e	F3/29e
Total Assets	740.7	763.2	762.8	802.3	846.7
Total Current Assets	494.9	500.7	455.8	477.3	503.7
Cash and Cash Equivalents	147.9	153.6	74.0	67.1	73.7
Marketable Securities	0.0	0.0	0.0	0.0	0.0
Account Receivables	166.8	169.4	177.8	193.8	204.8
Inventory	138.6	138.3	162.5	174.9	183.7
Other Current Assets	41.5	41.5	41.5	41.5	41.5
Total Long Term Assets	245.8	262.4	307.0	325.0	343.0
Net PP&E	136.7	158.0	197.9	215.9	233.9
Investments in Securities	60.0	67.7	60.0	60.0	60.0
Other assets	49.1	56.8	49.1	49.1	49.1
Total Current Liabilities	226.9	240.7	235.6	243.1	248.5
ST Debt & Current Portion of LT Debt	44.3	37.1	44.3	44.3	44.3
Accounts Payable	90.4	93.2	99.1	106.6	112.0
Other Current Liabilities	92.2	92.2	92.2	92.2	92.2
Total Long Term Liabilities	98.3	93.0	98.3	98.3	98.3
Long Term Debt	56.5	55.2	56.5	56.5	56.5
Other Long Term Liabilities	41.8	41.8	41.8	41.8	41.8
Minority Interests	1.5	1.9	1.5	1.5	1.5
Total Equity	415.5	449.4	428.9	460.8	499.9
Net Debt	-107.2	-129.0	-33.3	-26.3	0.0

Cash Flow Statement					
(Y bn)	F3/25	F3/26e	F3/27e	F3/28e	F3/29e
Net Cash from Operations	65.8	95.9	61.3	69.7	83.1
Net Income	37.5	26.9	36.3	48.6	55.5
Depreciation and Amortization	35.1	34.0	40.0	42.0	42.0
Changes in Working Capital:	9.0	16.0	-15.0	-20.9	-14.4
Other-net	-16.2	19.1	0.0	0.0	0.0
Net Cash from Investing Activities	-1.7	-58.4	-70.0	-60.0	-60.0
Purchase of PP&E	-43.0	-43.6	-70.0	-60.0	-60.0
Sale/Purchase of Investments	0.0	0.0	0.0	0.0	0.0
Other-net	41.4	-14.8	0.0	0.0	0.0
Free cash flow	64.1	37.5	-8.7	9.7	23.1
Net Cash from Financing Activities	-37.3	-41.1	-16.8	-16.6	-16.5
Change in Interest-bearing Debt	-28.5	-9.5	0.0	0.0	0.0
Change in Treasury Stock	0.0	0.0	-5.0	-5.0	-5.0
Payment of Cash Dividends	-9.2	-12.2	-11.8	-11.6	-11.5
Other-net	-0.5	-19.4	0.0	0.0	0.0
Net Effect of Exchange Rate Changes	-1.7	9.5	0.0	0.0	0.0
Net Change in Cash	25.2	5.9	-25.5	-7.0	6.7

Segment Information					
(Y bn)	F3/25	F3/26e	F3/27e	F3/28e	F3/29e
Net Sales	990.4	1,019.5	1,082.0	1,179.9	1,246.6
Components	348.0	358.4	422.6	479.9	502.1
Sensor & Communication	84.2	85.3	90.5	95.0	99.8
Mobility	537.2	555.1	548.9	584.9	624.6
Logistics/Others	21.0	20.8	20.0	20.0	20.0
Total Operating Profit	34.1	42.0	50.8	68.1	77.8
Components	30.4	30.2	41.8	56.1	64.8
Sensor & Communication	-3.4	-3.5	-2.0	0.0	1.0
Mobility	5.6	14.2	10.0	11.0	11.0
Logistics/Others	1.5	1.3	1.0	1.0	1.0
Operating Profit Margin	3.4%	4.1%	4.7%	5.8%	6.2%
Components	8.7%	8.4%	9.9%	11.7%	12.9%
Sensor & Communication	-4.0%	-4.1%	-2.2%	0.0%	1.0%
Mobility	1.0%	2.6%	1.8%	1.9%	1.8%
Logistics/Others	7.0%	6.0%	5.0%	5.0%	5.0%

Risk Reward – ALPS ALPINE (6770.T)

We forecast strong earnings growth above market consensus, but below the plan

PRICE TARGET ¥2,900

Derived from the base case. DCF model assumptions: risk-free rate 2.2%, equity beta 0.89, risk premium 3.7%, yielding WACC 5.0%. Zero growth from F3/36.



RISK REWARD CHART



OVERWEIGHT THESIS

- We expect a significant expansion in profits from smartphone camera actuators and components starting in F3/27.
- We expect only limited improvement in performance for the Module & System and Mobility segments.
- We expect continued earnings expansion that falls short of the mid-term plan's final-year (F3/28) OP target of ¥71.0bn, but significantly exceeds market consensus.

Consensus Rating Distribution



Risk Reward Themes

Contrarian: Positive

View descriptions of Risk Rewards Themes [here](#)

BULL CASE	¥4,000	BASE CASE	¥2,900	BEAR CASE	¥1,400
Derived from DCF					
Earnings improve at a pace exceeding the mid-term management plan. OP ¥41.9bn in F3/26, ¥69.3bn in F3/27, ¥91.6bn in F3/28, ¥101.3bn in F3/29. Same currency and DCF model assumptions as base case. Fair value equates to P/E 16.6x F3/27 diluted EPS ¥240.6, and 12.5x F3/28 diluted EPS ¥320.3, P/B 1.8x end-F3/27e diluted BPS of ¥227.3.					
Derived from DCF					
Although below the mid-term plan, we expect continued strong profit growth driven by camera actuators. We forecast OP ¥37.3bn in F3/26, ¥50.8bn in F3/27, ¥68.1bn in F3/28, ¥77.8bn in F3/29. Base case assumes ¥150/USD. Fair value equates to P/E 15.7x F3/27e diluted EPS ¥184.5 and 11.6x F3/28e EPS ¥250.3.					
F3/27e diluted BPS ¥1,995 x P/B 07x					
Sluggish output of high-end smartphones and declining market share cause smartphone camera actuator sales to fall 10% short of our base case. OP ¥31.4bn in F3/26, ¥25.7bn in F3/27, ¥25.7bn in F3/28, ¥27.7bn in F3/29. ROE 4.5% in F3/27, lower than the 5.5% cost of capital we assume, and the share drops to 0.7x of F3/27e diluted BPS.					

Note: e = Morgan Stanley Research estimates; Stock price as of the May 15 close for Risk Reward Chart
Source: Company data, Morgan Stanley Research

Nihon Dempa Kogyo: Key Financial Data and Risk Reward

Income Statement						Cashflow Statement					
(JPY bn)						(JPY bn)					
	F3/25	F3/26	F3/27e	F3/28e	F3/29e		F3/25	F3/26	F3/27e	F3/28e	F3/29e
Sales	53.1	54.6	56.1	58.0	59.8	Cash flow from operating activities	6.1	4.2	6.4	7.4	7.9
Costs of Sales	37.0	38.9	39.3	40.8	42.2	EBIT	3.0	2.6	3.3	3.9	3.9
Gross Profit	16.1	15.7	16.8	17.2	17.6	Depreciation	3.4	3.9	4.4	4.8	5.2
SG&A (incl. other income/losses)	11.4	12.4	13.0	12.8	13.2	Changes in Receivables	0.2	-0.5	-0.6	-0.5	-0.4
Operating Income	4.6	3.4	3.8	4.4	4.4	Changes in Inventory	0.6	-1.5	-0.4	-0.2	-0.2
Financial income/losses	-1.7	-0.8	-0.5	-0.5	-0.5	Changes in Payables	0.0	1.2	0.5	0.3	0.3
Profit before income taxes	3.0	2.6	3.3	3.9	3.9	Other	-1.1	-1.5	-0.7	-0.9	-0.9
Income taxes	1.2	0.6	1.0	1.2	1.2	Cash flow from investing activities	-4.5	-7.4	-6.8	-6.8	-6.5
(Effective tax rate)	39%	24%	30%	31%	31%	Purchase of PP&E	-3.7	-5.7	-6.5	-6.5	-6.2
Profit attributable to non-controlling interests	0.0	0.0	0.0	0.0	0.0	Sales of PP&E	0.1	0.0	0.0	0.0	0.0
Net Profit	1.8	2.1	2.3	2.7	2.7	Other	-0.9	-1.7	-0.3	-0.3	-0.3
Capex	4.6	7.4	6.3	6.3	6.0	Cash flow from financing activities	1.9	-2.4	-0.2	-2.3	-1.5
Depreciation & amortization	3.4	3.9	4.4	4.8	5.2	Net Change In Borrowings	3.3	-0.7	0.5	-1.4	-0.6
R&D	2.1	2.9	2.9	2.4	2.5	Purchase of treasury shares	0.0	-0.2	0.0	0.0	0.0
EPS (JPY)	77.8	89.7	100.0	117.3	117.3	Cash dividend paid	-0.7	-0.7	-0.7	-0.9	-0.9
USD/JPY (Average)	152.5	151.0	150.0	150.0	150.0	Other	-0.7	-0.9	0.0	0.0	0.0
						Effect of exchange rate change	0.0	0.5	0.0	0.0	0.0
						Changes in cash and cash equivalents	3.6	-5.1	-0.6	-1.8	-0.2
						Cash and cash equivalents at end of the period	15.9	10.8	12.3	10.6	10.4

Balance Sheet					
(JPY bn)					
	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Total assets	71.5	76.3	75.3	76.0	77.5
Current assets	43.5	44.0	41.9	40.8	41.3
Cash and cash equivalents	15.9	10.8	12.3	10.6	10.4
Accounts receivable	12.7	13.8	13.7	14.1	14.6
Inventories	10.5	12.4	11.5	11.8	12.0
Other financial assets	4.4	6.9	4.4	4.4	4.4
Fixed assets	28.0	32.3	33.3	35.1	36.2
Tangible fixed assets	19.7	23.2	24.8	26.5	27.5
Intangible fixed assets	2.3	3.2	2.5	2.6	2.7
Other non-current assets	6.0	5.9	6.0	6.0	6.0
Current liabilities	12.2	14.0	12.8	13.2	13.5
Short term borrowings	0.8	1.3	0.9	0.9	0.9
Accounts payable	9.5	10.3	10.0	10.3	10.7
Other current liabilities	1.9	2.3	1.9	1.9	1.9
Fixed liabilities	30.2	30.4	30.4	29.0	28.4
Long term borrowings	25.8	24.8	26.0	24.6	24.0
Other fixed liabilities	4.4	5.6	4.4	4.4	4.4
Net assets	29.2	31.9	32.1	33.8	35.6
Total liabilities and equity	71.5	76.3	75.3	76.0	77.5

Risk Reward – Nihon Dempa Kogyo (6779.T)

Ensuring profitability of DC, server products is key

PRICE TARGET ¥1,530

Our price target is ¥1,530, based on a DCF valuation (risk-free rate 2.2%; beta 1.2; risk premium 3.7%; WACC 4.6%); beyond our 3-year base earnings models, we also build forecasts for F3/29–30 into our DCF-based valuation, and our assumptions for F3/31 onward are that the firm maintains FCF/share of ¥125; our price target equates to 15.3x our F3/27E EPS of ¥100.0, and our forex rate assumption for the next quarter and thereafter is ¥150/\$.

RISK REWARD CHART



BULL CASE	¥2,530	BASE CASE	¥1,530	BEAR CASE	¥340
Accelerated sales growth for industrials/defence		Machinery/defence up but high mob comm competition		Weak demand for automotive, NA smartphones	
OP and EPS forecasts of ¥3.2bn/¥85.4 in F3/26, ¥4.0bn/¥104.3 in F3/27, ¥4.8bn/¥130.4 in F3/28, ¥5.6bn/¥152.1 in F3/29, and ¥6.6bn/¥182.5 in F3/30. Our calculations also assume that from F3/31 onward the firm can maintain FCF/share at ¥185. The fair value equates to 24.3x our F3/27E EPS of ¥104.3 and 1.8x BPS of ¥1,403.2.		OP and EPS forecasts of ¥3.2bn/¥85.4 in F3/26, ¥3.8bn/¥100.0 in F3/27, ¥4.4bn/¥117.3 in F3/28, ¥4.4bn/¥117.3 in F3/29, and ¥4.5bn/¥121.7 in F3/30. Our calculations also assume that from F3/31 onward the firm can maintain FCF/share at ¥125. Our PT equates to 15.3x our F3/27E EPS of ¥100.0 and 1.1x BPS of ¥1,398.8.		OP and EPS forecasts of ¥3.2bn/¥85.4 in F3/26, ¥2.9bn/¥73.9 in F3/27, ¥2.7bn/¥65.2 in F3/28, ¥2.3bn/¥56.5 in F3/29, and ¥2.3bn/¥56.5 in F3/30. Our calculations also assume that from F3/31 onward the firm can maintain FCF/share at ¥55. The fair value equates to 4.6x our F3/27E EPS of ¥73.9 and 0.2x BPS of ¥1,372.8.	

Note: e = Morgan Stanley Research estimates; Stock price as of the May 15 close for Risk Reward Chart
Source: Company data, Morgan Stanley Research

Meiko Electronics: Key Financial Data and Risk Reward

(JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e	(JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Sales	206.8	240.6	305.1	368.7	429.5	Cash flow from operating activities	21.7	27.5	29.2	46.4	63.0
Costs of Sales	167.0	190.1	239.2	284.6	327.6	EBIT	18.5	27.4	31.5	43.1	54.9
Gross Profit	39.8	50.4	65.9	84.1	101.9	Depreciation	12.5	13.7	20.0	26.0	31.5
SG&A	20.7	25.9	32.3	38.6	44.5	Changes in Receivables	-3.3	-8.3	-11.2	-9.5	-8.0
Operating Income	19.1	24.6	33.6	45.5	57.4	Change in inventory	2.9	6.5	9.5	9.1	8.7
Non-operating income/expenses	-0.3	1.9	-2.1	-2.4	-2.5	Changes in Payables	-3.4	-3.3	-6.3	-8.6	-11.0
Recurring Profit	18.8	26.5	31.5	43.1	54.9	Other	-5.5	-8.5	-14.3	-13.7	-13.1
Extraordinary income	-0.2	0.9	0.0	0.0	0.0	Cash flow from investing activities	-24.3	-55.5	-65.0	-65.0	-65.8
Profit before income taxes	18.5	27.4	31.5	43.1	54.9	Purchase of PP&E	-24.4	-49.8	-54.6	-54.6	-55.4
Income taxes	3.4	7.5	6.3	8.6	11.0	Sales of PP&E	0.2	0.0	0.0	0.0	0.0
(Effective tax rate)	19%	27%	20.0%	20.0%	20.0%	Other	-0.1	-5.7	-0.4	-0.4	-0.4
Profit attributable to non-controlling interests	0.2	0.2	0.2	0.2	0.2	Cash flow from financing activities	4.1	30.1	22.7	11.2	-2.6
Net Profit	14.9	19.8	25.0	34.3	43.7	Net Change In Borrowings	7.4	33.1	25.8	15.3	2.5
Capex	29.8	51.0	55.0	55.0	55.8	Purchase of treasury shares	0.0	0.0	0.0	0.0	0.0
Depreciation & amortization	12.5	13.7	20.0	26.0	31.5	Cash dividend paid	-2.4	-2.7	-3.1	-4.1	-5.1
R&D	6.3	7.2	9.2	11.1	12.9	Other	-0.8	-0.3	0.0	0.0	0.0
EPS (JPY)	569.5	758.6	974.1	1,336.5	1,702.8	Effect of exchange rate changes	0.1	1.4	0.0	0.0	0.0
USD/JPY (Average)	152.6	151.1	150.0	150.0	150.0	Changes in cash and cash equivalents	1.6	3.6	-3.1	2.6	4.6
						Cash and cash equivalents at end of the period	22.9	26.5	28.3	30.9	35.5

(JPY bn)	F3/25	F3/26e	F3/27e	F3/28e	F3/29e
Balance Sheet					
Total assets	256.4	335.3	376.2	431.1	481.2
Current assets	114.1	142.5	162.9	189.2	215.4
Cash and cash equivalents	23.2	27.5	28.3	30.9	35.5
Accounts receivable	46.2	56.6	68.1	82.3	95.9
Inventories	38.5	49.5	60.2	69.7	77.7
Other financial assets	6.3	8.9	6.3	6.3	6.3
Fixed assets	142.3	192.8	213.3	241.9	265.8
Tangible fixed assets	129.6	174.9	200.8	229.4	253.3
Intangible fixed assets	5.7	5.0	5.5	5.5	5.5
Other non-current assets	7.0	12.8	7.0	7.0	7.0
Current liabilities	100.3	128.3	134.7	151.5	159.6
Short term borrowings	52.1	63.7	72.0	79.4	78.5
Accounts payable	29.6	39.0	43.7	52.8	61.5
Other current liabilities	18.6	25.6	19.0	19.3	19.6
Fixed liabilities	40.5	63.2	86.1	93.9	97.4
Long term borrowings	34.7	57.6	80.3	88.1	91.6
Other fixed liabilities	5.8	5.5	5.8	5.8	5.8
Net assets	115.6	143.8	155.5	185.7	224.2
Total liabilities and equity	256.4	335.3	376.2	431.1	481.2

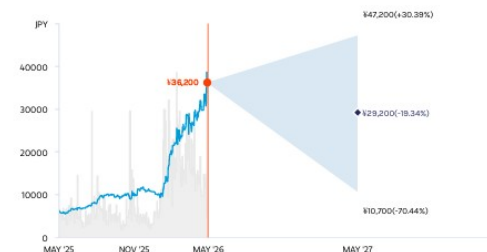
Risk Reward – Meiko Electronics (6787.T)

Rapid Earnings Growth Driven by Aggressive Investment Largely Priced in

PRICE TARGET ¥29,200

Our price target is ¥29,200, based on a DCF valuation (risk-free rate 2.2%; beta 1.5; risk premium 3.7%; WACC 4.9%). Beyond our 3-year base earnings models, we also build forecasts for F3/29–30 into our DCF-based valuation. Our assumption from F3/31 onward is that the firm maintains FCF/share of ¥2,300. Our price target equates to 30.0x our F3/27e EPS of ¥974.1. Our forex rate assumption for the next quarter and thereafter is ¥150/¥.

RISK REWARD CHART



Key: — Historical Stock Performance • Current Stock Price ◆ Price Target
Source: Refinitiv, Morgan Stanley Research

EQUAL-WEIGHT THESIS

Our scenario has the firm benefitting from production capacity enhancements undertaken to meet growing demand. We evaluate positively its proactive stance on expanding production capacity for build-up substrates, including those for satellite communications and high-end smartphones. In our view, Meiko's expertise in producing build-up substrates in Vietnam will enable it to benefit from growing demand for high layer/density build-up PCBs, as well as from efforts to move supply chains out of Greater China.

Risk Reward Themes

Secular Growth: Positive
View descriptions of Risk Rewards Themes [here](#)

BULL CASE	¥47,200	BASE CASE	¥29,200	BEAR CASE	¥10,700
Faster demand growth in buildup/semi PKG substrate		Continued growth in high layer/density buildup PCB		N. Am handset project stop, slowing sales for auto	
OP and EPS forecasts of ¥25.1bn/¥728.8 in F3/26, ¥37.2bn/¥1,079.4 in F3/27, ¥57.3bn/¥1,691.1 in F3/28, ¥77.1bn/¥2,299.0 in F3/29, and ¥98.1bn/¥2,953.6 in F3/30. Our calculations also assume that from F3/31 onward the firm can maintain FCF/share at ¥3,500.		OP and EPS forecasts of ¥25.1bn/¥760.0 in F3/26, ¥33.6bn/¥974.1 in F3/27, ¥45.5bn/¥1,336.5 in F3/28, ¥57.4bn/¥1,702.8 in F3/29, and ¥68.1bn/¥2,041.8 in F3/30. Our calculations also assume that from F3/31 onward the firm can maintain FCF/share at ¥2,300.		OP and EPS forecasts of ¥25.1bn/¥728.8 in F3/26, ¥27.6bn/¥794.9 in F3/27, ¥29.1bn/¥841.7 in F3/28, ¥30.1bn/¥884.5 in F3/29, and ¥32.1bn/¥950.8 in F3/30. Our calculations also assume that from F3/31 onward the firm can maintain FCF/share at ¥1,050.	
The fair value equates to 43.7x our F3/27e EPS of ¥1,079.4 and 8.1x BPS of ¥5,844.9.		Our PT equates to 30.0x our F3/27e EPS of ¥974.1 and 5.1x BPS of ¥5,770.9.		The fair value equates to 13.5x our F3/27e EPS of ¥794.9 and 1.9x BPS of ¥5,560.5.	

Note: e = Morgan Stanley Research estimates; Stock price as of the May 15 close for Risk Reward Chart
Source: Company data, Morgan Stanley Research

Hirose Electric: Key Financial Data and Risk Reward

Income Statement (JPY bn)						Cashflow Statement (JPY bn)					
	F3/25	F3/26	F3/27e	F3/28e	F3/29e		F3/25	F3/26	F3/27e	F3/28e	F3/29e
Sales	189.4	211.3	224.2	238.6	254.2	Cash flow from operating activities	55.7	46.9	53.7	57.8	62.0
Costs of Sales	104.0	122.3	128.6	137.3	145.8	EBIT	46.2	46.6	50.2	53.2	57.1
Gross Profit	85.4	89.0	95.6	101.3	108.4	Depreciation	17.9	19.5	22.0	24.2	26.0
SG&A (incl. other income/losses)	42.7	46.0	49.0	51.7	54.9	Changes in Receivables	-1.9	-7.2	-3.2	-3.1	-3.3
Operating Income	42.7	43.0	46.6	49.6	53.5	Changes in inventory	-1.4	-3.5	-1.3	-1.2	-1.2
Financial income/losses	3.5	3.6	3.6	3.6	3.6	Changes in Payables	2.0	3.5	1.0	1.0	1.1
Profit before income taxes	46.2	46.6	50.2	53.2	57.1	Other	-7.2	-13.0	-15.0	-16.3	-17.6
Income taxes	13.2	13.5	15.4	16.3	17.5	Cash flow from investing activities	-42.9	-9.2	-28.0	-28.6	-30.5
(Effective tax rate)	29%	29%	31%	31%	31%	Purchase of PP&E	-26.7	-20.0	-28.0	-28.6	-30.5
Profit attributable to non-controlling interests	0.0	0.0	0.0	0.0	0.0	Sales of PP&E	0.0	0.0	0.0	0.0	0.0
Net Profit	33.0	33.1	34.8	36.9	39.6	Other	-16.3	10.8	0.0	0.0	0.0
Capex	26.7	20.0	28.0	28.6	30.5	Cash flow from financing activities	-16.7	-37.9	-38.1	-31.2	-31.7
Depreciation & amortization	17.9	19.5	22.0	24.2	26.0	Net Change In Borrowings	0.0	0.0	0.0	0.0	0.0
R&D	13.5	13.5	15.9	16.9	18.1	Purchase of treasury shares	0.0	-20.4	-22.5	-15.0	-15.0
EPS (JPY)	978.3	991.9	1,096.5	1,198.4	1,315.5	Cash dividend paid	-15.7	-16.5	-15.6	-16.2	-16.7
USD/JPY (Average)	152.6	-	150.0	150.0	150.0	Other	-0.9	-1.1	0.0	0.0	0.0
						Effect of exchange rate change	-0.7	3.0	0.1	0.0	0.0
						Changes in cash and cash equivalents	-4.7	1.7	-12.3	-2.0	-0.2
						Cash and cash equivalents at end of the period	85.7	87.3	58.4	56.5	56.2

Balance Sheet (JPY bn)						Sales by Applications (JPY bn)					
	F3/25	F3/26	F3/27e	F3/28e	F3/29e		F3/25	F3/26	F3/27e	F3/28e	F3/29e
Total assets	416.9	431.2	412.8	419.5	428.3	General Industrial	46.9	62.0	68.4	73.9	79.8
Current assets	248.8	233.8	232.1	234.4	238.7	Smartphone	46.4	44.3	43.8	45.2	46.5
Cash and cash equivalents	85.7	87.3	58.4	56.5	56.2	Consumer/Mobile equipment	34.7	34.7	33.3	34.0	34.7
Marketable securities	87.6	57.6	87.6	87.6	87.6	Automotive/Mobility related	49.2	54.3	60.1	66.1	72.7
Accounts receivable	40.4	49.9	47.8	50.9	54.2	Telecommunications infrastructure equipment	5.4	6.7	9.3	10.3	11.3
Inventories	25.1	30.2	28.3	29.4	30.6	OA equipment and Others	6.8	9.3	9.2	9.2	9.2
Other financial assets	97.6	68.4	97.6	97.6	97.6	(YoY)					
Fixed assets	168.1	197.4	180.7	185.1	189.6	General Industrial	-7%	32%	11%	8%	8%
Tangible fixed assets	91.3	96.3	103.9	108.3	112.8	Smartphone	31%	-5%	2%	3%	3%
Intangible fixed assets	6.6	7.7	6.6	6.6	6.6	Consumer/Mobile equipment	37%	0%	-3%	2%	2%
Other non-current assets	70.2	93.4	70.2	70.2	70.2	Automotive/Mobility related	15%	10%	10%	10%	10%
						Telecommunications infrastructure equipment	20%	24%	43%	10%	10%
						OA equipment and Others	0%	36%	0%	0%	0%
Current liabilities	29.1	32.8	28.7	29.6	30.6						
Short term borrowings	0.0	0.0	0.0	0.0	0.0						
Accounts payable	12.9	16.7	15.3	16.2	17.3						
Other current liabilities	16.2	16.1	13.4	13.4	13.3						
Fixed liabilities	17.7	20.3	17.7	17.7	17.7						
Long term borrowings	0.0	0.0	0.0	0.0	0.0						
Other fixed liabilities	17.7	20.3	17.7	17.7	17.7						
Net assets	370.1	378.1	366.5	372.2	380.1						
Total liabilities and equity	416.9	431.2	412.8	419.5	428.3						

Risk Reward – Hirose Electric (6806.T)

Sustained high profitability and strong sales growth

PRICE TARGET ¥26,300

Our price target is ¥26,300, based on a DCF valuation (risk-free rate 2.2%; beta 0.9; risk premium 3.7%; WACC 5.2%). Beyond our three-year forecasts, we also build forecasts for F3/29–30 into our DCF-based valuation. Our assumptions for F3/31 onward are that the firm maintains FCF/share of ¥1,450. Our price target equates to 24.0x our F3/27e EPS of ¥1,096.5. Our forex rate assumption for the next quarter and thereafter is ¥150/\$.

Consensus Price Target Distribution



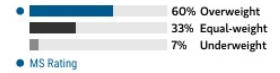
OVERWEIGHT THESIS

The equity story is the continued top-line growth across applications through the acquisition of high-margin projects. We like the continued high level of sales growth relative to historical levels, while maintaining a high-profitability structure enabled by its semi-fabless production model. We believe that if the equity market comes to view profit growth as a mid-term trend, there is further upside to valuations.

RISK REWARD CHART



Consensus Rating Distribution



Risk Reward Themes

Secular Growth: Positive
Technology Diffusion: Positive
View descriptions of Risk Rewards Themes [here](#)

BULL CASE	¥31,400	BASE CASE	¥26,300	BEAR CASE	¥15,600
Accelerated profit growth through new project wins					
OP and EPS forecasts of ¥42.5bn/¥952.1 in F3/26, ¥50.0bn/¥1,172.1 in F3/27, ¥56.4bn/¥1,351.0 in F3/28, ¥63.9bn/¥1,554.7 in F3/29, and ¥72.8bn/¥1,801.8 in F3/30. Our calculations also assume that from F3/31 onward the firm can maintain FCF/share at ¥1,800. The fair value equates to 26.8x our F3/27e EPS of ¥1,172.1 and 2.6x BPS of ¥11,913.2.					
Sales growth continues for industrial/automotive					
OP and EPS forecasts of ¥42.5bn/¥952.1 in F3/26, ¥46.6bn/¥1,096.5 in F3/27, ¥49.6bn/¥1,198.4 in F3/28, ¥53.5bn/¥1,315.5 in F3/29, and ¥58.0bn/¥1,455.1 in F3/30. Our calculations also assume that from F3/31 onward the firm can maintain FCF/share at ¥1,450. Our PT equates to 24.0x our F3/27e EPS of ¥1,096.5 and 2.2x BPS of ¥11,835.7.					
Lower NA phone share, slower growth of ind. sales					
OP and EPS forecasts of ¥42.5bn/¥952.1 in F3/26, ¥40.6bn/¥967.3 in F3/27, ¥38.4bn/¥945.1 in F3/28, ¥36.9bn/¥933.5 in F3/29, and ¥35.6bn/¥924.7 in F3/30. Our calculations also assume that from F3/31 onward the firm can maintain FCF/share at ¥700. The fair value equates to 16.1x our F3/27e EPS of ¥967.3 and 1.3x BPS of ¥11,703.3.					

Note: e = Morgan Stanley Research estimates; Stock price as of the May 15 close for Risk Reward Chart
Source: Company data, Morgan Stanley Research

Japan Aviation Electronics: Key Financial Data and Risk Reward

Income Statement (JPY bn)						Cashflow Statement (JPY bn)					
	F3/25	F3/26	F3/27e	F3/28e	F3/29e		F3/25	F3/26	F3/27e	F3/28e	F3/29e
Sales	221.6	227.9	231.4	241.4	252.0	Cash flow from operating activities	36.3	17.0	26.8	27.8	28.8
Costs of Sales	179.6	191.4	193.0	200.9	209.6	EBIT	15.2	10.2	10.5	11.8	12.7
Gross Profit	42.1	36.4	38.4	40.5	42.4	Depreciation	20.9	19.7	19.7	20.2	20.7
SG&A	26.4	27.5	28.1	29.0	30.0	Changes in Receivables	3.8	-0.1	-1.2	-1.7	-1.8
Operating Income	15.6	8.9	10.3	11.5	12.4	Change in inventory	-1.0	-2.6	-0.3	-0.7	-0.7
Non-operating income/expenses	-0.8	-0.7	0.2	0.3	0.3	Changes in Payables	1.6	-2.9	0.9	1.3	1.4
Recurring Profit	14.8	8.2	10.5	11.8	12.7	Other	-4.0	-7.2	-2.7	-3.1	-3.4
Extraordinary income	0.3	1.9	0.0	0.0	0.0	Cash flow from investing activities	-18.2	-24.4	-21.0	-21.9	-22.9
Profit before income taxes	15.2	10.2	10.5	11.8	12.7	Purchase of PP&E	-18.0	-23.7	-20.8	-21.7	-22.7
Income taxes	3.6	3.1	3.2	3.6	3.9	Sales of PP&E	0.4	0.0	0.0	0.0	0.0
(Effective tax rate)	24%	30%	30%	31%	31%	Other	-1.5	-0.7	-0.2	-0.2	-0.2
Profit attributable to non-controlling interests	0.0	0.0	0.0	0.0	0.0	Cash flow from financing activities	-31.6	0.5	-7.3	-5.7	-5.2
Net Profit	11.6	7.1	7.3	8.2	8.8	Net Change In Borrowings	-27.5	4.8	-3.3	-1.6	-1.1
Capex	18.0	23.7	20.8	21.7	22.7	Purchase of treasury shares	0.0	0.0	0.0	0.0	0.0
Depreciation & amortization	20.9	19.7	19.7	20.2	20.7	Cash dividend paid	-4.0	-4.0	-4.0	-4.0	-4.0
R&D	11.3	12.1	11.8	12.3	12.9	Other	0.0	-0.3	0.0	0.0	0.0
EPS (JPY)	172.0	104.9	108.3	121.6	130.5	Effect of exchange rate changes	-1.0	2.6	0.2	0.0	0.0
USD/JPY (Average)	152.6	150.7	150.0	150.0	150.0	Changes in cash and cash equivalents	-15.4	-4.4	-1.2	0.2	0.7
						Cash and cash equivalents at end of the period	52.9	48.5	50.7	50.9	51.6

Balance Sheet (JPY bn)						Sales by Application/Product (JPY bn)					
	F3/25	F3/26	F3/27e	F3/28e	F3/29e		F3/25	F3/26	F3/27e	F3/28e	F3/29e
Total assets	215.4	232.1	219.4	223.3	228.3	Connector	192.8	199.2	203.0	211.1	219.8
Current assets	125.3	130.7	126.7	129.2	132.5	Handhelds	66.9	65.2	65.2	66.6	68.1
Cash and cash equivalents	52.9	48.5	50.7	50.9	51.6	Automotive	107.1	111.0	113.8	119.4	125.4
Marketable securities	0.0	0.0	0.0	0.0	0.0	Industrial & Infrastructure	14.5	17.6	18.7	19.9	21.0
Accounts receivable	37.7	38.9	39.3	41.0	42.8	Others	4.3	5.4	5.3	5.3	5.3
Inventories	29.2	32.8	31.1	31.7	32.4	Aerospace	19.3	20.5	20.5	22.2	24.1
Other financial assets	5.6	10.4	5.6	5.6	5.6	Automotive	0.2	0.1	0.1	0.1	0.1
Fixed assets	90.2	101.5	92.8	94.0	95.9	Industrial & Infrastructure	10.7	9.3	9.5	10.1	10.7
Tangible fixed assets	70.3	77.6	73.4	74.9	76.9	Aviation & Space (Defence)	8.4	11.1	10.9	12.0	13.2
Intangible fixed assets	2.7	3.1	2.2	1.9	1.7	UIS	9.0	7.7	7.4	7.6	7.7
Other non-current assets	17.2	20.8	17.2	17.2	17.2	Automotive	5.5	4.6	4.5	4.5	4.5
Current liabilities	48.9	52.9	50.6	52.1	53.7	Industrial & Infrastructure	3.5	3.0	2.9	3.0	3.1
Short term borrowings	6.0	11.0	6.4	6.5	6.7	Others	0.5	0.5	0.5	0.5	0.5
Accounts payable	28.4	25.5	29.6	30.9	32.3	Connector	-1%	3%	3%	4%	4%
Other current liabilities	14.6	16.4	14.6	14.7	14.7	Handhelds	-11%	-2%	0%	2%	2%
Fixed liabilities	32.8	34.9	27.9	26.1	24.8	Automotive	7%	4%	3%	5%	5%
Long term borrowings	31.0	30.8	26.1	24.3	23.0	Industrial & Infrastructure	-2%	21%	8%	6%	6%
Other fixed liabilities	1.8	4.1	1.8	1.8	1.8	Others	-1%	24%	0%	0%	0%
Net assets	133.7	144.4	140.9	145.1	149.8	Aerospace	-4%	6%	9%	8%	8%
Total liabilities and equity	215.4	232.1	219.4	223.3	228.3	Automotive	-79%	-56%	1%	0%	0%
						Industrial & Infrastructure	-22%	-13%	8%	6%	6%
						Aviation & Space (Defence)	55%	31%	10%	10%	10%
						UIS	-11%	-14%	3%	2%	2%
						Automotive	7%	-15%	1%	0%	0%
						Industrial & Infrastructure	-30%	-14%	8%	4%	4%
						Others	-29%	-2%	0%	0%	0%

Risk Reward – Japan Aviation Electronics Industry (6807.T)

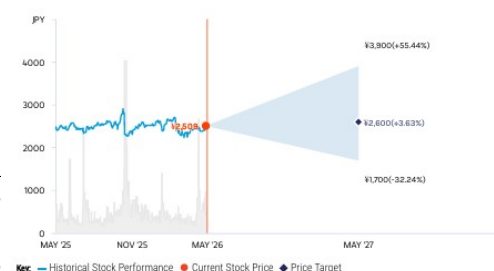
Key to profitability recovery is regaining North American smartphone share

PRICE TARGET ¥2,600

Our price target is ¥2,600, based on a DCF valuation (risk-free rate 2.2%; beta 1.0; risk premium 3.7%; WACC 4.8%). Beyond our three-year forecasts, we also build forecasts for F3/29–30 into our DCF-based valuation. Our assumptions for F3/31 onward are that the firm maintains FCF/share of ¥130. Our price target equates to 24.0x our F3/27e EPS of ¥108.3. Our forex rate assumption for the next quarter and thereafter is ¥150/¥.



RISK REWARD CHART



Source: Refinitiv, Morgan Stanley Research

BULL CASE ¥3,900 BASE CASE ¥2,600 BEAR CASE ¥1,700

Growth accelerates led by auto/phone connectors

OP and EPS forecasts of ¥9.0bn/¥78.8 in F3/26, ¥11.9bn/¥124.6 in F3/27, ¥14.2bn/¥148.3 in F3/28, ¥16.1bn/¥169.1 in F3/29, and ¥18.4bn/¥192.8 in F3/30. Our calculations also assume that from F3/31 onward the firm can maintain FCF/share at ¥210. The fair value equates to 31.3x our F3/27e EPS of ¥124.6 and 1.9x BPS of ¥2,106.6.

Sales grow, but midterm margin flat

OP and EPS forecasts of ¥9.0bn/¥78.8 in F3/26, ¥10.3bn/¥108.3 in F3/27, ¥11.5bn/¥121.6 in F3/28, ¥12.4bn/¥130.5 in F3/29, and ¥12.7bn/¥133.5 in F3/30. Our calculations also assume that from F3/31 onward the firm can maintain FCF/share at ¥130. Our PT equates to 24.0x our F3/27e EPS of ¥108.3 and 1.2x BPS of ¥2,090.3.

OP down on weak auto and smartphone sales

OP and EPS forecasts of ¥9.0bn/¥78.8 in F3/26, ¥8.3bn/¥89.0 in F3/27, ¥7.9bn/¥84.5 in F3/28, ¥7.6bn/¥81.6 in F3/29, and ¥7.0bn/¥75.6 in F3/30. Our calculations also assume that from F3/31 onward the firm can maintain FCF/share at ¥75. The fair value equates to 19.1x our F3/27e EPS of ¥89.0 and 0.8x BPS of ¥2,071.0.

EQUAL-WEIGHT THESIS

The equity story centers on margin improvement driven by a recovery in North American smartphone connector share. Top-line growth is expected to continue, driven not only by increased sales of wire harnesses for ADAS cameras but also by connectors for BEV chargers and expanded connector sales in India. However, for the stock to rise, we believe profitability must improve through a rebound in North American B-to-B connector market share and expansion of the relatively high-margin aerospace business.

Consensus Rating Distribution



Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Secular Growth: Negative

View descriptions of Risk Rewards Themes [here](#)

Note: e = Morgan Stanley Research estimates; Stock price as of the May 15 close for Risk Reward Chart

Source: Company data, Morgan Stanley Research

CMK: Key Financial Data and Risk Reward

Income Statement						Cashflow Statement					
(JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e	(JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Sales	95.5	100.2	104.9	109.9	115.3	Cash flow from operating activities	9.1	7.8	9.3	10.5	11.3
Costs of Sales	79.9	85.3	89.2	92.6	96.3	EBIT	5.3	5.3	2.4	3.5	4.4
Gross Profit	15.6	14.9	15.7	17.3	19.0	Depreciation	5.9	6.5	8.2	8.5	8.7
SG&A	11.8	12.1	12.8	13.4	14.1	Changes in Receivables	-0.2	0.5	-0.7	-0.6	-0.7
Operating Income	3.8	2.8	2.9	3.9	4.9	Change in Inventory	0.7	0.3	0.8	0.7	0.8
Nonoperating income/expenses	1.7	1.3	-0.5	-0.4	-0.5	Changes in Payables	-0.8	-1.0	-0.7	-1.1	-1.3
Recurring Profit	5.5	4.1	2.4	3.5	4.4	Other	-2.0	-3.8	-0.6	-0.5	-0.6
Extraordinary income	-0.2	1.2	0.0	0.0	0.0	Cash flow from investing activities	-18.8	-6.9	-6.5	-6.0	-6.3
Profit before income taxes	5.3	5.3	2.4	3.5	4.4	Purchase of PP&E	-18.3	-8.6	-6.3	-5.8	-6.1
Income taxes	1.4	1.9	0.7	1.1	1.3	Sales of PP&E	0.3	0.0	0.0	0.0	0.0
(Effective tax rate)	27%	36%	29%	31%	30%	Other	-0.8	1.8	-0.2	-0.2	-0.2
Profit attributable to non-controlling interests	0.1	0.7	0.1	0.1	0.1	Cash flow from financing activities	4.7	-4.1	-5.2	-3.3	0.3
Net Profit	3.8	4.0	1.6	2.3	3.0	Net Change in Borrowings	8.2	-1.8	-3.7	-1.8	1.8
						Purchase of treasury shares	0.0	0.0	0.0	0.0	0.0
Capex	18.9	9.2	6.5	6.0	6.3	Cash dividend paid	-1.4	-1.4	-1.4	-1.4	-1.4
Depreciation & amortization	5.9	6.5	8.2	8.5	8.7	Other	-2.2	-0.9	-0.1	-0.1	-0.1
R&D	0.6	0.5	0.5	0.5	0.6	Effect of exchange rate changes	1.8	0.4	0.4	0.0	0.0
EPS (JPY)	53.2	56.5	22.4	32.3	42.1	Changes in cash and cash equivalents	-3.1	-2.8	-2.0	1.1	5.3
USD/JPY (Average)	151.8	149.8	150.0	150.0	150.0	Cash and cash equivalents at end of the period	22.2	19.4	17.3	18.5	23.8

Balance Sheet						Sales by Product					
(JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e	(JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Total assets	148.5	151.5	146.3	146.4	151.0	Double-sided/Multi-layer	61.1	63.4	63.7	64.9	66.1
Current assets	62.4	61.7	60.7	63.5	70.5	Double-sided	10.0	9.5	9.7	9.7	9.7
Cash and cash equivalents	22.2	19.5	17.3	18.5	23.8	Multi-layer	51.0	53.9	54.0	55.2	56.4
Accounts receivable	17.8	19.9	20.1	21.1	22.1	HDI	30.2	33.0	37.4	41.2	45.4
Inventories	18.6	18.4	19.5	20.2	20.8	Others	4.2	3.8	3.8	3.8	3.8
Other financial assets	3.8	3.8	3.8	3.8	3.8	(YoY)					
Fixed assets	86.1	89.8	85.5	83.0	80.5	Double-sided/Multi-layer	2%	4%	2%	2%	2%
Tangible fixed assets	76.5	80.3	75.6	72.9	70.4	Double-sided	0%	-5%	-2%	0%	0%
Intangible fixed assets	1.0	1.5	1.3	1.3	1.4	Multi-layer	2%	6%	2%	2%	2%
Other non-current assets	8.6	8.1	8.7	8.7	8.7	HDI	16%	9%	14%	10%	10%
						Others	-8%	-10%	-1%	0%	0%
Current liabilities	33.7	38.0	36.1	37.3	38.6						
Short term borrowings	12.8	10.3	13.4	13.6	13.9						
Accounts payable	14.0	12.5	15.4	16.1	16.9						
Other current liabilities	6.9	15.3	7.3	7.6	7.9						
Fixed liabilities	33.4	27.4	30.1	28.1	29.7						
Long term borrowings	31.3	25.3	28.0	26.0	27.6						
Other fixed liabilities	2.1	2.2	2.1	2.1	2.1						
Net assets	81.4	86.0	80.0	81.0	82.6						
Total liabilities and equity	148.5	151.5	146.3	146.4	151.0						

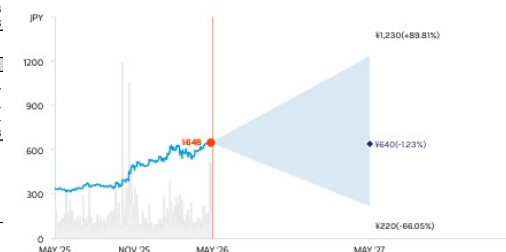
Risk Reward – CMK (6958.T)

Further vehicle electrification and growth in new business areas awaited

PRICE TARGET ¥640

Our price target is ¥640, based on a DCF valuation (risk-free rate 2.2%; beta 1.3; risk premium 3.7%; WACC 5.1%). Beyond our 3-year base earnings models, we also build forecasts for F3/29–30 into our DCF-based valuation, and our assumptions for F3/31 onward are that the firm maintains FCF/share of ¥55. Our price target equates to 28.5x our F3/27e EPS of ¥22.4. Our forex rate assumption for the next quarter and thereafter is ¥150/\$.

RISK REWARD CHART



Key: — Historical Stock Performance • Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

BULL CASE	¥1,230	BASE CASE	¥640	BEAR CASE	¥220
Improves productivity to meet rising demand		BU sales for Integrated ECU in earnest from F3/27		Weak automotive production, slow uptake for ADAS	
OP and EPS forecasts of ¥2.8bn/¥33.8 in F3/26, ¥3.5bn/¥28.1 in F3/27, ¥5.3bn/¥46.3 in F3/28, ¥7.7bn/¥68.7 in F3/29, and ¥10.5bn/¥95.4 in F3/30. Our calculations also assume that from F3/31 onward the firm can maintain FCF/share at ¥95. The fair value equates to 43.8x our F3/27E EPS of ¥28.1 and 1.1x BPS of ¥1,092.3.		OP and EPS forecasts of ¥2.8bn/¥33.8 in F3/26, ¥2.9bn/¥22.4 in F3/27, ¥3.9bn/¥32.3 in F3/28, ¥4.9bn/¥42.1 in F3/29, and ¥5.7bn/¥49.1 in F3/30. Our calculations also assume that from F3/31 onward the firm can maintain FCF/share at ¥55. Our PT equates to 28.5x our F3/27E EPS of ¥22.4 and 0.6x BPS of ¥1,086.7.		OP and EPS forecasts of ¥2.8bn/¥33.8 in F3/26, ¥1.6bn/¥9.8 in F3/27, ¥2.1bn/¥15.4 in F3/28, ¥2.4bn/¥18.2 in F3/29, and ¥2.8bn/¥22.4 in F3/30. Our calculations also assume that from F3/31 onward the firm can maintain FCF/share at ¥25. The fair value equates to 22.4x our F3/27E EPS of ¥9.8 and 0.2x BPS of ¥1,074.1.	

Note: e = Morgan Stanley Research estimates; Stock price as of the May 15 close for Risk Reward Chart

Source: Company data, Morgan Stanley Research

Daishinku: Key Financial Data and Risk Reward

Income Statement (JPY bn)						Cashflow Statement (JPY bn)					
	F3/25	F3/26	F3/27e	F3/28e	F3/29e		F3/25	F3/26	F3/27e	F3/28e	F3/29e
Sales	38.6	39.6	41.3	43.3	45.3	Cash flow from operating activities	2.3	-1.8	6.6	6.5	7.3
Costs of Sales	29.3	30.4	31.6	32.9	34.0	EBIT	1.6	1.1	0.9	1.4	2.2
Gross Profit	9.3	9.2	9.7	10.4	11.3	Depreciation	4.0	4.2	5.0	5.2	5.4
SG&A	8.4	8.1	8.4	8.6	8.8	Changes in Receivables	0.5	-0.2	-0.5	-0.5	-0.5
Operating Income	0.9	1.1	1.3	1.8	2.5	Change in inventory	-1.9	-5.6	0.1	0.2	0.3
Non-operating income/expenses	-0.5	-0.4	-0.4	-0.4	-0.3	Changes in Payables	1.0	0.1	0.2	0.2	0.2
Recurring Profit	0.4	0.7	0.9	1.4	2.2	Other	-2.8	-1.5	0.9	0.0	-0.3
Extraordinary income	1.2	0.3	0.0	0.0	0.0	Cash flow from investing activities	-6.3	-3.1	-4.0	-4.3	-4.5
Profit before income taxes	1.6	1.1	0.9	1.4	2.2	Purchase of PP&E	-6.9	-3.0	-4.0	-4.3	-4.5
Income taxes	0.6	0.5	0.3	0.4	0.7	Sales of PP&E	0.0	0.1	0.0	0.0	0.0
(Effective tax rate)	36%	42%	33%	29%	32%	Other	0.5	-0.2	0.0	0.0	0.0
Profit attributable to non-controlling interests	0.7	0.2	0.0	0.0	0.0	Cash flow from financing activities	-1.7	1.2	-1.7	-3.8	-3.9
Net Profit	0.3	0.4	0.6	1.0	1.5	Net Change In Borrowings	-0.1	2.7	-0.4	-2.6	-2.6
Capex	1.0	8.2	4.0	4.3	4.5	Purchase of treasury shares	-0.3	-0.2	0.0	0.0	0.0
Depreciation & amortization	4.0	4.2	5.0	5.2	5.4	Cash dividend paid	-0.9	-0.9	-0.9	-1.0	-1.0
R&D	0.6	2.3	2.4	2.5	2.6	Other	-0.5	-0.4	-0.3	-0.3	-0.3
EPS (JPY)	8.9	13.2	18.9	31.4	47.2	Effect of exchange rate changes	-0.1	1.2	0.1	0.0	0.0
USD/JPY (Average)	152.2	150.7	150.0	150.0	150.0	Changes in cash and cash equivalents	-5.9	-2.5	1.0	-1.7	-1.2
						Cash and cash equivalents at end of the period	18.5	16.0	16.3	14.6	13.5

Balance Sheet (JPY bn)					
	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Total assets	89.9	98.8	91.4	89.2	87.3
Current assets	48.9	53.4	50.5	49.2	48.3
Cash and cash equivalents	18.7	18.3	16.3	14.6	13.5
Accounts receivable	9.1	10.2	10.2	10.7	11.2
Inventories	18.0	24.6	20.9	20.8	20.5
Other financial assets	3.1	2.3	3.1	3.1	3.1
Fixed assets	41.0	45.4	40.9	40.0	39.1
Tangible fixed assets	34.9	38.4	34.8	33.9	33.0
Intangible fixed assets	1.0	0.8	1.0	1.0	1.0
Other non-current assets	5.1	6.2	5.1	5.1	5.1
Current liabilities	24.9	32.9	25.9	24.9	23.9
Short term borrowings	16.9	22.6	17.9	16.6	15.2
Accounts payable	3.4	4.1	3.6	3.8	4.0
Other current liabilities	4.6	6.3	4.4	4.6	4.7
Fixed liabilities	19.8	17.9	21.7	20.5	19.2
Long term borrowings	16.3	14.1	18.3	17.0	15.8
Other fixed liabilities	3.5	3.8	3.5	3.5	3.5
Net assets	45.2	48.0	43.7	43.8	44.2
Total liabilities and equity	89.9	98.8	91.4	89.2	87.3

Risk Reward – Daishinku (6962.T)

Demand for new devices accelerating and production system upgrades awaited

PRICE TARGET ¥610

Our price target is ¥610, based on a DCF valuation (risk-free rate 2.2%; beta 1.2; risk premium 3.7%; WACC 4.8%). Beyond our 3-year base earnings models, we also build forecasts for F3/29–30 into our DCF-based valuation, and our assumptions for F3/31 onward are that the firm maintains FCF/share of ¥70. Our price target equates to 32.3x our F3/27e EPS of ¥18.9. Our forex rate assumption for the next quarter and thereafter is ¥150/¥.

RISK REWARD CHART



BULL CASE

¥1,220

Telecom/auto growth speeds up

OP and EPS forecasts of ¥0.9bn/¥7.1 in F3/26, ¥1.8bn/¥31.4 in F3/27, ¥2.8bn/¥53.5 in F3/28, ¥4.2bn/¥81.8 in F3/29, and ¥5.3bn/¥110.0 in F3/30. Our calculations also assume that from F3/31 onward the firm can maintain FCF/share at ¥110. The fair value equates to 38.8x our F3/27E EPS of ¥31.4 and 1.1x BPS of ¥1,124.8.

BASE CASE

¥610

Earnings steadily recover but miss MTP target

OP and EPS forecasts of ¥0.9bn/¥7.1 in F3/26, ¥1.3bn/¥18.9 in F3/27, ¥1.8bn/¥31.4 in F3/28, ¥2.5bn/¥47.2 in F3/29, and ¥2.9bn/¥56.6 in F3/30. Our calculations also assume that from F3/31 onward the firm can maintain FCF/share at ¥70. Our PT equates to 32.3x our F3/27E EPS of ¥18.9 and 0.5x BPS of ¥1,124.8.

BEAR CASE

¥270

Investment burden, slowing topline growth

OP and EPS forecasts of ¥0.9bn/¥7.1 in F3/26, ¥0.6bn/¥3.1 in F3/27, ¥0.4bn/¥3.1 in F3/28, ¥0.4bn/¥3.1 in F3/29, and ¥0.4bn/¥3.1 in F3/30. Our calculations also assume that from F3/31 onward the firm can maintain FCF/share at ¥50. The fair value equates to 85.9x our F3/27E EPS of ¥3.1 and 0.2x BPS of ¥1,124.8.

Note: e = Morgan Stanley Research estimates; Stock price as of the May 15 close for Risk Reward Chart

Source: Company data, Morgan Stanley Research

Kyocera: Key Financial Data and Risk Reward

Income Statement					
(Y bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Sales	2,014.5	2,070.2	1,895.2	1,891.5	2,107.4
COGS	1,455.3	1,462.6	1,267.2	1,326.1	1,468.8
Gross Profit	559.2	607.6	628.0	565.4	638.6
SG&A	531.9	489.5	506.5	517.4	471.8
Operating Profit	27.3	118.1	121.5	138.0	166.8
Non-operating Income/Loss	36.3	50.9	9.0	1.7	-7.2
Pretax Profit	63.6	169.0	130.5	139.7	159.6
Income Tax	36.2	24.1	33.9	36.3	41.5
(Effective Tax Rate)	1%	0%	0%	0%	0%
Minority Interest	3.4	4.0	3.6	3.6	3.6
Net Profit	24.1	141.0	92.9	99.7	114.5
Capex	161.7	149.1	160.0	160.0	160.0
Depreciation and Amortization	111.7	110.9	115.0	120.0	120.0
R&D	104.3	115.7	120.0	125.0	125.0
EBITDA	139.0	229.1	236.5	258.0	286.8
ROE	0.7%	4.4%	3.2%	3.7%	4.6%
Average Share Outstanding (Diluted)	1,412.4	1,372.6	1,307.0	1,272.0	1,237.0
EPS (Diluted, Yen)	17.1	102.7	71.1	78.4	92.6
BPS (Yen)	2,149.4	2,264.8	2,059.5	1,928.8	1,843.1
DPS (Yen)	50.0	52.0	65.0	70.0	70.0
USD/JPY: Average Exchange Rate	151.8	150.8	150.0	150.0	150.0
EUR/JPY: Average Exchange Rate	163.8	174.8	170.0	170.0	170.0
YoY % Change					
Sales	1%	3%	-6%	5%	6%
Operating Profit	-71%	333%	19%	14%	21%
Net Profit	-76%	485%	-23%	7%	15%
As % of Sales					
Gross Profit	27.8%	29.4%	33.1%	33.1%	30.3%
Operating Profit	1.4%	5.7%	6.4%	7.0%	7.9%
Net Profit	1.2%	6.8%	4.9%	5.0%	5.4%
Capex	8.0%	7.2%	8.4%	8.1%	7.6%
Depreciation and Amortization	5.5%	5.4%	6.1%	6.1%	5.7%
Balance Sheet					
(Y bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Total Assets	4,406.5	4,476.3	3,992.8	3,761.4	3,605.6
Total Current Assets	1,330.5	1,362.1	1,243.8	1,182.4	1,196.6
Cash and Cash Equivalents	331.9	310.3	347.5	259.5	211.8
Marketable Securities	0.0	0.0	0.0	0.0	0.0
Account Receivables	378.9	389.4	356.5	372.7	396.4
Inventory	537.6	580.3	457.7	468.0	506.3
Other Current Assets	82.1	82.1	82.1	82.1	82.1
Total Long Term Assets	3,076.0	3,114.2	2,749.0	2,579.0	2,409.0
Net PP&E	716.0	754.1	809.0	849.0	889.0
Investments and Advances	1,702.6	1,702.6	1,282.6	1,072.6	892.6
Goodwill	282.9	282.9	282.9	282.9	282.9
Intangible Assets	152.2	152.2	152.2	152.2	152.2
Other Assets	222.4	222.4	222.4	222.4	222.4
Total Current Liabilities	461.2	462.2	435.0	443.2	463.1
ST Debt and Current Portion of LT Debt	15.3	15.3	15.3	15.3	15.3
Accounts Payable	203.2	204.2	176.9	185.1	205.0
Other Current Liabilities	242.7	242.7	242.7	242.7	242.7
Total Long Term Liabilities	742.3	742.3	742.3	742.3	742.3
Long Term Debt	199.8	199.8	199.8	199.8	199.8
Other Long Term Liabilities	542.6	542.6	542.6	542.6	542.6
Minority Interests	23.9	20.0	16.7	13.1	9.5
Total Equity	3,179.1	3,251.8	2,798.8	2,567.7	2,390.7
Net Debt (excl. KDDI Shares)	-129.8	-216.4	-883.6	-821.0	-834.2

Cash Flow Statement					
(Y bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Net Cash from Operations	139.4	195.7	266.7	197.6	188.9
Net Income	24.1	141.0	92.9	99.7	114.5
Depreciation and Amortization	111.7	110.9	115.0	120.0	120.0
Changes in Working Capital:	7.0	-52.2	62.4	-18.4	-42.1
Other-net	-3.4	-4.0	-3.6	-3.6	-3.6
Net Cash from Financing Activities	-161.7	-149.1	50.0	50.0	50.0
Purchase of PP&E	-161.7	-149.1	-160.0	-160.0	-160.0
Sale/Purchase of Investments	0.0	0.0	210.0	210.0	210.0
Other-net	0.0	0.0	0.0	0.0	0.0
Free cash flow	-22.2	46.6	316.7	247.8	238.9
Net Cash from Financing Activities	-70.6	-68.2	-325.0	-335.8	-286.6
Change in Interest-bearing Debt	0.0	0.0	0.0	0.0	0.0
Change in Treasury Stock	0.0	0.0	-250.0	-250.0	-200.0
Payment of Cash Dividends	-70.6	-68.2	-75.0	-85.8	-86.6
Other-net	0.0	0.0	0.0	0.0	0.0
Net Effect of Exchange Rate Changes	0.0	0.0	0.0	0.0	0.0
Net Change in Cash	-22.9	21.6	-8.3	-88.1	-47.7
Cash at End of Period	331.9	310.3	347.5	259.5	211.8
Segment Information					
(Y bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Net Sales	2,014.5	2,070.2	1,895.2	1,891.5	2,107.4
Core Components Business	867.1	853.4	851.3	893.9	742.5
Industrial & Automotive Components Unit	233.1	250.1	265.5	278.7	298.3
Semiconductor Components Unit	300.8	379.4	365.9	395.1	422.8
Others	33.3	23.9	20.0	20.0	21.4
Electronic Components Business	354.6	363.5	373.8	396.2	421.1
Solutions Business	1,111.0	1,070.9	892.1	913.4	967.5
Industrial Tools Unit	305.9	285.9	148.3	152.0	162.7
Document Solutions Unit/Information Eq.	480.0	478.5	482.4	494.5	519.2
Communications Unit	225.5	219.2	221.4	226.9	242.8
Others	99.7	87.3	40.0	40.0	42.8
Others	17.7	17.7	17.7	17.7	17.7
Elimination	-35.4	-31.8	-38.0	-38.0	-40.7
Net OP	24.0	133.2	130.5	147.0	175.8
Core Components Business	-1.1	63.1	78.0	86.5	99.7
Industrial & Automotive Components Unit	25.0	18.7	30.0	35.0	40.4
Semiconductor Components Unit	-27.8	46.9	47.5	51.0	58.8
Others	1.7	-2.6	0.5	0.5	0.5
Electronic Components Business	-0.8	7.3	15.0	18.0	23.1
Solutions Business	72.9	103.9	80.5	85.5	96.0
Others	0.0	0.0	0.0	0.0	0.0
Segment Margin					
Core Components Business	-0.2%	6.4%	6.9%	7.4%	8.3%
Industrial & Automotive Components Unit	10.7%	7.5%	11.3%	12.6%	13.6%
Semiconductor Components Unit	-9.3%	12.4%	13.0%	12.9%	13.9%
Electronic Components Business	-0.2%	2.0%	4.0%	4.5%	5.5%
Solutions Business	6.6%	9.7%	9.0%	9.4%	9.9%

Risk Reward – Kyocera (6971.T)

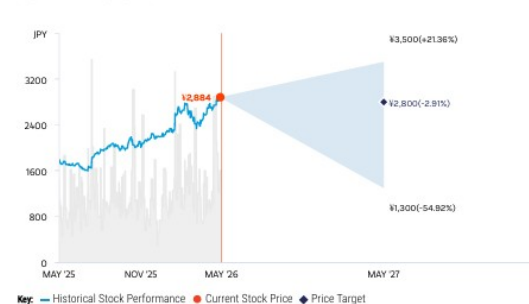
Growth areas still uncertain, but prospects of balance sheet reform

PRICE TARGET ¥2,800

Derived from the base case. DCF model assumptions: 2.2% risk-free rate, 0.88 equity beta, 3.7% risk premium, for WACC of 5.2%; zero growth from F3/36.



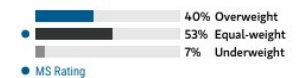
RISK REWARD CHART



EQUAL-WEIGHT THESIS

- Needs to further strengthen core components business and restructure solution business.
- Sale of KDDI shares and improve shareholder return should improve ROE. However, it is not clear which businesses will be the earnings drivers and whether or not earnings improvement will continue in the mid/longer term.
- We expect ROE to improve to 4.6% in F3/29 from 0.7% in F3/25, which is still lower than cost of capital of 5.4% and management's target of 7.0%.

Consensus Rating Distribution



BULL CASE

¥3,500

Derived from DCF

Sales top the base case by 5% as parts demand recovers: OP ¥131.4bn in F3/26, ¥160.5bn in F3/27, ¥197.0bn in F3/28, ¥225.9bn in F3/29. Same FX & DCF model assumptions as the base case. Bull case fair value equates to P/E of 35.4x on F3/27 diluted EPS of ¥98.9.

BASE CASE

¥2,800

Derived from DCF

Expect BS structure change by sales of KDDI shares and improvement of return to shareholders. However, it is not clear which business will be earnings drivers. OP ¥102.4bn in F3/26, ¥121.5bn in F3/27, ¥138.0bn in F3/28, ¥166.9bn in F3/29. FX assumptions: ¥150/US\$, ¥170/€. Base case fair value equates to 39.4 x F3/27e diluted EPS of ¥71.1, 35.7x F3/28e diluted EPS of ¥78.4.

BEAR CASE

¥1,300

F3/27e diluted BPS ¥2,050 x P/B 0.65

Limited improvement in components business earnings. Improvement of capital efficiency limited, although shareholder return improvement continues. OP ¥77.4bn in F3/26, OP ¥96.5bn in F3/27, OP ¥103.0bn in F3/28, OP ¥118.5bn in F3/29. Same FX assumptions as the base case. ROE of 2.5% in F3/27 and 2.5% in F3/28 remains below capital costs of 5.4%, prompting the stock to trade on P/B of 0.65x.

Note: e = Morgan Stanley Research estimates; Stock price as of the May 15 close for Risk Reward Chart
Source: Company data, Morgan Stanley Research

Taiyo Yuden: Key Financial Data and Risk Reward

Income Statement

(¥ bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Net Sales	341.4	355.3	383.0	414.7	449.3
COGS	269.9	273.4	288.9	312.2	340.7
Gross Profit	71.6	81.9	94.2	102.4	108.6
SG&A Expenses	61.1	61.9	62.4	63.4	64.4
Operating Profit	10.5	20.0	31.8	39.1	44.2
Non-operating Income/Loss	0.1	4.1	3.4	3.4	3.4
Recurring Profit	10.5	24.1	35.2	42.4	47.6
Extraordinary Income/Loss	-2.2	-4.1	-2.0	-2.0	-2.0
Pretax Profit	8.4	20.1	33.2	40.4	45.6
Income Tax	6.0	5.1	8.3	10.1	11.4
(Effective Tax Rate)	72%	25%	25%	25%	25%
Minority Interest	0.0	0.0	0.0	0.0	0.0
Net Profit	2.3	14.8	24.9	30.3	34.2

Capex	64.2	40.4	50.0	60.0	60.0
Depreciation and Amortization	46.3	49.1	50.0	52.0	52.0
R&D Expenses	15.0	14.5	16.5	17.0	17.0
EBITDA	56.7	69.1	81.8	91.1	96.2
ROE	0.7%	4.5%	7.5%	8.8%	9.4%
Average Share Outstanding (Diluted)	144.3	137.7	137.2	137.2	137.2
EPS (Diluted, Yen)	16.1	107.6	181.3	221.1	249.1
BPS (Yen)	2,206.4	2,501.8	2,462.3	2,588.0	2,732.8
DPS (Yen)	90.0	90.0	100.0	110.0	120.0

USD/JPY: Average Exchange Rate	152.6	150.0	150.0	150.0	150.0
YoY % Change					
Net Sales	6%	4%	8%	8%	8%
Operating Profit	15%	91%	41%	23%	13%
Net Profit	-72%	536%	46%	22%	13%

As % of Sales					
Gross Profit	21.0%	23.1%	24.6%	24.7%	24.2%
Operating Profit	3.1%	5.8%	8.3%	9.4%	9.8%
Net Profit	0.7%	4.2%	6.5%	7.3%	7.6%
Capex	18.8%	11.4%	13.1%	14.5%	13.4%
Depreciation and Amortization	13.5%	13.8%	13.1%	12.5%	11.8%

Balance Sheet

(¥ bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Total Assets	573.2	615.5	694.4	614.0	636.7
Total Current Assets	279.3	318.3	305.9	317.5	332.2
Cash and Cash Equivalents	78.2	100.1	97.2	98.0	100.3
Marketable Securities	0.0	0.0	0.0	0.0	0.0
Account Receivables	80.5	86.4	88.3	94.4	101.1
Inventory	110.0	125.8	109.9	114.5	120.2
Other Current Assets	10.5	6.0	10.5	10.5	10.5
Total Long Term Assets	293.9	297.2	288.6	296.6	304.6
Net PP&E	286.8	290.0	281.5	289.5	297.5
Investments in Securities	0.0	0.0	0.0	0.0	0.0
Other Assets	7.0	7.3	7.0	7.0	7.0
Total Current Liabilities	79.6	93.8	81.6	83.9	86.8
ST Debt and Current Portion of LT Debt	19.7	35.2	19.7	19.7	19.7
Accounts Payable	27.2	26.5	29.1	31.4	34.3
Other Current Liabilities	32.8	32.1	32.8	32.8	32.8
Total Long Term Liabilities	174.4	177.4	174.4	174.4	174.4
Long Term Debt	144.7	136.0	144.7	144.7	144.7
Other Long Term Liabilities	29.7	41.4	29.7	29.7	29.7
Net Assets	318.2	344.4	338.5	355.8	375.6
Minority Interests	0.0	0.0	0.0	0.0	0.0
Stock Acquisition Rights	0.7	0.0	0.7	0.7	0.7
Net Debt	86.2	71.1	67.2	66.4	64.1

Cash Flow Statement

(¥ bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Net Cash from Operations	33.9	59.1	67.3	73.9	76.6
Net Income	2.3	14.8	24.9	30.3	34.2
Depreciation and Amortization	46.3	49.1	50.0	52.0	52.0
Changes in Working Capital	-13.5	-9.0	-7.6	-8.4	-9.6
Other-net	-1.2	3.2	0.0	0.0	0.0
Net Cash from Investing Activities	-63.5	-25.7	-50.0	-60.0	-60.0
Purchase of PP&E	-62.6	-40.4	-50.0	-60.0	-60.0
Sale/Purchase of Investments	1.3	0.0	0.0	0.0	0.0
Other-net	-2.2	14.7	0.0	0.0	0.0
Free Cash Flow	-29.6	32.4	17.3	13.9	16.6
Net Cash from Financing Activities	3.0	-6.8	-11.8	-13.1	-14.3
Change in Interest-bearing Debt	15.7	6.5	0.0	0.0	0.0
Change in Treasury Stock	0.0	0.0	0.0	0.0	0.0
Payment of Cash Dividends	-11.2	-11.2	-11.8	-13.1	-14.3
Other-net	-1.5	-2.1	0.0	0.0	0.0
Net Effect of Exchange Rate Changes	-0.9	4.9	0.0	0.0	0.0
Net Change in Cash	-27.4	30.5	5.5	0.8	2.3
Cash at End of Period	67.5	98.1	87.2	88.1	90.3

Segment Information

(¥ bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Net Sales	341.4	355.3	383.0	414.7	449.3
Capacitors	232.1	251.8	275.4	303.0	333.3
Elna	0.0	0.0	0.0	0.0	0.0
Ferrite and Applied Products	61.5	64.3	68.1	72.2	76.5
Complex Devices	23.0	14.8	14.5	14.5	14.5
Others	24.8	24.5	25.0	25.0	25.0
Total Operating Profit	10.5	20.0	31.8	39.1	44.2
Operating Profit Margin	3.1%	5.6%	8.3%	9.4%	9.8%

Risk Reward – Taiyo Yuden (6976.T)

While earnings are improving, a new record high is unlikely.

PRICE TARGET ¥4,700

Derived from the base case. DCF assumptions: 2.2% risk-free rate, 1.22 equity beta, 3.7% risk premium, for 5.6% WACC; zero growth from F3/35.



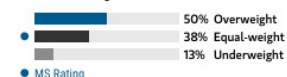
RISK REWARD CHART



EQUAL-WEIGHT THESIS

- Earnings continue to recover, but surpassing the F3/22 record high will likely take considerable time.
- We expect continued earnings recovery in the core MLCC business.
- We expect the firm to maintain its tech competitiveness in mainstay MLCCs.
- We expect sales growth for power inductors used in 5G smartphones and base stations to continue.

Consensus Rating Distribution



BULL CASE	¥6,500	BASE CASE	¥4,700	BEAR CASE	¥2,000
Derived from DCF		Derived from DCF		F3/27e diluted BPS ¥2,403 x P/B 0.83	
MLCC sales exceed the base case. OP ¥25.1bn in F3/26, ¥44.0bn in F3/27, ¥56.1bn in F3/28. Same FX & DCF model assumptions as the base case. Fair value equates to P/E 26.2x on F3/27e diluted EPS of ¥248.0 and 20.7x on F3/28e diluted EPS of ¥314.3.		While higher MLCC utilization is driving earnings improvement, a full recovery will take time: OP ¥22.5bn in F3/26, ¥31.8bn in F3/27, ¥39.1bn in F3/28. FX assumption: ¥150/US\$. Base case fair value equates to P/E 25.9x on F3/27e diluted EPS of ¥181.3, and 21.3x on F3/28e diluted EPS of ¥221.1.		Stays competitive in core products, but slowing end demand drags profits down. OP ¥18.4bn in F3/26, OP ¥23.0bn in F3/27, ¥25.0bn in F3/28. Same FX and DCF assumptions as in the base case. ROE 4.4% in F3/26, 5.8% in F3/27, below capital costs of 6.7% and with earnings lower than market expectation, P/B falls to 0.83x.	

Note: e = Morgan Stanley Research estimates; Stock price as of the May 15 close for Risk Reward Chart
Source: Company data, Morgan Stanley Research

Murata Manufacturing: Key Financial Data and Risk Reward

Income Statement					
(Y bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Net Sales	1,743.4	1,830.9	1,944.9	2,195.1	2,485.0
Cost of Goods Sold	1,025.7	1,056.0	1,018.8	1,165.0	1,359.4
Gross Profit	717.7	774.8	926.2	1,030.2	1,125.6
SG&A Expenses	438.0	493.0	553.3	570.8	588.3
Operating Profit	279.7	281.8	372.8	459.3	537.3
Non-Operating Income/Loss	24.7	30.0	12.0	12.0	12.0
Pretax profit	304.4	308.6	384.8	471.3	549.3
Income Tax	71.4	74.9	92.4	113.1	131.8
(Effective Tax Rate)	23%	24%	24%	24%	24%
Minority Interest	-0.8	-0.1	0.0	0.0	0.0
Net Profit	233.8	233.8	292.5	358.2	417.5
Capex	180.5	247.8	220.0	210.0	220.0
Depreciation and Amortization	173.3	178.2	180.0	185.0	190.0
R&D	149.3	158.9	165.0	170.0	173.0
EBITDA	453.0	460.0	552.8	644.3	727.3
ROE	9.1%	8.8%	11.1%	13.1%	14.4%
Average Share Outstanding (Diluted)	1,869.3	1,831.4	1,828.3	1,828.3	1,828.3
EPS (Diluted, Yen)	125.08	127.65	159.98	195.92	228.34
BPS (Yen)	1,380.1	1,484.0	1,461.4	1,532.6	1,636.3
DPS (Yen)	57.0	70.0	65.0	70.0	75.0
USD/JPY: Average Exchange Rate	152.6	150.8	150.0	150.0	150.0
YoY % Change					
Net Sales	6%	5%	8%	13%	13%
Operating Profit	30%	1%	38%	23%	17%
Net Profit	29%	0%	33%	22%	17%
As % of Sales					
Gross Profit	41.2%	42.3%	47.6%	46.9%	45.3%
Operating Profit	16.0%	15.4%	19.2%	20.9%	21.6%
Net Profit	13.4%	12.8%	15.0%	16.3%	16.8%
Capex	10.4%	13.5%	11.3%	9.6%	8.9%
Depreciation and Amortization	9.9%	9.7%	9.3%	8.4%	7.6%
Balance Sheet					
(Y bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Total Assets	3,028.2	3,199.1	3,132.4	3,274.6	3,479.9
Total Current Assets	1,497.9	1,581.4	1,486.9	1,604.0	1,779.3
Cash and Cash Equivalents	625.1	653.7	515.0	528.1	574.0
Marketable Securities	0.0	0.0	0.0	0.0	0.0
Account Receivables	294.4	328.2	394.6	445.3	504.1
Inventory	482.8	520.5	481.8	535.0	605.6
Other Current Assets	95.5	79.1	95.5	95.5	95.5
Total Long Term Assets	1,530.3	1,617.7	1,645.6	1,670.6	1,700.6
Net PP&E	1,183.7	1,300.9	1,299.0	1,324.0	1,354.0
Investments in Securities	40.3	50.4	40.3	40.3	40.3
Other assets	306.3	266.4	306.3	306.3	306.3
Total Current Liabilities	278.5	316.5	289.8	301.6	317.4
ST Debt & Current Portion of LT Debt	0.8	1.7	0.8	0.8	0.8
Accounts Payable	69.6	79.8	82.9	94.7	110.5
Other Current Liabilities	206.1	234.9	206.1	206.1	206.1
Total Long Term Liabilities	171.7	164.8	171.7	171.7	171.7
Long Term Debt	1.7	1.5	1.7	1.7	1.7
Other Long Term Liabilities	170.0	163.3	170.0	170.0	170.0
Minority Interests	-0.8	-0.9	-0.8	-0.8	-0.8
Total Equity	2,580.0	2,717.8	2,671.8	2,802.1	2,991.5
Net Debt	-663.0	-700.8	-552.8	-566.0	-611.8

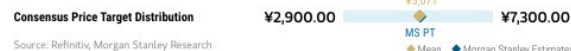
Cash Flow Statement					
(Y bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Net Cash from Operations	451.8	425.2	450.1	451.1	493.8
Net Income	233.8	233.8	292.5	358.2	417.5
Depreciation and Amortization	173.3	178.2	180.0	185.0	190.0
Changes in Working Capital:	24.4	-18.0	-22.4	-92.1	-113.6
Other-net	20.3	31.3	0.0	0.0	0.0
Net Cash from Investing Activities	-208.1	-193.8	-220.0	-210.0	-220.0
Purchase of PP&E	-179.4	-243.0	-220.0	-210.0	-220.0
Sale/Purchase of Investments	3.4	2.2	0.0	0.0	0.0
Other-net	-32.1	46.9	0.0	0.0	0.0
Free cash flow	243.8	231.4	230.1	241.1	273.8
Net Cash from Financing Activities	-242.7	-221.8	-210.2	-228.0	-228.0
Change in Interest-bearing Debt	-50.4	0.8	0.0	0.0	0.0
Change in Treasury Stock	-80.0	-100.0	-100.0	-100.0	-100.0
Payment of Cash Dividends	-101.6	-110.7	-110.2	-128.0	-128.0
Other-net	-10.7	-11.9	0.0	0.0	0.0
Net Effect of Exchange Rate Changes	2.0	19.0	0.0	0.0	0.0
Net Change in Cash	3.1	28.6	19.9	13.2	45.9
Segment Information					
(Y bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Net Sales	1,793.3	1,830.9	1,944.9	2,195.1	2,485.0
Components Total	1,033.1	1,159.7	1,245.3	1,371.4	1,505.6
Capacitors	831.8	936.4	1,016.7	1,133.6	1,258.3
Inductors and EMI Filters	201.3	223.3	228.6	237.8	247.3
Devices/Modules	697.2	656.0	686.6	810.7	966.4
High Frequency Devices/Communications Module	443.6	394.8	415.7	498.8	598.6
Battery/Power supply	155.7	154.1	159.3	193.6	242.4
Functional Devices	97.8	107.1	111.6	118.3	125.4
Others	13.1	15.2	13.0	13.0	13.0
Total Operating Profit	279.7	281.8	372.8	459.3	537.3
Components Total	275.2	44.5	329.8	388.1	441.1
Devices/Modules	10.0	-26.5	49.0	77.3	102.2
Others	-5.4	-6.8	-6.0	-6.0	-6.0
Operating Profit Margin	15.6%	15.4%	19.2%	20.9%	21.6%
Components Total	26.6%	3.8%	26.5%	28.3%	29.3%
Devices/Modules	2.3%	-6.7%	11.8%	15.5%	17.1%
Sales by Application					
(Y bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Net Sales (excl. Non-product Sales)	1,793.3	1,830.9	1,944.9	2,195.1	2,485.0
AV	674.2	-	-	-	-
Communication	453.1	-	-	-	-
Computer and Peripheral	281.9	-	-	-	-
Automobile Electronics	150.4	-	-	-	-
Home Appliances and Others	183.7	-	-	-	-
As % of Total					
AV	38.7%	-	-	-	-
Communication	26.0%	-	-	-	-
Computer and Peripheral	16.2%	-	-	-	-
Automobile Electronics	8.6%	-	-	-	-
Home Appliances and Others	10.5%	-	-	-	-

Risk Reward – Murata Manufacturing (6981.T)

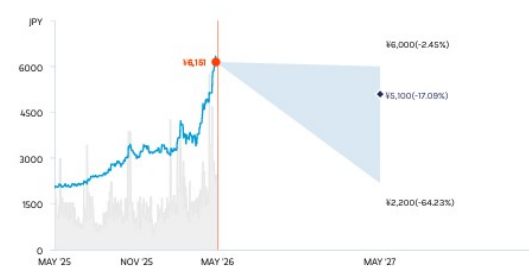
Expect MLCC utilization rates to remain high and earnings to continue to expand

PRICE TARGET ¥5,100

Derived from our base case and a DCF model, with the following assumptions: 2.2% risk-free rate, 1.09 equity beta, 3.7% risk premium, yielding 6.2% WACC; zero growth from F3/36.



RISK REWARD CHART



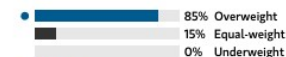
Key: — Historical Stock Performance • Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

OVERWEIGHT THESIS

- In F3/26, we expect MLCC sales to increase and LiB losses to shrink.
- Utilization rates in the MLCC space are likely to remain high as demand for use in AI servers continues to rise, and we thus expect currently high profits to expand further.
- We expect Murata to remain competitive in MLCCs and other products over the long term.
- In F3/27 onward, we expect module earnings to improve as sales of RF modules rise.

Consensus Rating Distribution



MS Rating

Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

New Data Era: Positive

Secular Growth: Positive

View descriptions of Risk Rewards Themes [here](#)

BULL CASE	¥6,000	BASE CASE	¥5,100	BEAR CASE	¥2,200
Derived from DCF					
Demand for end products such as smartphones and autos is stronger than in the base case. OP assumptions: ¥318.5bn in F3/26, ¥430.2bn in F3/27, ¥526.7bn in F3/28. Same FX & DCF model assumptions as the base case. Bull case fair value equates to P/E of 34.3x on F3/27 diluted EPS of ¥175.1 and 28.1x on F3/28 diluted EPS of ¥213.3.					
Derived from DCF					
Earnings continue to grow and beat consensus over the medium/long-term due to technological advantages & competitiveness, especially in MLCCs. Assumptions: OP ¥270.6bn in F3/26, ¥372.8bn in F3/27, ¥459.3bn in F3/28; FX ¥150/US\$, ¥170/€; DCF model (same as valuation methodology): 2.2% risk-free rate, 1.09 equity beta, 3.7% risk premium, yielding 6.2% WACC; zero growth from F3/36. Base case fair value equates to P/E 31.9x on F3/27e diluted EPS of ¥160.0 and 26.0x on F3/28e diluted EPS of ¥195.9.					
F3/27e BPS ¥1,448x P/B 1.5					
Murata remains competitive, but final demand for its components is lower than we envision. OP assumptions: ¥240.7bn in F3/26, ¥279.1bn in F3/27, ¥310.2bn in F3/28. Same FX assumptions as the base case. Assumes ROE moves from 9.1% in F3/25 to 7.5% in F3/26 and 8.2% in F3/27, higher than our capital cost estimate of 6.2%, but with earnings missing market expectations, P/B drops to 1.5x.					

Note: e = Morgan Stanley Research estimates; Stock price as of the May 15 close for Risk Reward Chart

Source: Company data, Morgan Stanley Research

Nichicon: Key Financial Data and Risk Reward

Income Statement						Cashflow Statement					
(JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e	(JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Sales	175.6	168.7	162.6	167.8	192.7	Cash flow from operating activities	16.3	8.2	14.1	16.2	17.1
Costs of Sales	147.4	139.7	140.0	152.0	155.2	EBIT	8.5	7.2	9.0	10.3	11.1
Gross Profit	28.3	30.0	33.6	35.8	37.5	Depreciation	8.5	8.1	9.3	9.4	9.6
SG&A	23.1	23.5	25.2	26.1	27.0	Changes in Receivables	3.2	-1.2	-2.2	-1.3	-1.3
Operating Income	5.2	6.5	8.4	9.7	10.5	Change in inventory	3.7	1.8	-0.8	-0.1	0.1
Non-operating income/expenses	2.3	1.9	0.6	0.6	0.6	Changes in Payables	-2.2	-4.0	1.4	0.8	0.8
Recurring Profit	7.5	8.3	9.0	10.3	11.1	Other	-3.3	-1.7	-2.6	-3.0	-3.2
Extraordinary income	1.0	-1.1	0.0	0.0	0.0	Cash flow from investing activities	-8.4	-6.6	-9.1	-9.4	-12.5
Profit before income taxes	8.5	7.2	9.0	10.3	11.1	Purchase of PP&E	-10.7	-7.1	-9.1	-9.4	-12.5
Income taxes	2.1	6.3	2.8	3.2	3.4	Sales of PP&E	0.0	0.0	0.0	0.0	0.0
(Effective tax rate)	25%	88%	31%	31%	31%	Other	2.3	0.5	0.0	0.0	0.0
Profit attributable to non-controlling interests	0.5	0.0	0.0	0.0	0.0	Cash flow from financing activities	-14.3	-3.8	-3.1	-4.0	-2.4
Net Profit	5.9	6.3	6.2	7.1	7.7	Net Change in Borrowings	2.2	-0.8	-0.7	-1.4	0.4
						Purchase of treasury shares	-1.6	0.0	0.0	0.0	0.0
Capex	11.1	7.6	9.1	9.4	12.5	Cash dividend paid	-2.5	-2.4	-2.4	-2.6	-2.8
Depreciation & amortization	8.5	8.1	9.3	9.4	9.6	Other	-12.5	-0.6	0.0	0.0	0.0
R&D	7.2	7.3	9.3	9.6	9.8	Effect of exchange rate changes	0.5	1.2	0.3	0.0	0.0
EPS (JPY)	86.0	94.0	92.3	105.7	114.7	Changes in cash and cash equivalents	-3.9	-1.1	2.2	2.8	2.2
USD/JPY	151.6	150.0	150.0	150.0	150.0	Cash and cash equivalents at end of the period	25.5	24.4	31.5	34.3	36.5

Balance Sheet						Sales by Product					
(JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e	(JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Total assets	192.6	195.6	203.0	207.0	213.1	Capacitor	99.2	102.7	109.9	113.0	115.7
Current assets	108.3	107.7	120.7	124.9	128.3	Aluminum capacitor	79.3	82.0	85.8	87.7	88.8
Cash and cash equivalents	25.5	24.5	31.5	34.3	36.5	Film capacitor	19.8	20.7	24.1	25.3	26.9
Accounts receivable	44.8	47.4	46.6	47.9	49.2	NECST	76.6	67.0	72.7	74.8	77.0
Inventories	33.2	32.3	38.0	38.1	38.0	Applied system and equipment	7.3	7.2	7.2	7.4	7.5
Other financial assets	4.7	3.5	4.6	4.6	4.6	EV	5.6	3.5	3.6	3.6	3.6
Fixed assets	84.3	87.9	82.3	82.1	84.9	Power source	26.3	21.7	23.9	24.4	24.9
Tangible fixed assets	53.1	52.5	51.4	51.4	54.3	Storage	35.6	32.4	35.8	37.2	38.8
Intangible fixed assets	1.9	2.1	1.6	1.4	1.3	Others	1.9	2.2	2.2	2.2	2.2
Other non-current assets	29.3	33.2	29.3	29.3	29.3	Capacitor	-7%	4%	6%	3%	2%
Current liabilities	51.6	47.2	53.2	53.6	53.9	Aluminum capacitor	-9%	3%	4%	2%	1%
Short term borrowings	9.2	10.9	9.7	9.2	8.7	Film capacitor	-19%	6%	11%	8%	5%
Accounts payable	28.5	23.3	29.6	30.4	31.2	NECST	2%	-13%	3%	3%	3%
Other current liabilities	13.9	13.0	13.9	14.0	14.0	Applied system and equipment	6%	-1%	2%	2%	2%
Fixed liabilities	27.2	24.8	25.1	24.1	25.0	EV	-31%	-37%	0%	0%	0%
Long term borrowings	15.8	13.3	13.6	12.7	13.5	Power source	11%	-17%	2%	2%	2%
Other fixed liabilities	11.5	11.6	11.5	11.5	11.5	Storage	4%	-9%	4%	4%	4%
						Others	-20%	15%	0%	0%	0%
Net assets	113.8	123.5	124.7	128.3	134.2	OP by Product					
Total liabilities and equity	192.6	195.6	203.0	207.0	213.1	(JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
						Capacitor	1.6	4.6	5.2	6.0	6.4
						Aluminum capacitor	0.5	3.1	2.9	3.5	3.7
						Film capacitor	1.0	1.5	2.2	2.5	2.7
						NECST	3.7	1.9	3.2	3.7	4.1
						Applied system and equipment	1.1	0.8	0.7	0.7	0.8
						EV	-0.3	-0.8	-0.2	-0.2	-0.2
						Power source	0.5	0.1	-0.2	-0.1	0.0
						Storage	2.2	1.5	2.9	3.2	3.5
						Others/Adjustment (OPM)	0.2	0.2	0.0	0.0	0.0
						Capacitor	1.6%	4.5%	4.7%	5.3%	5.5%
						Aluminum capacitor	0.7%	3.8%	3.4%	4.0%	4.1%
						Film capacitor	5.1%	7.2%	9.3%	9.8%	10.1%
						NECST	4.8%	2.8%	4.4%	5.0%	5.4%
						Applied system and equipment	14.4%	11.3%	9.5%	9.9%	10.3%
						EV	-5.4%	-21.3%	-5.5%	-5.1%	-4.7%
						Power source	1.9%	0.3%	-0.8%	-0.3%	0.0%
						Storage	6.2%	4.7%	8.2%	8.7%	9.0%

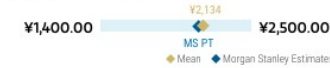
Risk Reward – Nichicon (6996.T)

AEC earnings to recover gradually

PRICE TARGET ¥2,100

Our price target is ¥2,100, based on a DCF valuation (risk-free rate 2.2%; beta 1.0; risk premium 3.7%; WACC 4.7%). Beyond our 3-year base earnings models, we also build forecasts for F3/29–30 into our DCF-based valuation, and our assumptions for F3/31 onward are that the firm maintains FCF/share of ¥105. Our price target equates to 22.7x our F3/27E EPS of ¥92.3. Our forex rate assumption for the next quarter and thereafter is ¥150/\$.

Consensus Price Target Distribution



RISK REWARD CHART



EQUAL-WEIGHT THESIS

The equity story is centered on increased sales of high value-added, high-capacity capacitors and the stable growth of the NECST business. From F3/27, we expect the capacitor business to expand, driven primarily by large-size products for AI servers and film capacitors for xEV applications, while competition in general-purpose products is likely to remain intense amid rivalry with Chinese players. We expect NECST business to deliver stable growth, supported by the wider adoption of residential energy storage systems.

Consensus Rating Distribution



Risk Reward Themes

Electric Vehicles: Positive

View descriptions of Risk Rewards Themes [here](#)

BULL CASE	¥3,700	BASE CASE	¥2,100	BEAR CASE	¥900
Strong sales growth continues in AEC/NECST					
OP and EPS forecasts of ¥6.2bn/¥91.6 in F3/26, ¥10.2bn/¥111.7 in F3/27, ¥13.2bn/¥141.5 in F3/28, ¥16.1bn/¥171.2 in F3/29, and ¥18.4bn/¥195.1 in F3/30. Our calculations also assume that from F3/31 onward the firm can maintain FCF/share at ¥200. The fair value equates to 33.1x our F3/27E EPS of ¥111.7 and 2.0x BPS of ¥1,826.2.					
High-value aluminum electrolytics drive recovery					
OP and EPS forecasts of ¥6.2bn/¥91.6 in F3/26, ¥8.4bn/¥92.3 in F3/27, ¥9.7bn/¥105.7 in F3/28, ¥10.5bn/¥114.7 in F3/29, and ¥10.6bn/¥116.1 in F3/30. Our calculations also assume that from F3/31 onward the firm can maintain FCF/share at ¥105. Our PT equates to 22.7x our F3/27E EPS of ¥92.3 and 1.2x BPS of ¥1,806.8.					
OP declines on slower growth and higher costs					
OP and EPS forecasts of ¥6.2bn/¥91.6 in F3/26, ¥5.7bn/¥65.5 in F3/27, ¥5.3bn/¥61.0 in F3/28, ¥4.9bn/¥56.6 in F3/29, and ¥4.7bn/¥44.7 in F3/30. Our calculations also assume that from F3/31 onward the firm can maintain FCF/share at ¥35. The fair value equates to 13.7x our F3/27E EPS of ¥65.5 and 0.5x BPS of ¥1,780.0.					

Note: e = Morgan Stanley Research estimates; Stock price as of the May 15 close for Risk Reward Chart

Source: Company data, Morgan Stanley Research

Nippon Chemi-con: Key Financial Data and Risk Reward

Income Statement						Cashflow Statement					
(JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e	(JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Sales	122.7	136.8	147.6	153.5	157.5	Cash flow from operating activities	-0.5	7.6	6.4	8.9	10.3
Costs of Sales	99.0	112.4	119.3	123.7	126.4	EBIT	0.6	3.5	4.1	4.9	5.6
Gross Profit	23.7	24.4	28.3	29.8	31.1	Depreciation	7.1	7.3	6.7	6.8	7.0
SG&A	20.0	21.1	22.7	23.4	24.0	Changes in Receivables	2.7	-4.3	-2.6	-1.4	-0.9
Operating Income	3.7	3.4	5.6	6.4	7.1	Change in inventory	-3.8	3.1	-2.3	-1.0	-0.5
Non-operating income/expenses	-2.2	-1.3	-1.5	-1.5	-1.5	Changes in Payables	-1.0	-0.5	1.2	0.6	0.4
Recurring Profit	1.6	2.1	4.1	4.9	5.6	Other	-6.1	-1.5	-0.8	-1.0	-1.3
Extraordinary income	-1.0	1.4	0.0	0.0	0.0	Cash flow from investing activities	-9.8	-5.3	-7.4	-7.7	-7.9
Profit before income taxes	0.6	3.5	4.1	4.9	5.6	Purchase of PP&E	-9.3	-4.4	-7.1	-7.4	-7.6
Income taxes	0.5	1.1	1.3	1.5	1.7	Sales of PP&E	0.0	0.0	0.0	0.0	0.0
(Effective tax rate)	89%	31%	32%	31%	30%	Other	-0.4	-0.9	-0.3	-0.3	-0.3
Profit attributable to non-controlling interests	0.0	0.1	0.0	0.0	0.0	Cash flow from financing activities	-11.9	-6.2	5.3	-1.8	-2.4
Net Profit	0.0	2.4	2.8	3.4	3.9	Net Change in Borrowings	-10.6	-2.7	6.4	-0.5	-1.1
Capex	7.6	5.9	7.4	7.7	7.9	Purchase of treasury shares	0.0	0.0	0.0	0.0	0.0
Depreciation & amortization	7.1	7.3	6.7	6.8	7.0	Cash dividend paid	-0.2	-0.6	-1.1	-1.3	-1.3
R&D	4.2	3.9	4.6	4.8	4.9	Other	-1.2	-3.0	0.0	0.0	0.0
EPS (JPY)	1.0	68.8	102.3	124.2	142.4	Effect of exchange rate changes	0.7	1.3	0.3	0.0	0.0
USD/JPY	152.6	150.8	150.0	150.0	150.0	Changes in cash and cash equivalents	-21.4	-2.6	4.5	-0.6	-0.1
						Cash and cash equivalents at end of the period	23.9	21.3	25.3	24.7	24.7

Balance Sheet						Sales by Product					
(JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e	(JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Total assets	162.7	166.0	171.0	173.3	175.2	Capacitor	118.0	132.5	142.8	148.7	152.7
Current assets	86.6	87.6	96.9	98.7	100.1	Aluminum capacitor	106.1	122.0	131.3	137.1	141.0
Cash and cash equivalents	23.9	21.3	25.3	24.7	24.7	DLCAP	3.8	3.3	3.7	3.8	3.9
Accounts receivable	28.9	29.2	34.8	36.2	37.1	Other capacitor	2.5	2.3	2.6	2.6	2.6
Inventories	33.1	31.2	36.0	37.0	37.5	Capacitor material	5.7	4.8	5.3	5.3	5.3
Other financial assets	0.8	5.8	0.8	0.8	0.8	Mechanical part	3.4	2.7	2.9	3.0	2.9
Fixed assets	76.1	79.4	74.2	74.6	75.2	Other	1.2	1.6	1.8	1.8	1.8
Tangible fixed assets	48.3	47.5	47.3	48.2	49.1	Capacitor	-19%	12%	8%	4%	3%
Intangible fixed assets	2.3	2.6	1.4	1.0	0.6	Aluminum capacitor	-20%	15%	9%	4%	3%
Other non-current assets	25.4	28.3	25.4	25.4	25.4	DLCAP	-28%	-12%	3%	3%	3%
Current liabilities	60.6	58.5	68.0	70.0	71.3	Other capacitor	-28%	-4%	0%	0%	0%
Short term borrowings	36.7	34.6	41.3	42.7	43.5	Capacitor material	24%	-16%	0%	0%	0%
Accounts payable	13.1	8.7	15.7	16.4	16.8	Mechanical part	-2%	-21%	0%	0%	0%
Other current liabilities	10.8	15.1	10.9	11.0	11.0	Other	-21%	34%	0%	0%	0%
Fixed liabilities	45.4	44.4	38.9	37.1	35.1						
Long term borrowings	38.1	35.9	31.6	29.8	27.8						
Other fixed liabilities	7.3	8.5	7.3	7.3	7.3						
Net assets	56.7	63.1	64.1	66.2	68.8						
Total liabilities and equity	162.7	166.0	171.0	173.3	175.2						

Risk Reward – Nippon Chemi-Con (6997.T)

Earnings to recover gradually, but be careful about risk of share dilution

PRICE TARGET ¥880

Our price target is ¥880, based on a DCF valuation (risk-free rate 2.2%; beta 1.1; risk premium 3.7%; WACC 4.2%). Beyond our 3-year base earnings models, we also build forecasts for F3/29–30 into our DCF-based valuation, and our assumptions for F3/31 onward are that the firm maintains FCF/share of ¥130. Our price target equates to 8.6x our F3/27E EPS of ¥102.3. Our forex rate assumption for the next quarter and thereafter is ¥150/\$.

RISK REWARD CHART



Key: — Historical Stock Performance • Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

UNDERWEIGHT THESIS

The equity story centers on earnings growth from higher sales of aluminum electrolytic capacitors for industrial and xEV applications. Although demand for high value-added capacitors for AI servers and xEVs is driving an earnings upturn, we think earnings are unlikely to return to the F3/23 peak in the near term, and profitability improvement will remain limited amid stronger overseas and MLCC competition. There is also a risk that Class B shares may be converted on favorable terms, or that Class A shares may need to be redeemed in cash with a premium added to the ¥10bn investment amount, as OP fell short of ¥8.8bn in F3/25.

Risk Reward Themes

Pricing Power: Negative
Secular Growth: Negative

View descriptions of Risk Rewards Themes [here](#)

BULL CASE	¥2,460	BASE CASE	¥880	BEAR CASE	¥400
Significant demand growth in AEC					
OP and EPS forecasts of ¥3.6bn/¥56.3 in F3/26, ¥5.8bn/¥105.9 in F3/27, ¥7.8bn/¥157.1 in F3/28, ¥9.8bn/¥204.5 in F3/29, and ¥12.0bn/¥259.3 in F3/30. Our calculations also assume that from F3/31 onward the firm can maintain FCF/share at ¥230. The fair value equates to 23.2x our F3/27E EPS of ¥105.9 and 1.0x BPS of ¥2,346.3.					
Earnings recover but factoring in dilution					
OP and EPS forecasts of ¥3.6bn/¥56.3 in F3/26, ¥5.6bn/¥102.3 in F3/27, ¥6.4bn/¥124.2 in F3/28, ¥7.1bn/¥142.4 in F3/29, and ¥7.4bn/¥149.7 in F3/30. Our calculations also assume that from F3/31 onward the firm can maintain FCF/share at ¥130. Our PT equates to 8.6x our F3/27E EPS of ¥102.3 and 0.4x BPS of ¥2,322.6.					
Earnings remain subdued at the trough					
OP and EPS forecasts of ¥3.6bn/¥56.3 in F3/26, ¥4.3bn/¥73.0 in F3/27, ¥4.4bn/¥76.7 in F3/28, ¥4.4bn/¥76.7 in F3/29, and ¥4.3bn/¥73.0 in F3/30. Our calculations also assume that from F3/31 onward the firm can maintain FCF/share at ¥100. The fair value equates to 5.5x our F3/27E EPS of ¥73.0 and 0.2x BPS of ¥2,313.4.					

Note: e = Morgan Stanley Research estimates; Stock price as of the May 15 close for Risk Reward Chart

Source: Company data, Morgan Stanley Research

KOA: Key Financial Data and Risk Reward

Income Statement (JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Sales	64.1	72.3	78.0	82.2	85.0
Costs of Sales	46.0	50.4	53.4	56.2	57.8
Gross Profit	18.2	21.9	24.6	26.0	27.2
SG&A	17.0	18.2	19.4	20.2	20.6
Operating Income	1.2	3.6	5.2	5.8	6.6
Non-operating income/expenses	0.1	1.6	-0.5	-0.4	-0.4
Recurring Profit	1.2	5.2	4.7	5.4	6.2
Extraordinary income	0.0	0.0	0.0	0.0	0.0
Profit before income taxes	1.3	5.3	4.7	5.4	6.2
Income taxes	1.0	1.3	1.4	1.7	1.9
(Effective tax rate)	79%	25%	30%	31%	31%
Profit attributable to non-controlling interests	0.0	0.0	0.0	0.0	0.0
Net Profit	0.3	4.0	3.3	3.7	4.3
Capex	24.2	5.9	7.4	7.8	8.1
Depreciation & amortization	5.9	7.0	7.4	7.6	7.8
R&D	3.6	3.6	3.9	4.1	4.3
EPS (JPY)	7.0	106.4	88.9	99.6	115.8
USD/JPY (Average)	152.6	151.1	150.0	150.0	150.0

Balance Sheet (JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Total assets	141.4	151.6	154.0	154.1	155.8
Current assets	57.5	62.7	69.0	69.2	70.6
Cash and cash equivalents	26.6	29.0	33.5	32.5	33.6
Accounts receivable	13.8	16.0	16.8	17.7	18.3
Inventories	13.4	13.5	15.2	15.3	15.1
Other financial assets	3.6	4.3	3.6	3.6	3.6
Fixed assets	83.9	88.8	84.9	85.0	85.1
Tangible fixed assets	71.5	74.5	72.6	72.7	72.8
Intangible fixed assets	3.2	3.0	3.1	3.1	3.1
Other non-current assets	9.2	11.3	9.2	9.2	9.2
Current liabilities	15.9	24.8	22.2	21.7	20.8
Short term borrowings	3.3	9.9	7.8	6.7	5.4
Accounts payable	5.0	3.2	6.0	6.4	6.6
Other current liabilities	7.6	11.7	8.3	8.7	8.9
Fixed liabilities	47.3	38.1	51.3	49.4	48.9
Long term borrowings	41.2	35.5	45.2	43.3	42.8
Other fixed liabilities	6.1	2.6	6.1	6.1	6.1
Net assets	78.1	88.6	80.5	83.1	86.1
Total liabilities and equity	141.4	151.6	154.0	154.1	155.8

Cashflow Statement (JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Cash flow from operating activities	8.1	8.8	9.8	11.0	12.2
EBIT	1.3	5.3	4.7	5.4	6.2
Depreciation	5.9	7.0	7.4	7.6	7.8
Changes in Receivables	0.3	-1.6	-1.3	-0.9	-0.6
Change in inventory	0.7	0.5	-0.6	-0.1	0.2
Changes in Payables	-0.5	-0.9	0.5	0.3	0.2
Other	0.5	-1.4	-0.8	-1.2	-1.5
Cash flow from investing activities	-23.9	-6.8	-7.4	-7.8	-8.1
Purchase of PP&E	-25.1	-7.4	-7.3	-7.7	-8.0
Sales of PP&E	0.0	0.0	0.0	0.0	0.0
Other	1.1	0.6	-0.1	-0.1	-0.1
Cash flow from financing activities	11.3	-0.5	3.0	-4.2	-3.1
Net Change in Borrowings	13.2	0.7	4.1	-3.0	-1.8
Purchase of treasury shares	0.0	0.0	0.0	0.0	0.0
Cash dividend paid	-1.8	-1.1	-1.1	-1.1	-1.3
Other	-0.1	-0.1	0.0	0.0	0.0
Effect of exchange rate changes	0.2	1.1	0.1	0.0	0.0
Changes in cash and cash equivalents	-4.4	2.6	5.5	-0.9	1.0
Cash and cash equivalents at end of the period	26.6	29.0	33.5	32.5	33.6

Risk Reward – KOA (6999.T)

Medium-term plan earnings targets set a high bar

PRICE TARGET ¥1,540

Our DCF-based price target is ¥1,540 (risk-free rate 2.2%, beta 1.2, risk premium 3.7%, WACC 5.0%). In addition to our three-year forecasts, we reference our forecasts for F3/29-30. For F3/31 onward, we assume FCF per share of ¥120. Our PT is equivalent to 17.3x our F3/27 EPS estimate of ¥88.9. Our forex rate assumption for the next quarter and thereafter is ¥150/\$.

RISK REWARD CHART



Key: — Historical Stock Performance • Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

BULL CASE	¥3,170	BASE CASE	¥1,540	BEAR CASE	¥610
Achieves medium-term earnings targets					
OP and EPS forecasts of ¥3.7bn/¥86.2 in F3/26, ¥6.0bn/¥102.3 in F3/27, ¥8.2bn/¥142.7 in F3/28, ¥10.5bn/¥188.5 in F3/29, and ¥12.4bn/¥223.5 in F3/30. Our calculations also assume that from F3/31 onward the firm can maintain FCF/share at ¥225. The fair value equates to 31.0x our F3/27E EPS of ¥102.3 and 15x BPS of ¥2,180.4.					
Earnings recovery is only gradual					
OP and EPS forecasts of ¥3.7bn/¥86.2 in F3/26, ¥5.2bn/¥88.9 in F3/27, ¥5.8bn/¥99.6 in F3/28, ¥6.6bn/¥115.8 in F3/29, and ¥6.8bn/¥121.2 in F3/30. Our calculations also assume that from F3/31 onward the firm can maintain FCF/share at ¥120. Our PT equates to 17.3x our F3/27E EPS of ¥88.9 and 0.7x BPS of ¥2,166.9.					
Earnings remain sluggish as sales growth slows					
OP and EPS forecasts of ¥3.7bn/¥86.2 in F3/26, ¥4.1bn/¥67.3 in F3/27, ¥4.1bn/¥70.0 in F3/28, ¥4.1bn/¥70.0 in F3/29, and ¥4.1bn/¥70.0 in F3/30. Our calculations also assume that from F3/31 onward the firm can maintain FCF/share at ¥70. The fair value equates to 9.1x our F3/27E EPS of ¥67.3 and 0.3x BPS of ¥2,145.4.					

Note: e = Morgan Stanley Research estimates; Stock price as of the May 15 close for Risk Reward Chart

Source: Company data, Morgan Stanley Research

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(as of April 30, 2026)

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Stock Rating Category	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
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Equal-weight/Hold	1568	43%	358	39%	23%	715	44%
Not-Rated/Hold	4	0%	0	0%	0%	1	0%
Underweight/Sell	555	15%	84	9%	15%	202	12%
Total	3,673		909			1627	

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INDUSTRY COVERAGE: Electronic Components

COMPANY (TICKER)	RATING (AS OF)	PRICE* (05/18/2026)
Shoji Sato		
ALPS ALPINE (6770.T)	O (03/17/2026)	¥2,204
Hamamatsu Photonics (6965.T)	U (09/17/2025)	¥2,867
Ibiden (4062.T)	U (02/04/2026)	¥15,635
Kyocera (6971.T)	E (06/25/2020)	¥2,834
Mabuchi Motor (6592.T)	E (11/03/2022)	¥1,566
Minebea Mitsumi (6479.T)	E (10/23/2025)	¥3,815
Murata Manufacturing (6981.T)	O (11/26/2025)	¥6,171
Nidec (6594.T)	NR (09/05/2025)	¥2,527
Niterra (5334.T)	O (01/17/2024)	¥9,855
Taiyo Yuden (6976.T)	E (10/19/2023)	¥7,194

TDK (6762.T)	O (08/02/2022)	¥2,978
Sota Harashima		
CMK (6958.T)	E (02/28/2025)	¥671
Daishinku (6962.T)	E (03/07/2024)	¥779
Hirose Electric (6806.T)	O (07/10/2024)	¥23,840
IRISO Electronics (6908.T)	E (08/02/2022)	¥3,375
Japan Aviation Electronics Industry (6807.T)	E (01/17/2024)	¥2,560
KOA (6999.T)	U (11/04/2025)	¥2,115
Meiko Electronics (6787.T)	E (04/03/2026)	¥39,800
Nichicon (6996.T)	E (11/10/2021)	¥2,872
Nihon Dempa Kogyo (6779.T)	E (03/07/2024)	¥2,342
Nippon Chemi-Con (6997.T)	U (09/20/2024)	¥2,944

Stock Ratings are subject to change. Please see latest research for each company.

* Historical prices are not split adjusted.